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Enhanced oil recovery

Enhanced oil recovery (EOR), also called improved oil recovery, covers a number of injection schemes aimed at improving hydrocarbon recovery compared to that achieved using the primary production mechanisms such as solution gas drive and water drive. All of the schemes add energy to the reservoir, for example using water injection to supplement natural water influx and gas injection to displace oil or to maintain gas cap pressure. As well as maintaining pressure drive, the physical and chemical properties of the injection fluids and gases can also improve the recovery efficiency.

Two major factors determine the recovery efficiency. The sweep efficiency determines the amount of the reservoir contacted by the injection fluids. The volumetric sweep efficiency comprises the areal sweep efficiency which is determined by the placement of the injecting and producing wells and the vertical sweep efficiency which is influenced by permeability distribution in the reservoir. The displacement efficiency is a measure of how much oil is displaced by the injection fluids. This is influenced by a number of factors, including the mobility ratio of the injection or displacing fluid and the oil which is the displaced fluid. If the oil moves faster than the injection fluid, which occurs if the mobility ratio is less than 1, this leads to a good recovery as the injection fluid pushes the oil ahead of it, leaving less oil isolated in the formation, reducing the amount of residual oil.

The EOR techniques aim to improve the recovery efficiency by changing the fluid viscosities and relative permeabilities and by reducing or eliminating the interfacial forces. The three main types of EOR are miscible, chemical and thermal.

Miscible EOR

In miscible injection, the boundaries between the injected displacing fluid and the displaced oil disappear and the oil dissolves into the injected fluid. Miscible injection fluids include hydrocarbons, liquified gases and carbon dioxide. As well as the miscible effects, the injected fluids or gases can improve recovery in different ways, for example, when gas dissolves in the oil, the oil swells. Eclipse has several models that you can use to model miscible displacement:

- [Miscible flood modeling](#) uses the Todd-Longstaff mixing parameter to model miscibility in a three-component system.
- The [Solvent model](#) is a four-component extension of the miscible flood model.

There are several models available for modeling carbon dioxide injection and storage. They deal with the properties of carbon dioxide, including its solubility in water, which has an impact on its behavior when injected into the reservoir.

- [CO2 injection: geological storage, EOR and ECBM](#)
- [CO2/H2S solution in the aqueous phase](#)
- [GASSOL solution in the aqueous phase](#)

Chemical EOR

In chemical enhanced oil recovery, chemicals are added to the injection fluid to improve the recovery from the injection scheme.

- In polymer flooding, a polymer is added to the injection water which increases the viscosity of the injected water. This improves the mobility ratio and the volumetric sweep efficiency which together increase the recovery efficiency. In a typical flooding scheme, a slug of the polymer is injected into the well and is pushed through the formation by conventional water injection. The [Polymer flood model](#) describes how Eclipse models polymer injection, taking into account factors such as polymer adsorption onto the rock surface.
- The reaction of alkaline solution such as sodium hydroxide with the crude oil in the reservoir produces surfactants in the reservoir (in situ surfactants). These surfactant products reduce interfacial tension and the residual oil saturation. For more information, see the [Alkaline model](#).
- Surfactants reduce the interfacial tension between oil and water in the reservoir. This reduces the residual oil saturation and improves the displacement efficiency. A typical method involves injecting a water-oil surfactant (a micellar solution) into the reservoir, followed by a polymer slug and then by conventional water injection. The [Surfactant model](#) describes how Eclipse models surfactants.
- Foams are typically produced by adding surfactant to the aqueous phase and using this mixture to trap gas bubbles. Foams decrease the mobility of the gas phase which can improve gas injection. The [Foam model](#) describes how Eclipse models foams, accounting for effects including the foam's degradation and adsorption onto the reservoir rocks.

Some of the chemical techniques are affected by salinity, for example the viscosity of some polymers is affected by the presence of sodium chloride. You can model the effects of salinity with chemical EOR using the [Brine tracking](#) facility.

Thermal EOR

Thermal techniques include injecting steam or hot water into the reservoir, or burning some of the crude oil in the formation. This creates water and steam zones which push the oil ahead of them. The heat reduces the oil viscosity and improves (decreases) the mobility ratio. It also reduces the interfacial tension, improves the relative permeability of the oil and reduces the residual oil saturation. The heated oil expands in the water zone and some oil is vaporized in the steam zone which again improves its mobility.

There are several thermal injection techniques. For example, in steam flooding, steam is injected continuously to improve recovery. In huff puff injection, steam is injected into a well which is then shut in. The area around the well is heated and then the well is reopened to recover the oil. Steam assisted gravity drainage is used to recover heavy oils. Steam is injected into a horizontal well which lies above a lower well. The heated oil becomes mobile and falls under the effect of gravity to be recovered in the lower well.

You can model thermal effects in Eclipse using the [Temperature](#) and [Thermal](#) options.

Miscible flood modeling

x	Eclipse 100
	Eclipse 300

This chapter describes the Eclipse 100 miscible flood model.

The miscible flood model is an implementation of the empirical treatment suggested by [Todd and Longstaff \(1972\)](#).

The Eclipse 100 solvent model also provides a four-component extension of the black oil model to enable modeling of reservoir recovery mechanisms in which injected fluids are miscible with the hydrocarbons in the reservoir to model gas injection projects without going to the complexity and expense of using a compositional model. The solvent extension implements the Todd and Longstaff empirical model for miscible floods and is flexible enough to model a wide range of gas injection schemes. See "[Solvent model](#)" for more information.

An alternative approach to miscibility is adopted in Eclipse 300, which is based on the use of surface tensions based on the Macleod-Sugden correlation; see "[Surface tension effects](#)" for details. This chapter refers only to the Todd-Longstaff model adopted in Eclipse 100.

Model formulation

The model is a three-component system consisting of reservoir oil, injection gas (solvent) and water. The reservoir oil component consists of stock tank oil together with the associated solution gas. The solvent and reservoir oil components are assumed to be miscible in all proportions and consequently only one hydrocarbon phase exists in the reservoir. The relative permeability requirements of the model are those for a two-phase system (water/hydrocarbon). The Todd-Longstaff mixing parameter technique requires modification of the viscosity and density calculations in a black oil simulator.

The Todd-Longstaff model is an empirical treatment of the effects of physical dispersion between the miscible components in the hydrocarbon phase. The model introduces an empirical parameter, ω , whose value lies between 0 and 1, to represent the size of the dispersed zone in each grid cell. The value of ω thus controls the degree of fluid mixing within each grid cell. A value of $\omega=1$ models the case when the size of the dispersed zone is much greater than a typical grid cell size and the hydrocarbon components can be considered to be fully mixed in each cell. In this case the miscible components have the same value for the viscosity and density, as given by the appropriate mixing rule formulae. A value of $\omega=0$ models the effect of a negligibly thin dispersed zone between the gas and oil components, and the miscible components should then have the viscosity and density values of the pure components. In practical applications an intermediate value of ω would be needed to model incomplete mixing of the miscible components.

In this miscible flood model, the injected gas is identified as a miscible agent (solvent) with the oil. The flow equations are a special case of the standard three-phase, three-component flow (water, oil and gas) where the gas-oil capillary pressure is set to zero because of the miscibility $P_{cog} = 0$.

A non-zero capillary pressure between the water and hydrocarbon phases, $P_{cnw}(S_w)$, which depends on the water phase saturation is however possible. The component pressures are then given by:

$$P_g = P_o \quad \text{Eq. 6.1}$$

$$P_w = P_o - P_{cnw}(S_w) \quad \text{Eq. 6.2}$$

Relative permeability model

The injected gas and reservoir oil are considered to be miscible components of the hydrocarbon (non-wetting) phase. The flow is two-phase in character and two-phase relative permeability curves need to be defined for the water and hydrocarbon phases, say $k_{rw}(S_w)$ and $k_{rn}(S_n)$ (where S_w is the water saturation

and $S_n = S_o + S_g$ is the hydrocarbon phase saturation). k_{rn} can be considered to be the relative permeability of oil as measured in a water flood test. The oil and gas relative permeabilities are then computed as saturation weighted fraction that is:

$$k_{rg} = \left(\frac{S_g - S_{gc}}{S_n - S_{gc} - S_{or}} \right) k_{rn}(S_n) \quad \text{Eq. 6.3}$$

$$k_{ro} = \left(\frac{S_o - S_{or}}{S_n - S_{gc} - S_{or}} \right) k_{rn}(S_n) \quad \text{Eq. 6.4}$$

It is possible to modify the straight line miscible relative permeabilities using the **MSFN** keyword. In this case the miscible gas and oil relative permeabilities are:

$$k_{rg} = M_{krsg} \left(\frac{S_g - S_{gc}}{S_n - S_{gc} - S_{or}} \right) k_{rn}(S_n) \quad \text{Eq. 6.5}$$

$$k_{ro} = M_{kro} \left(\frac{S_o - S_{or}}{S_n - S_{gc} - S_{or}} \right) k_{rn}(S_n) \quad \text{Eq. 6.6}$$

where the M_{krsg} and M_{kro} functions are supplied with the **MSFN** keyword. The saturations supplied in this keyword are normalized and honor the end-point supplied in the **SORWMIS** and **SGCWMIS** keywords. The critical oil-to-water saturation S_{or} is entered with the **SORWMIS** keyword and the critical oil to gas saturation S_{gc} is supplied with the **SGCWMIS** keyword.

Effect of water saturation

A feature of miscible gas injection processes that may also be modeled is the screening effect of high water saturation on the contact between the miscible gas and the oil in-place in each grid cell. The effective residual oil saturation to a miscible gas drive is found to increase with increasing water saturation and correct modeling of the effect is important since it may reduce the efficiency of the miscible displacement. The process is modeled by introducing an effective residual oil saturation, S_{or} , which depends on the water saturation ($S_{or} + S_{or}(S_w)$). A mobile oil saturation is then calculated by:

$$S_o^* = \text{MAX}(S_o - S_{or}, 0.0) \quad \text{Eq. 6.7}$$

where S_o is the oil saturation as above. The mobile oil saturation S_o^* is then used to determine the relative permeabilities of miscible components and the effective gas and oil viscosities and densities in each grid cell.

Note: Equations 6.5, 6.6 and 6.7 can produce a discontinuity in the oil and gas relative permeabilities at the oil saturation $S_o = S_{or}$ if the maximum value of the effective residual oil saturation exceeds the residual saturation of the oil to water flooding. Eclipse may experience non-linear convergence problems if this situation arises, but it issues a warning message during data checking.

Viscosity model

The effective oil and miscible gas viscosities follow the Todd-Longstaff model. They are given by:

$$\mu_{o \text{ eff}} = \mu_o^{1-\omega} \mu_m^\omega \quad \text{Eq. 6.8}$$

$$\mu_{g \text{ eff}} = \mu_g^{1-\omega} \mu_m^\omega \quad \text{Eq. 6.9}$$

The mixture viscosity μ_m is given by a 1/4-power fluid mixing rule

$$\frac{1}{\mu_m}^{1/4} = \frac{S'_g}{S'_n} \left(\frac{1}{\mu_g} \right)^{1/4} + \frac{S'_o}{S'_n} \left(\frac{1}{\mu_o} \right)^{1/4} \quad \text{Eq. 6.10}$$

which can be rewritten as

$$\mu_m = \frac{\mu_o \mu_g}{\left(\frac{S'_g}{S'_n} \cdot \mu_o^{1/4} + \frac{S'_o}{S'_n} \cdot \mu_g^{1/4} \right)^4} \quad \text{Eq. 6.11}$$

where the mobile saturations are defined by:

$$S'_o = S_o - S_{or}$$

$$S'_g = S_g - S_{gc}$$

$$S'_n = S'_o + S'_g$$

For a fully mixed state, that is $\omega = 1$, the effective viscosities are equal to the mixture viscosities, $\mu_{o \text{ eff}} = \mu_{g \text{ eff}} = \mu_m$. This case models a large dispersed oil-gas zone. The Todd-Longstaff model treats this case as a local unit mobility ratio displacement.

For a fully segregated case, that is $\omega = 0$, the effective viscosities are equal to the ‘pure’ phase viscosities, $\mu_{o \text{ eff}} = \mu_o$, $\mu_{g \text{ eff}} = \mu_g$. Such a case corresponds to a local high adverse mobility ratio displacement and models a negligibly thin oil-gas dispersed zone.

The mixing parameter approach allows the case of a partial mixing zone to be modeled by choosing an intermediate value of ω .

Density model

The treatment of effective oil and gas densities is based on the same 1/4-power rule as the effective viscosities. By default the density calculation will use the same mixing parameter as the viscosity. However, a separate mixing parameter may optionally be specified for the effective density calculation.

The densities are computed as follows. First, the partially mixed or effective viscosities are calculated using equations 6.8, 6.9 and 6.11. The value of each effective component viscosity is then used to yield an effective saturation fraction to be used in oil and gas density calculations:

$$\left(\frac{S_o}{S_n} \right)_{oe} = \frac{\mu_o^{1/4} (\mu_o^{1/4} - \mu_g^{1/4})}{\mu_o^{1/4} \cdot (\mu_o^{1/4} - \mu_g^{1/4})} \quad \text{Eq. 6.12}$$

$$\left(\frac{S_o}{S_n} \right)_{ge} = \frac{\mu_g^{1/4} (\mu_g^{1/4} - \mu_o^{1/4})}{\mu_g^{1/4} \cdot (\mu_o^{1/4} - \mu_g^{1/4})} \quad \text{Eq. 6.13}$$

The effective oil and gas densities ($\rho_{o \text{ eff}}$, $\rho_{g \text{ eff}}$) are now computed from the effective saturations fractions in expressions in equations 6.12 and 6.13 and the pure component densities (ρ_o , ρ_g) using the following formulae:

$$\rho_{o \text{ eff}} = \rho_o \left(\frac{S_o}{S_n} \right)_{oe} + \rho_g \left[1 - \left(\frac{S_o}{S_n} \right)_{oe} \right] \quad \text{Eq. 6.14}$$

$$\rho_{g \text{ eff}} = \rho_o \left(\frac{S_o}{S_n} \right)_{ge} + \rho_g \left[1 - \left(\frac{S_o}{S_n} \right)_{ge} \right] \quad \text{Eq. 6.15}$$

For the fully mixed case, that is $\omega = 1$, the effective saturations are equal:

$$\left(\frac{S_o}{S_n}\right)_{oe} = \left(\frac{S_o}{S_n}\right)_{ge} = \frac{\mu_o^{1/4}(\mu_m^{1/4} - \mu_g^{1/4})}{\mu_m^{1/4} \cdot (\mu_o^{1/4} - \mu_g^{1/4})}$$

and the effective densities are therefore equal, $\rho_{o\text{ eff}} = \rho_{g\text{ eff}} = \rho_m$.

For the unmixed, fully segregated case, that is $\omega = 0$, the effective saturations simplify to

$$\left(\frac{S_o}{S_n}\right)_{oe} = 1, \left(\frac{S_o}{S_n}\right)_{ge} = 0$$

and the effective densities are equal to the “pure” phase densities $\rho_{o\text{ eff}} = \rho_o$, $\rho_{g\text{ eff}} = \rho_g$.

The model allows you to specify a different surface density for the solvent gas to that for the solution gas. In equations 6.14 and 6.15, the pure solvent gas density at reservoir conditions, ρ_g , is computed from the solvent gas surface density input with the `SDENSITY` keyword. The oil density at reservoir conditions, ρ_o , is calculated from the stock tank oil and solution gas surface densities.

Note that equations 6.12 and 6.13 are singular for unit mobility ratio displacements. In that case, where $\mu_o = \mu_g$, the effective densities are computed as follows. First a mixture density is defined as:

$$\rho_m = \rho_o \left(\frac{S_o}{S_n}\right) + \rho_g \left(\frac{S_g}{S_n}\right) \quad \text{Eq. 6.16}$$

Then:

$$\rho_{o\text{ eff}} = (1-\omega)\rho_o + \omega\rho_m \quad \text{Eq. 6.17}$$

$$\rho_{g\text{ eff}} = (1-\omega)\rho_g + \omega\rho_m \quad \text{Eq. 6.18}$$

where ω is the Todd and Longstaff parameter input with the `TLMIXPAR` keyword.

Todd and Longstaff mixing parameter

The mixing parameter model would be of limited use unless the mixing parameter could itself be modeled over a wide range of operating conditions. A value of $\omega=1$ results in a piston-like displacement of oil by the injected gas. If $\omega=0$ the displacement is similar to an immiscible displacement (except for the treatment of relative permeability). An intermediate value of ω results in a continuous gas saturation increase behind the gas front. Todd-Longstaff accounted for the effects of viscous fingering in two dimensional studies by setting $\omega=2/3$ independent of mobility ratio. For field scale simulations they suggested setting $\omega=1/3$. However, in general history matching applications the mixing parameter may be regarded as a useful history matching variable to account for any reservoir process inadequately modeled.

The control of miscibility with pressure

In many miscible displacements, the gas is only miscible with the reservoir oil at high pressure. Typically the gas-oil capillary pressure reduces with increasing pressure, and only when it has reduced to zero can the two fluids be considered to be miscible. It is possible to model the transition between miscibility and immiscibility as a function of pressure by using the `PMISC` keyword.

The `PMISC` function interpolates between the immiscible and miscible values of the PVT properties, relative permeability data and capillary pressure data.

The PVT properties are interpolated as follows (taking gas properties as an example):

$$\frac{1}{B_g} = \frac{1}{B_{gm}} M_p + \frac{1}{B_{gi}} (1-M_p) \quad \text{Eq. 6.19}$$

$$\frac{1}{B\mu_g} = \frac{1}{B\mu_{gm}}M_p + \frac{1}{B\mu_{gi}}(1-M_p) \quad \text{Eq. 6.20}$$

where

B_g = gas formation volume factor

B_{gi} = immiscible gas formation volume factor

B_{gm} = miscible gas formation volume factor

$B\mu_g$ = gas FVF * viscosity

$B\mu_{gi}$ = gas FVF * viscosity (immiscible)

$B\mu_{gm}$ = gas FVF * viscosity (miscible)

M_p = pressure dependent miscibility function.

The miscible formation volume factor is calculated from the Todd-Longstaff density:

$$B_{gm} = \frac{\rho_{g,s}}{\rho_{g\text{ eff}}} \quad \text{Eq. 6.21}$$

where

$\rho_{g,s}$ = surface density of solvent gas (**SDENSITY** keyword)

$\rho_{g\text{ eff}}$ = effective solvent density in equations 6.15 or 6.18.

The relative permeability interpolation algorithm has two steps:

1. Scale the relative permeability end-points by the miscibility function. For example, the residual oil saturation:

$$S_{or} = S_{orm}M_p + S_{ori}(1-M_p)$$

where

S_{orm} = miscible residual oil

S_{ori} = immiscible residual oil

M_p = pressure dependent miscibility function.

2. Calculate the miscible and immiscible relative permeability scaling for the new end points. Then the relative permeability is again an interpolation between the two using the miscibility function:

$$k_r = k_{rm}M_p + k_{ri}(1-M_p)$$

where

k_{rm} = scaled miscible k_r

k_{ri} =scaled immiscible k_r .

Because the miscible capillary pressure is zero, the capillary pressure is treated as follows:

$$P_{cog} = P_{cog}(\text{immisc}) (1 - M_p) \quad \text{Eq. 6.22}$$

Control of numerical diffusion in miscible flow simulation

The relative permeability of the oil and solvent gas components in the Todd-Longstaff model are specified by equations 6.3 and 6.4 or 6.5 and 6.6. The relative permeability of each component is essentially proportional to the saturation of each component for low saturations. In such cases the influence of numerical diffusion in spreading the miscible components can be unacceptably high, and some control of the numerical diffusion errors is required.

A two-point upstream algorithm operating only on the miscible component relative permeabilities is available for use in the Eclipse Miscible Flood option to control the numerical diffusion of oil and solvent gas. The two-point upstream projection technique is illustrated in the following figure.

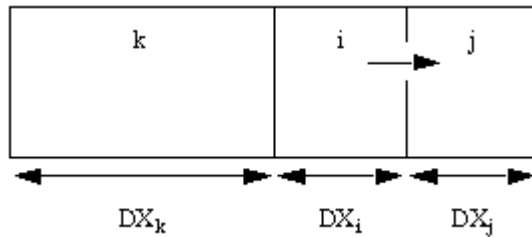


Figure 6.1. The two-point upstream projection technique

Component C flows between cells i and j , which are aligned along the X -axis. Cell i is the upstream cell for the flow of component C between i and j . Cell k is the second upstream cell to cell j and K_{ck} , K_{ci} and K_{cj} denote the component relative permeabilities in cells k , i and j respectively. DX_k , DX_i and DX_j are the X -direction cell sizes.

For single point upstream weighting, the flow rate of component C across the i - j cell interface is $F_{ij,c}$ where:

$$F_{ij,c} = T_{ij} \frac{K_{ci}}{B_{ci} \mu_{ci}} DP_{ij,c} \quad \text{Eq. 6.23}$$

where

T_{ij} is the transmissibility between cells i and j

$DP_{ij,c}$ is the potential difference between cells i and j for component C

B_c , μ_c are the formation volume factor and viscosity of component C

The two-point upstreaming algorithm operating on component relative permeabilities replaces K_{ci} in the equation 6.23 by a linearly extrapolated relative permeability, K_{ci}^* , obtained from cells i and k ,

$$K_{ci}^* = \frac{2DX_i + DX_k}{DX_i + DX_k} K_{ci} - \frac{DX_i}{DX_i + DX_k} K_{ck} \quad \text{Eq. 6.24}$$

The above expression for K_{ci}^* can yield negative values or values greater than unity, and in such cases needs to be restricted to physical values. This is most simply achieved by constraining K_{ci}^* to lie between the values K_{ci} and K_{cj} . Thus:

$$\text{MIN}(K_{ci}, K_{cj}) < K_{ci}^* < \text{MAX}(K_{ci}, K_{cj}) \quad \text{Eq. 6.25}$$

In some cases a single point upstream calculation of K_{ci}^* (that is $K_{ci}^* = K_{ci}$) should be used instead of the two-point upstream value. Such cases are

- Cell k lies outside the reservoir mesh (that is the k-i cell boundary is an edge of the reservoir).
- Cell k is inactive.
- The k-i cell boundary contains a zero transmissibility multiplier.
- Cells i and j are coupled by a non-neighbor connection.

Using the Eclipse miscible flood option

The Eclipse Miscible Flood option can only be used in oil-gas and oil-gas-water combinations (that is gas and oil are treated as the miscible components). The keyword **MISCIBLE** in the RUNSPEC section activates the model. The two-point upstreaming calculation of component relative permeabilities is achieved by setting item 3 in the **MISCIBLE** keyword to **TWOPOINT**, to allow comparison of runs with and without numerical diffusion control.

The keyword **MISCNUM** should be used in the REGIONS section to define the miscibility region to which each grid cell belongs. This keyword is not obligatory, and a default region number of 1 is assumed. The maximum region number that can be defined (**NTMISC**) must be entered in item 1 of the keyword **MISCIBLE** in the RUNSPEC section. Echo of the **MISCNUM** region data can be obtained using the mnemonic **MISCNUM** in the **RPTREGS** keyword.

It is possible to override the miscible flow calculation of the gas and oil properties in certain cells in the grid. If a cell is allocated a **MISCNUM** region number of zero, then the gas and oil will be assumed to be immiscible under all conditions in the cell.

The mixing parameter, ω , should be input in the PROPS section under the keyword **TLMIXPAR**. A single real number in the range 0 to 1 must be supplied for each miscible region. If a different mixing parameter is required for the effective densities, a second mixing parameter can be supplied for each miscible region. If the grid contains any cells whose **MISCNUM** region is zero, the mixing parameter is automatically set to zero in those cells.

For gas-oil runs, the keywords **SGFN** and **SOF2** should only be entered if there is at least one cell which has a **MISCNUM** region number of zero. For those cells that are allocated a positive **MISCNUM** region number, Eclipse assumes that the gas-oil capillary pressure is zero and the relative permeabilities are equal to their respective saturations.

In oil-water-gas runs the keywords **SOF2** and **SWFN** are obligatory. In the miscible regions, the **SWFN** data defines the water relative permeability and water-hydrocarbon capillary pressure in the usual way. The **SOF2** keyword represents the relative permeability of the hydrocarbon phase with respect to hydrocarbon phase saturation (the function $k_m(S_n)$ in equations 6.4 and 6.3). If there are any immiscible regions in the grid, the **SOF3** and **SGFN** keywords must be used to enter the immiscible gas-oil relative permeabilities.

The hysteresis model is available in both the miscible and the immiscible zones. In the miscible zones, a two phase treatment (water/hydrocarbon) of the hysteresis effect is invoked on the water-hydrocarbon capillary pressures and relative permeabilities.

The Saturation Table Rescaling option is also available in both the miscible and immiscible zones of the grid. In the miscible regions, the saturation table endpoints for the hydrocarbon phase should be entered as oil-in-water data using the appropriate scaling keywords. Data entered for gas end-points will be ignored in the miscible zones.

The miscible residual oil saturation and critical gas saturation versus water saturation tables can be defined in the PROPS section using the keywords `SORWMIS` and `SGCWMIS`. A maximum of `NTMISC` tables, each with a maximum of `NSMISC` rows of data, are allowed. The `SORWMIS`/`SGCWMIS` keywords are not obligatory and an effective residual oil saturation of 0.0 and critical gas value of 0.0 will be assumed if the appropriate keyword is omitted. The miscible gas-oil relative permeability curves can be modified using the `MSFN` keyword. If the `MSFN` keyword is omitted, straight line relative permeability curves are used.

The calculation of the initial reservoir state in the miscible zones is performed using capillary pressures defined in equations 6.1 and 6.2 with end points implied from equations 6.3 and 6.4 or 6.5 and 6.6.

The surface densities of the stock-tank oil, water and solution gas components should be defined using the existing `DENSITY` or `GRAVITY` keyword. The surface density of the injection gas must be entered using the `SDENSITY` keyword in the PROPS section. In the immiscible zones, the injection gas surface density is taken from the `DENSITY` or `GRAVITY` keyword to equal that of the solution gas. An echo of the surface density of each component can be obtained using the mnemonic `DENSITY` in the `RPTPROPS` keyword.

Specification of pressure function data for the pure components is unaltered from the usual data requirements of Eclipse. If the oil is defined as dead (no `DISGAS` keyword in `RUNSPEC`) then the initial solution gas is omitted from the GOR reports in the results file and the GOR refers to the solvent gas-oil ratio. It is possible to include the initial solution gas in the GOR reports by defining the oil component to be live (keyword `DISGAS` in `RUNSPEC`). In either case, the reported saturation values for oil and solvent gas refer to the relative volume occupied by each component as if they were not mixed.

The pressure dependence of miscibility can be controlled by using the `PMISC` keyword. This keyword is optional; if it is not present then miscibility is assumed at all pressures.

If `DISGAS` has been specified in the `RUNSPEC` section, the value of `DRSDT` refers to any immiscible zones in the grid. Its default value in these regions will be infinity. In the miscible zones, the value of `DRSDT` is always zero.

No new keywords have been added in the `SUMMARY` file section. The existing graphics facilities allow line graphs to be obtained of total gas injection / production rates for connections, wells, groups and the field; also for example, for cumulative totals, inter-region gas flows and gas-in-place. If the injected gas needs to be distinguished from the initial solution gas, then the existing tracer option allows separate line graphs of the above quantities to be produced to distinguish the two types of gas.

The two-point upstreaming technique is intended for use in dispersed flow models and is not compatible with the calculation of relative permeability in the Vertical Equilibrium option. The Todd-Longstaff model is based on a dispersed fluid distribution throughout each grid cell. It is therefore suggested that the Vertical Equilibrium model should not be used with the Miscible Flood option.

The two-point upstreaming algorithm produces substantial reductions in the level of numerical diffusion associated with component relative permeabilities. However, the algorithm inevitably involves an increase in the property evaluation and matrix element computation time as well as an increase in the work performed by the linear solver. Experience on small problems has indicated an approximate increase of around 50% in run times due to the use of two-point upstreaming.

Summary of keywords

RUNSPEC

- `MISCIBLE` Initiates the mixing calculation.

PROPS

Keyword	Description
<code>MSFN</code>	Miscible gas/oil saturation functions.
<code>PMISC</code>	Pressure dependent miscibility tables.
<code>RPTPROPS</code>	Controls output from the PROPS section. Argument PMISC outputs the miscibility function and the pressure miscibility function.
<code>SDENSITY</code>	Solvent density at surface conditions.
<code>SGCWMIS</code>	Miscible critical gas saturation tables.
<code>SORWMIS</code>	Miscible residual oil saturation tables.
<code>TLMIXPAR</code>	Todd-Longstaff mixing parameter.

REGIONS

Keyword	Description
<code>MISCNUM</code>	Miscibility region numbers.
<code>RPTREGS</code>	Controls output from the REGIONS section. Argument MISCNUM outputs miscibility region numbers.

SOLUTION

Keyword	Description
<code>RPTSOL</code>	Controls output from the SOLUTION section. Use as for gas phase output.

SUMMARY

No specific keyword is required for the miscible option. The outputs are those available for the gas phase.

SCHEDULE

- `RPTSCHED` Controls output from the SCHEDULE section. Use as for gas phase output.

Solvent model

x	Eclipse 100
	Eclipse 300

This chapter describes a four-component extension of the Eclipse black-oil model. Please refer to "[Miscible flood modeling](#)" for more detail on the three-component modeling. The aim of the extension is to enable modeling of reservoir recovery mechanisms in which injected fluids are miscible with the hydrocarbons in the reservoir.

An injected fluid sets up a miscible displacement if there is no phase boundary or interface between the injected fluid and the reservoir oil. A miscible displacement has the advantage over immiscible displacements such as water flooding, of enabling very high recoveries. An area swept by a miscible fluid typically leaves a very small residual oil saturation.

In miscible gas injection processes, total displacement of the oil in the swept regions does not guarantee a high overall recovery efficiency. This is because the high adverse mobility ratio inherent in gas injection means that the flow is unstable and leads to the growth of viscous fingers that leave regions of bypassed oil in the reservoir at gas breakthrough. Gravitational fingering may also be present as a result of the large density differences between the injected gas and the in-place oil. The flow is unstable to finger formation at mobility ratios greater than unity, or in places where a dense fluid overlies a less dense fluid, and the fingers grow rapidly from small fluctuations in reservoir homogeneity. Reservoir simulation models must account for the growth of fingers whenever they might arise in the reservoir.

A further feature of miscible fluid displacement to be included in a simulation model is the mixing of the miscible components. Component mixing occurs on a microscopic scale and results from molecular diffusion and fluid velocity variations within the porous medium on the scale of the pore size. The mixing process is represented by a diffusion term in the component equations. Correct treatment of this mixing effect within the simulation model is important since the mixing term exerts a considerable damping effect on the growth of viscous and gravity fingers.

The simulation of incompressible miscible displacement was initially based on a direct solution of the convection-diffusion equations for the local concentration of each of the miscible components. However, such models are unreliable unless the fine scale structure of the fingers can be resolved, otherwise mixing cell effects will dampen the finger growth rates and produce optimistic oil recovery forecasts. The numerical diffusion inherent in finite difference models also masks the true mixing of the miscible components unless the grid is sufficiently fine. The number of grid cells required for accurate field-scale study of miscible displacement is unacceptably high and the cost of computing time prohibitive.

The Eclipse implementation

The model is optionally a four-component system consisting of water, oil, reservoir gas and an injected solvent, or a three-component system of water, oil and solvent gas or an oil/solvent system.

In regions of the reservoir containing only solvent and reservoir oil (possibly containing dissolved gas), the solvent and reservoir oil components are assumed to be miscible in all proportions and consequently only one hydrocarbon phase exists in the reservoir. The relative permeability requirements of the model are those for a two-phase system (water/hydrocarbon).

In regions of the reservoir containing only oil and reservoir gas, the gas and oil components will be immiscible and will behave in a traditional black oil manner. In regions containing both dry gas and solvent, an intermediate behavior is assumed to occur. The modeling of the immiscible/miscible transition is described in subsequent sections.

The Todd-Longstaff model is an empirical treatment of the effects of physical dispersion between the miscible components in the hydrocarbon phase. The model introduces an empirical parameter, ω , whose value lies between 0 and 1, to represent the size of the dispersed zone in each grid cell. The value of ω thus

controls the degree of fluid mixing within each grid cell. A value of $\omega = 1$ models the case where the size of the dispersed zone is much greater than a typical grid cell size and the hydrocarbon components can be considered to be fully mixed in each cell. In this case the miscible components have the same value for the viscosity and density, as given by the appropriate mixing rule formulae. A value of $\omega = 0$ models the effect of a negligibly thin dispersed zone between the gas and oil components, and the miscible components should then have the viscosity and density values of the pure components. In practical applications an intermediate value of ω would be needed to model incomplete mixing of the miscible components. By default the viscosity and density values will be calculated using the same mixing parameter, ω . However, if the mixing parameter is being used as a history matching parameter, it is sometimes useful to specify two separate mixing parameters, one for the viscosities and one for the densities.

Applications

There is a wide variety of secondary and tertiary recovery schemes in which the aim is to enhance the reservoir sweep by using a miscible injection fluid.

Typically the miscible injection fluid is expensive compared with traditional flooding fluids, such as dry gas and water. Miscible fluids are often referred to as ‘solvents’; examples of solvent schemes are listed below:

- High pressure dry gas processes, in which mass transfer effects at the gas-oil contact generate miscible flow conditions between the gas and the oil.
- A solvent such as LPG or propane may be injected as a ‘slug’ to be followed by an extended period of lean gas injection. The slug fluid is miscible with both the gas and oil, and the success of the method depends upon maintenance of both the reservoir pressure and the solvent slug itself.
- Certain non-hydrocarbon gases such as carbon dioxide produce miscible displacement of oil at pressures above a threshold value.
- All-liquid miscible displacements by fluids such as alcohol, normally injected as a slug between the in-place oil and the injected chase water.

In many of the schemes, the cost of the solvent is such that the recovery scheme needs to be ‘engineered’ to allow enough solvent to be injected to increase the total oil recovery while keeping the solvent costs low enough to make the project economic.

In applications where the reservoir pressure falls below the bubble point, and in schemes where a solvent slug is chased by a lean gas flood, there is a requirement to model four components within the reservoir (water, oil, solvent and lean gas). The solvent model in Eclipse provides a simple method of modeling these recovery mechanisms.

Model formulation

The solvent model is an extension of the standard three-phase, three-component model (water, oil, and gas), whereby an extra conservation equation is solved for the solvent. Below we detail the modeling of the different flow parts and the mixing rules used which are based on the empirical treatment suggested by *Todd and Longstaff (1972)*.

Relative permeability model: immiscible case

In regions of the reservoir where the saturation of solvent is small, for example in the region swept by lean chase gas, the displacement is immiscible. In the usual black-oil model the relative permeabilities for the three phases water, oil and gas are specified as follows:

$k_{rw} = k_{rw}(S_w)$ as a function of water saturation

$k_{rg} = k_{rg}(S_g)$ as a function of gas saturation

$k_{ro} = k_{ro}(S_w, S_g)$ as a function of both water and gas saturations.

When two gas components are present, the assumption is made that the total relative permeability of the gas phase is a function of the total gas saturation,

$$k_{rgt} = k_{rg}(S_g + S_s) \quad \text{Eq. 6.26}$$

where S_s is the solvent saturation.

This total relative permeability is obtained from the **SGFN** keyword. Then the relative permeability of each gas component is taken as a function of the fraction of each gas component within the gas phase as:

Fraction of solvent:

$$F_{sol} = \frac{S_s}{S_g + S_s} \quad \text{Eq. 6.27}$$

Fraction of reservoir gas:

$$F_{gas} = \frac{S_g}{S_g + S_s} \quad \text{Eq. 6.28}$$

$$\begin{aligned} k_{rs} &= k_{rgt} \cdot k_{rfs}(F_{sol}) \\ k_{rg} &= k_{rgt} \cdot k_{rfg}(F_{gas}) \end{aligned} \quad \text{Eq. 6.29}$$

Typically the functions $k_{rfs} = k_{rs}/k_{rgt}$ and $k_{rfg} = k_{rg}/k_{rgt}$ (equation 6.29) are ‘straight line’ functions such that

$$k_{rfs}(0.0) = 0.0 \quad k_{rfg}(0.0) = 0.0$$

$$k_{rfs}(1.0) = 1.0 \quad k_{rfg}(1.0) = 1.0$$

However, Eclipse allows you to specify these gas-solvent functions using the **SSFN** keyword.

Relative permeability model: miscible case

In regions where solvent is displacing oil, the hydrocarbon displacement is miscible. However, the two-phase character of the water/hydrocarbon displacement needs to be taken into account. The relative permeabilities are given by:

$$k_{ro} = \frac{S_o}{S_n} k_{rn}(S_n) \quad \text{Eq. 6.30}$$

$$k_{rgt} = \frac{S_s + S_g}{S_n} k_{rn}(S_n) \quad \text{Eq. 6.31}$$

where

$S_n = S_g + S_s + S_o$ is the total hydrocarbon saturation,

k_{rgt} is the total relative permeability of the gas and solvent,

$k_{rn}(S_n)$ is the relative permeability of hydrocarbon to water obtained from the [SOF2](#) keyword.

Then, the gas and solvent component relative permeability are computed from k_{rgt} as above, equations [6.27](#) - [6.29](#). It is possible to modify the straight line miscible relative permeabilities, using the [MSFN](#) keyword to supply multipliers M_{kro} and M_{krsg} . The oil and total gas relative permeabilities will then be:

$$k_{ro} = M_{kro} \left(\frac{S_o}{S_n} \right) k_{rn}(S_n) \quad \text{Eq. 6.32}$$

$$k_{rgt} = M_{krsg} \left(\frac{S_s + S_g}{S_n} \right) k_{rn}(S_n) \quad \text{Eq. 6.33}$$

Note that saturations end-points are taken into account in the previous equations as explained below.

Transition between miscible and immiscible cases

The descriptions above are the treatments of the extremes when the displacement behavior is either fully miscible or fully immiscible. In practice there will be regions of transition where the displacement is changing from miscible to immiscible in character. Typically this occurs when a lean chase gas breaks through the solvent slug. Todd and Longstaff recommend that the displacement will be miscible until the solvent saturation becomes very small (of the order of 0.01). However, this transition is likely to vary from case to case.

The transition is handled by a user-specified ‘Miscibility function’. This is a function of the solvent fraction in the gas phase. The function is tabulated between 0 and 1, where 0 implies immiscible displacement and 1 implies miscible. Note that in an implicit simulator like Eclipse, the transition needs to occur over a range of solvent fraction rather than at a ‘cut-off’ point. The miscibility function is supplied using the [MISC](#) keyword.

The transition algorithm has two steps:

1. Scale the relative permeability end points by the miscibility function. The residual oil saturation is

$$S_{or} = S_{orm} M + S_{ori}(1-M) \quad \text{Eq. 6.34}$$

where

S_{orm} is the miscible residual oil saturation (this is taken from [SORWMIS](#), or defaulted to zero if [SORWMIS](#) is not present)

S_{ori} is the immiscible residual oil saturation (this is taken from S_{ogcr} in the [SOF3](#) table, or [SOGCR](#) if entered)

M is the miscibility function entered using the [MISC](#) keyword

Similarly, for the solvent gas

$$S_{sgr} = S_{sgrm} M + S_{sgri}(1-M) \quad \text{Eq. 6.35}$$

where

S_{sgrm} is the miscible residual solvent + gas saturation (taken from [SGCWMIS](#) or defaulted to zero)

S_{sgr} is the immiscible residual solvent + gas saturation (taken from S_{gcr} in the [SGFN](#) table, or [SGCR](#) if entered)

2. Calculate the miscible and immiscible relative permeabilities at the scaled saturations using the new end points. Then the relative permeability is again an interpolation between the two using the miscibility function:

$$k_r = k_{rm} M + k_{ri}(1-M) \quad \text{Eq. 6.36}$$

where

k_{rm} the scaled miscible relative permeability

k_{ri} is the scaled immiscible relative permeability

The formulae above, equations 6.30 to 6.33, are modified to take into account these new saturation end points, so that:

$$k_{rgt} = \left(\frac{S_g + S_s - S_{sgr}}{S_n - S_{sgr} - S_{or}} \right) k_{rn}(S_n) \quad \text{Eq. 6.37}$$

$$k_{ro} = \left(\frac{S_o - S_{or}}{S_n - S_{sgr} - S_{or}} \right) k_{rn}(S_n) \quad \text{Eq. 6.38}$$

Modifying the straight line miscible functions using the [MSFN](#) keyword will result in

$$k_{rgt} = M_{krsg} \left(\frac{S_g + S_s - S_{sgr}}{S_n - S_{sgr} - S_{or}} \right) k_{rn}(S_n) \quad \text{Eq. 6.39}$$

$$k_{ro} = M_{kro} \left(\frac{S_o - S_{or}}{S_n - S_{sgr} - S_{or}} \right) k_{rn}(S_n) \quad \text{Eq. 6.40}$$

where the M_{kro} and M_{krsg} functions are supplied with the [MSFN](#) keyword.

The saturations supplied in this keyword are normalized and honor the end-point supplied in the [SORWMIS](#) and [SGCWMIS](#) keywords. The critical oil to water saturation S_{or} is entered with the [SORWMIS](#) keyword and the critical oil to gas saturation S_{gc} is supplied with the [SGCWMIS](#) keyword as a function of the water saturation. See "[Effect of water saturation](#)".

Pressure effect on relative permeability and capillary pressure miscibility

In many miscible displacements, the solvent is only miscible with the reservoir oil at high pressure. Typically the solvent/oil capillary pressure will reduce with increasing pressure, and only when it has reduced to zero can the two fluids be considered to be miscible.

The pressure dependence can be modeled by using the [PMISC](#) keyword to supply a pressure dependent miscibility function M_p . This is used to interpolate between the immiscible and miscible values of the PVT, relative permeability and capillary pressure data. Here we described its usage with the relative permeability and capillary pressure.

For the relative permeabilities and saturation end points, the effect of pressure miscibility is combined with the solvent saturation [MISC](#) effects, so that the multiplier M coefficient in equations 6.34 to 6.36 is now replaced by:

$$M_T = M M_p \quad \text{Eq. 6.41}$$

The capillary pressure is interpolated as follows:

$$P_{cog} = M_p P_{cog\ m} + (1 - M_p) P_{cog\ i} \quad \text{Eq. 6.42}$$

$$\begin{aligned} P_{cog\ m} &= P_{cog}(S_g) \\ P_{cog\ i} &= P_{cog}(S_g + S_s) \end{aligned} \quad \text{Eq. 6.43}$$

Effect of water saturation

A feature of miscible gas injection processes that may also be modeled is the screening effect of high water saturation which reduces the contact between the miscible gas and the oil in-place. The effective residual oil saturation to a miscible gas drive is found to increase with increasing water saturation, and correct modeling of the effect is important since it may reduce the efficiency of the miscible displacement. The process is modeled by introducing an effective residual oil saturation, S_{or} , which depends on the water saturation ($S_{or} = S_{or}(S_w)$) and is input with the **SORWMIS** keyword. A mobile oil saturation is then calculated by

$$S_o^* = \text{MAX}(S_o - S_{or}, 0.0) \quad \text{Eq. 6.44}$$

For completeness, a corresponding mobile gas saturation has been defined in which a critical gas saturation ($S_{gc} = (S_{gc})(S_w)$) is specified for the miscible flood and input with the **SGCWMIS** keyword. Then the mobile gas saturation is taken as:

$$S_g^* = \text{MAX}(S_g - S_{gc}, 0.0) \quad \text{Eq. 6.45}$$

The mobile oil and gas saturations S_o^* , S_g^* are used to determine the miscible component relative permeabilities and the effective viscosities and densities in each grid cell. When end-point scaling is used in conjunction with **SORWMIS**, then S_{orm} is replaced by the scaled value of the critical hydrocarbon to water saturation taken from the **SOF2** table (or keyword **SOGCR** if entered). When end-point scaling is used in conjunction with **SGCWMIS**, then S_{sgrm} is replaced by the scaled value of **SGCR**.

Viscosity model

The full three hydrocarbon components mixing calculation can be thought of as two separate miscible displacements: gas/solvent and solvent/oil. The effective viscosities of the hydrocarbon components follow the Todd and Longstaff model:

$$\begin{aligned} \mu_{o\ \text{eff}} &= \mu_o^{1-\omega} \cdot \mu_{mos}^\omega \\ \mu_{s\ \text{eff}} &= \mu_s^{1-\omega} \cdot \mu_m^\omega \\ \mu_{g\ \text{eff}} &= \mu_g^{1-\omega} \cdot \mu_{msg}^\omega \end{aligned} \quad \text{Eq. 6.46}$$

where

μ_o , μ_s , μ_g are the component viscosities of oil, solvent and gas.

μ_{mos} is the fully mixed viscosity of oil + solvent

μ_{msg} is the fully mixed viscosity of solvent + gas

μ_m is the fully mixed viscosity of oil + solvent + gas

ω is the Todd-Longstaff parameter.

The mixture viscosities μ_{mos} , μ_{msg} and μ_m are defined using the 1/4th-power fluid mixing rule, as follows:

$$\mu_{mos} = \frac{\mu_o \mu_s}{\left(\frac{S'_o}{S'_{os}} \mu_s^{1/4} + \frac{S'_s}{S'_{os}} \mu_o^{1/4} \right)^4} \quad \text{Eq. 6.47}$$

$$\mu_{msg} = \frac{\mu_s \mu_g}{\left(\frac{S'_s}{S'_{sg}} \mu_g^{1/4} + \frac{S'_g}{S'_{sg}} \mu_s^{1/4} \right)^4} \quad \text{Eq. 6.48}$$

$$\mu_m = \frac{\mu_o \mu_s \mu_g}{\left(\frac{S'_o}{S'_n} \mu_s^{1/4} \mu_g^{1/4} + \frac{S'_s}{S'_n} \mu_o^{1/4} \mu_g^{1/4} + \frac{S'_g}{S'_n} \mu_o^{1/4} \mu_s^{1/4} \right)^4} \quad \text{Eq. 6.49}$$

where the mobile saturations are defined by

$$S'_o = S_o - S_{or}$$

$$S'_g = S_g - S_{gc}$$

$$S'_s = S_s - S_{gc}$$

$$S'_n = S'_o + S'_g + S'_s,$$

$$S'_{os} = S'_o + S'_s \text{ and}$$

$$S'_{sg} = S'_s + S'_g$$

In the special case of no mixing, $\omega = 0$, the effective viscosities, equation 6.46 are equal to the pure phase viscosities, while in the case of full mixing $\omega = 1$ the effective viscosities are equal to the mixed viscosities defined by equations 6.47 - 6.49.

These effective viscosities are also used to compute the effective densities as detailed below.

Density model

The treatment of effective oil, gas and solvent densities is based on the fourth power law for the effective viscosities above. By default, the density calculation can be based on the same mixing parameter as the viscosity, however a separate mixing parameter may be specified for the effective density calculation if required.

The densities are computed by the following procedure. First, the effective viscosities are calculated using equations 6.46 to 6.49 above. The value of each effective component viscosity is then used in turn in equations 6.47 to 6.49 to yield an effective saturation fraction to be used in oil, gas and solvent density calculations as:

$$\left(\frac{S_o}{S_n} \right)_{oe} = \frac{\mu_o^{1/4} (\mu_o^{1/4} - \mu_s^{1/4})}{\mu_o^{1/4} (\mu_o^{1/4} - \mu_s^{1/4})} \quad \text{Eq. 6.50}$$

$$\left(\frac{S_o}{S_n} \right)_{ge} = \frac{\mu_s^{1/4} (\mu_g^{1/4} - \mu_g^{1/4})}{\mu_g^{1/4} (\mu_s^{1/4} - \mu_g^{1/4})} \quad \text{Eq. 6.51}$$

$$\left(\frac{S_s}{S_n}\right)_{se} = \frac{\mu_s^{1/4}(S_{gf}\mu_o^{1/4} + S_{of}\mu_g^{1/4}) - \mu_o^{1/4}\mu_g^{1/4}\left(\frac{\mu_s^{1/4}}{\mu_{se}^{1/4}}\right)}{\mu_s^{1/4}(S_{gf}\mu_o^{1/4} + S_{of}\mu_g^{1/4}) - \mu_o^{1/4}\mu_g^{1/4}} \quad \text{Eq. 6.52}$$

where

$$S_{of} = S'_o/S'_{og}$$

$$S_{gf} = S'_g/S'_{og}$$

$$S'_{og} = S'_o + S'_g$$

The effective oil, gas and solvent densities ($\rho_{o\text{ eff}}$, $\rho_{g\text{ eff}}$, $\rho_{s\text{ eff}}$) are now computed from the effective saturation fractions in equations 6.50 to 6.52, and the pure component densities (ρ_o , ρ_g , ρ_s) using the following formulae

$$\rho_{o\text{ eff}} = \rho_o\left(\frac{S_o}{S_n}\right)_{oe} + \rho_s\left[1 - \left(\frac{S_o}{S_n}\right)_{oe}\right] \quad \text{Eq. 6.53}$$

$$\rho_{g\text{ eff}} = \rho_s\left(\frac{S_o}{S_n}\right)_{ge} + \rho_g\left[1 - \left(\frac{S_o}{S_n}\right)_{ge}\right] \quad \text{Eq. 6.54}$$

$$\rho_{s\text{ eff}} = \rho_s\left(\frac{S_s}{S_n}\right)_{se} + \rho_g S_{gf}\left[1 - \left(\frac{S_s}{S_n}\right)_{se}\right] + \rho_o S_{of}\left[1 - \left(\frac{S_s}{S_n}\right)_{se}\right] \quad \text{Eq. 6.55}$$

In the special case of no mixing, $\omega = 0$, the effective saturation fractions defined in equations 6.50 - 6.52 simplify to:

$$\left(\frac{S_o}{S_n}\right)_{oe} = 1, \left(\frac{S_o}{S_n}\right)_{ge} = 0, \left(\frac{S_s}{S_n}\right)_{se} = 1$$

so that the effective densities, equations 6.53 - 6.55 are equal to their pure phase values.

Note that equations 6.50 and 6.51 are singular for unit mobility ratio solvent displacements, $\mu_o = \mu_s$ or $\mu_g = \mu_s$. In that case, the effective densities are computed as follows. First a mixture density is defined as:

$$\rho_m = \rho_o\left(\frac{S_o}{S_n}\right) + \rho_g\left(\frac{S_g}{S_n}\right) + \rho_s\left(\frac{S_s}{S_n}\right) \quad \text{Eq. 6.56}$$

Then

$$\rho_{o\text{ eff}} = (1-\omega)\rho_o + \omega\rho_m \quad \text{Eq. 6.57}$$

$$\rho_{g\text{ eff}} = (1-\omega)\rho_g + \omega\rho_m \quad \text{Eq. 6.58}$$

$$\rho_{s\text{ eff}} = (1-\omega)\rho_s + \omega\rho_m$$

Where ω is the Todd and Longstaff parameter input with the [TLMIXPAR](#) keyword.

Pressure effect on viscosity and density miscibility

As for the relative permeability and capillary pressure, the pressure dependence of the transition between the immiscible and the miscible flow can be modeled for PVT data. This is achieved by using the [PMISC](#) keyword to supply a pressure dependent miscibility function M_p . This is used to interpolate between the immiscible and miscible values of the PVT as follows, with the gas component taken as an example:

$$\frac{1}{B_g} = \frac{1}{B_{gm}}M_p + \frac{1}{B_{gi}}(1-M_p) \quad \text{Eq. 6.59}$$

$$\frac{1}{(B\mu)_g} = \frac{1}{(B\mu)_{gm}}M_p + \frac{1}{(B\mu)_{gi}}(1-M_p) \quad \text{Eq. 6.60}$$

where

B_g is the effective gas formation volume factor

B_{gi} is the immiscible gas formation volume factor

B_{gm} is the miscible gas formation volume factor

$(B\mu)_g$ is the effective gas FVF $\times$ viscosity

$(B\mu)_{gi}$ is the gas FVF $\times$ viscosity (immiscible)

$(B\mu)_{gm}$ is the gas FVF $\times$ viscosity (miscible)

M_p is the pressure dependent miscibility function entered using the `PMISC` keyword.

Note that for the relative permeability the interpolation can be a combination of both `MISC` and `PMISC` keywords.

Choice of the mixing parameter

The mixing-parameter model would be of limited use unless the mixing parameter could itself be modeled over a wide range of operating conditions. A value of $\omega = 1$ results in a piston-like displacement of oil by the injected solvent. If $\omega = 0$ the displacement is similar to an immiscible displacement (except for the treatment of relative permeability). An intermediate value of ω results in a continuous solvent saturation increase behind the solvent front. Todd and Longstaff accounted for the effects of viscous fingering in 2D studies by setting $\omega = 2/3$ independently of mobility ratio. For field-scale simulations they suggested setting $\omega = 1/3$. However, in general history matching applications, the mixing parameter may be regarded as a useful history matching variable to account for any reservoir process inadequately modeled.

Using the solvent model

Initiating the model

Two keywords in `RUNSPEC` control the Solvent and Miscible models:

`SOLVENT` Activates the separate solvent component

`MISCIBLE` Initiates the mixing calculation.

If `SOLVENT` is specified, then the four-component model is used. The `SOLVENT` keyword should only be used with the other three components/phases (Water, Oil, Gas) present.

If the Solvent Model is run in immiscible mode, the mixing is not performed, and the four components take their pure PVT properties. The relative permeabilities are calculated as for a traditional black oil case, with the solvent and gas taking their respective fraction of the total 'gas' relative permeability.

PVT data

This consists of the PVDS and SDENSITY keywords:

SDENSITY Surface density of the solvent

PVDS Pressure dependent data for each PVT region (analogous to **PVDG**)

These PVT data can be used with the mixing parameter ω , input in the PROPS section with the keyword **TLMIXPAR** as described in the "[Viscosity model](#)" and "[Density model](#)". If the grid contains cells whose **MISCNUM** region is zero, this mixing parameter is automatically set to zero in those cells. Use the mnemonic **TLMIXPAR** in the **RPTPROPS** keyword to echo the **TLMIXPAR** data. The miscibility region numbers are specified using keyword **MISCNUM** in the **REGIONS** section. The first argument **NTMISC** of the keyword **MISCIBLE** sets the maximum number of regions which can be defined.

Relative permeability data

The relative permeability data needs to be specified as follows:

SGFN For k_{rg} in immiscible regions (either **MISCNUM**=0 or $S_g \gg 0$)

SWFN For any case containing water.

SOF2 Three-component or miscible case: cases with **MISCNUM**=0 in at least one cell, **SOF2** gives k_{rog} .

Four-component case: **SOF2** gives k_{rhc} (k_r of the miscible hydrocarbon phase to water).

SOF3 Three-component cases: Three-phase cases with **MISCNUM**=0 in at least one cell.

Four-component cases: immiscible displacement regions

SSFN In immiscible or miscible four-component cases.

In four-component cases, the miscibility function needs to be defined to control the transition from miscible to immiscible relative permeabilities. This is done using the **MISC** keyword in the **PROPS** section.

The miscible solvent-oil relative permeability curves can be modified using the **MSFN** keyword. If the **MSFN** keyword is omitted, straight line relative permeability curves will be used.

Transition between immiscibility and miscibility

The transition between an immiscible and a miscible displacement is controlled by the following keywords:

MISC Tables of the miscibility function versus local solvent fraction (four-component)

PMISC Tables of the miscibility function versus oil pressure.

MISC is required for all four-component runs, but **PMISC** is optional. If **PMISC** is not specified, the solvent/oil displacement is assumed to be miscible at all pressures.

Injecting solvent into the reservoir

Solvent can be injected by specifying a gas injection well. A separate keyword `WSOLVENT` can then be used to specify the fraction of the gas flow as solvent.

It is also possible to initialize the model by enumeration and to include a Solvent saturation. This is achieved using the `SSOL` keyword in the `SOLUTION` section.

Control of numerical diffusion in miscible flow simulation

The relative permeability of the oil and solvent components in the Todd- Longstaff model are specified by equation 6.29. The relative permeability of each component is essentially proportional to the saturation of each component for low saturations. In such cases the influence of numerical diffusion in spreading the miscible components can be unacceptably high and some control of the numerical diffusion errors is required.

A two point upstream algorithm, operating only on the miscible component relative permeabilities, is available for use in the Eclipse Miscible Flood option to control the numerical diffusion of oil and solvent. In the present version of Eclipse the two-point upstreaming algorithm is only available for three-phase cases and hence cannot be used with the separate solvent option.

The two-point upstream projection technique is illustrated in the following figure.

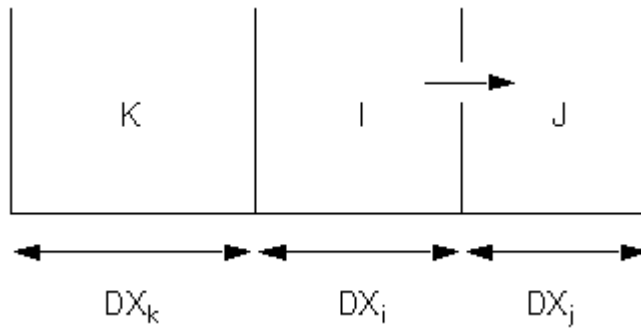


Figure 6.2. The two point upstream projection technique

We need to compute the flow of component C between cells I and J which we take to be aligned along the X-axis. Suppose that cell I is the upstream cell for the flow of component C between I and J. Let cell K be the second upstream cell to cell J and K_{ck} , K_{ci} and K_{cj} denote the component relative permeabilities in cells K, I and J respectively. DX_k , DX_i and DX_j are the X-direction cell sizes.

For single point upstream weighting, the flow rate of component C across the I-J cell interface is $F_{ij,c}$ where

$$F_{ij,c} = T_{ij} \cdot \frac{K_{ci}}{B_{ci} \mu_{ci}} \cdot \delta\Phi_{ij,c} \quad \text{Eq. 6.61}$$

where

T_{ij} is the transmissibility between cells I and J

$\delta\Phi_{ij,c}$ is the potential difference between cells I and J for component C

B_c, μ_c are the formation volume factor and viscosity of component C

The two point upstreaming algorithm operating on component relative permeabilities replaces K_{ci} in the equation 6.61 by a linearly extrapolated relative permeability, K_{ci}^* , obtained from cells I and K,

$$K_{ci}^* = \frac{2DX_i + DX_k}{DX_i + DX_k} K_{ci} - \frac{DX_i}{DX_i + DX_k} K_{ck} \quad \text{Eq. 6.62}$$

The above expression for K_{ci}^* can yield negative values or values greater than unity, and in such cases needs to be restricted to physical values. This is most simply achieved by constraining K_{ci}^* to lie between the values K_{ci} and K_{cj} . Thus,

$$\text{MIN}(K_{ci}, K_{cj}) < K_{ci}^* < \text{MAX}(K_{ci}, K_{cj}) \quad \text{Eq. 6.63}$$

In some cases a single point upstream calculation of K_{ci}^* (that is, $K_{ci}^* = K_{ci}$) should be used instead of the two point upstream value. Such cases are

- Cell K lies outside the reservoir mesh (that is the K-I cell boundary is an edge of the reservoir)
- Cell K is inactive
- The K-I cell boundary contains a zero transmissibility multiplier
- Cells I and J are coupled by a non-neighbor connection.

Summary of keywords

RUNSPEC section

Keyword	Description
SOLVENT	Activates the separate solvent component.
MISCIBLE	Initiates the mixing calculation.

PROPS section

Keyword	Description
MISC	Solvent fraction dependent miscibility tables.
MSFN	Miscible gas/oil saturation functions.
PMISC	Pressure dependent miscibility tables.
PVDS	Pressure dependent PVT properties for solvent gas.
RPTPROPS	Controls output from the PROPS section. • MISC or PMISC outputs the miscibility function and the pressure miscibility function. • PVDS outputs solvent PVT properties. • SSFN outputs solvent/gas relative permeabilities.
SDENSITY	Solvent density at surface conditions.
SGCWMIS	Miscible critical gas saturation tables versus water saturation.
SORWMIS	Miscible residual oil saturation tables versus water saturation.
SSFN	Gas/solvent relative permeability functions.

Keyword	Description
TLMIXPAR	Todd-Longstaff mixing parameter.

SOLUTION section

Keyword	Description
RPTSOL	Controls on output from SOLUTION section. - FIPSOL outputs solvent fluid in place reports. - FLOSOL outputs solvent inter-block flow rates. - KRN outputs grid block solvent relative permeabilities - SSOL outputs initial solvent saturations.
SSOL	Initial solvent saturations.

SUMMARY section

Refer to [Solvent model](#) in the *Eclipse Reference Manual*.

SCHEDULE section

Keyword	Description
RPTSCHED	Controls on output from SCHEDULE section. - FIPSOL outputs solvent fluid in place reports. - FLOSOL outputs solvent inter-block flow rates. - KRN outputs grid block solvent relative permeabilities. - SSOL outputs current solvent saturations.
WSOLVENT	Sets solvent concentrations for gas injection wells.

Example

RUNSPEC

```

RUNSPEC
TITLE
SPE Fifth Comparison Test Problem - Scenario 2 - 4 Phase

DIMENS
  7      7      3  /

NONNC
OIL
WATER
GAS
DISGAS
SOLVENT
FIELD

MISCIBLE
  1      20      'NONE'  /

```



```
EQLDIMS
1 100 10 1 1 /

WELLDIMS
3 1 1 3 /

START
1 'JAN' 1986 /

NSTACK
24 /

FMTIN
UNIFOUT
UNIFIN
```

GRID

```
GRID =====
EQUALS
'DX' 500 /
'DY' 500 /
'PORO' 0.3 /
'DZ' 20 1 7 1 7 1 1 /
'PERMX' 500 /
'PERMZ' 50 /
'TOPS' 8325 /
'DZ' 30 1 7 1 7 2 2 /
'PERMX' 50 /
'PERMZ' 50 /
'DZ' 50 1 7 1 7 3 3 /
'PERMX' 200 /
'PERMZ' 25 /

/

COPY
'PERMX' 'PERMY' 1 7 1 7 1 3 /

/

RPTGRID
'DX' 'DY' 'DZ' 'PERMX' 'PERMY' 'PERMZ' 'MULTZ'
'PORO' 'TOPS' 'PORV' 'TRANX' 'TRANY' 'TRANZ' /
```

PROPS

```
PROPS =====
SWFN
0.20 0.0 45.00
0.2899 0.0022 19.03
0.3778 0.0180 10.07
0.4667 0.0607 4.90
0.5556 0.1438 1.80
0.6444 0.2809 0.50
0.7000 0.4089 0.05
0.7333 0.4855 0.01
0.8222 0.7709 0.00
0.9111 1.0000 0.00
1.0000 1.0000 0.00 /

SGFN
0.0000 0.0 0.0
0.0500 0.0 0.0
0.0889 0.001 0.0
0.1778 0.01 0.0
0.2667 0.03 0.001
0.3556 0.05 0.001
0.4444 0.1 0.03
0.5333 0.2 0.8
0.6222 0.35 3.0
```

```

0.6500 0.39 4.0
0.7111 0.56 8.0
0.8000 1.0 30.0 /

SOF2
0.0 0.0
0.0889 0.0
0.1778 0.0
0.2667 0.0
0.3000 0.0
0.3556 0.0123
0.4444 0.0835
0.5333 0.2178
0.6222 0.4153
0.7101 0.6769
0.8000 1.0000 /

SOF3
-- KROW KROG
0.0 0.0 0.0
0.0889 0.0 0.0
0.15 1* 0.0
0.1778 0.0 0.0110
0.2667 0.0 0.0370
0.3000 0.0 1*
0.3556 0.0123 0.0878
0.4444 0.0835 0.1715
0.5333 0.2178 0.2963
0.6222 0.4153 0.4705
0.7101 0.6769 0.7023
0.75 1* 0.8800
0.8000 1.0000 1.0000 /

SSFN
-- KRG* KRS*
0 0.0 0.0
1.0 1.0 1.0
/

MISC
0.0 0.0
0.1 1.0
1.0 1.0 /

PVTW
4000.0 1.000 3.3D-6 0.70 0 /

ROCK
4000.0 5.0D-6 /

DENSITY
38.53 62.40 0.06867 /

SDENSITY
0.06243 /

TLMIXPAR
0.7 /

PVDG
14.7 211.4160 0.0107
500.0 5.9242 0.0127
1000.0 2.8506 0.0134
1200.0 2.3441 0.0138
1500.0 1.8457 0.0145
1800.0 1.5202 0.0153
2000.0 1.3602 0.0159
2302.3 1.1751 0.0170
2500.0 1.1025 0.0177
3000.0 0.9852 0.0195
3500.0 0.9116 0.0214
4000.0 0.8621 0.0232
4500.0 0.8224 0.0250
4800.0 0.8032 0.0261 /

```

PVDS				
	14.7	223.2140	0.011	
	500.0	5.6022	0.012	
	1000.0	2.5310	0.013	
	1200.0	2.0354	0.014	
	1500.0	1.5593	0.016	
	1800.0	1.2657	0.018	
	2000.0	1.1296	0.019	
	2302.3	0.9803	0.022	
	2500.0	0.9085	0.023	
	3000.0	0.7807	0.027	
	3500.0	0.6994	0.031	
	4000.0	0.6430	0.034	
	4500.0	0.6017	0.037	
	4800.0	0.5817	0.038	/
PVTO				
	0.0000	14.7	1.0348	0.310 /
	0.1176	500.0	1.1017	0.295 /
	0.2226	1000.0	1.1478	0.274 /
	0.2677	1200.0	1.1677	0.264 /
	0.3414	1500.0	1.1997	0.249 /
	0.4215	1800.0	1.2350	0.234 /
	0.4790	2000.0	1.2600	0.224 /
	0.5728	2302.3	1.3010	0.208
		3302.3	1.2988	0.235
		4302.3	1.2966	0.260 /
	0.6341	2500.0	1.3278	0.200 /
	0.7893	3000.0	1.3956	0.187 /
	0.9444	3500.0	1.4634	0.175 /
	1.0995	4000.0	1.5312	0.167 /
	1.2547	4500.0	1.5991	0.159 /
	1.3478	4800.0	1.6398	0.155
		5500.0	1.6305	0.168 /
/				
RPTPROPS				
	'SOF2'	'SWFN'	'SGFN'	'PVTW'
	'PVTG'	'DENSITY'	'ROCK'	/

SOLUTION

SOLUTION =====										
EQUIL										
	8400	4000	9900	0	1000	0	1	0	0	/
RSVD										
	8200	0.5728								
	8500	0.5728	/							
RPTSOL										
	'PRES'	'SOIL'	'SWAT'	'SGAS'	'FIP'	/				

SUMMARY

SUMMARY =====										
FOPR										
FNPR										
FGPR										
BOSAT										
	4	4	1	/						
/										
BNSAT										
	4	4	1	/						
/										

SCHEDULE

```

=====
SCHEDULE
RPTSCHED
  'PRES'  'SOIL'  'SWAT'  'SGAS'  'SUMMARY=2'
  'CPU=2'  /

WELSPECS
  'PRODUCER' 'G'      7  7      8400 'OIL'  /
  'INJ-G'    'G'      1  1      8335 'GAS'  /
  'INJ-W'    'G'      1  1      8335 'WAT'  /
/

COMPDAT
  'PRODUCER'  7  7  3  3 'OPEN' 0  -1  0.5  10000.0 /
  'INJ-G'     1  1  1  1 'OPEN' 0  -1  0.5  10000.0 /
  'INJ-W'     1  1  1  1 'OPEN' 0  -1  0.5  10000.0 /
/

WCONPROD
  'PRODUCER' 'OPEN' 'ORAT' 12000 4* 3000 /
/

WECON
  'PRODUCER' 1* 1* 5.0 10.0 1* 'WELL' 'YES' /
/

WCONINJE
  'INJ-W' 'WAT' 'OPEN' 'RATE' 45000 1* 4500 /
  'INJ-G' 'GAS' 'OPEN' 'RATE' 20000 1* 4500 /
/

WSOLVENT
  'INJ-G' 1.0 /
/

-- YEAR 1 -----
INCLUDE
  'CASE2.ONEYEAR' /

-- YEAR 2 -----
INCLUDE
  'CASE2.ONEYEAR' /

END

```

Included file

```

-- INCLUDED FILE: CASE2.ONEYEAR
-- FIRST 3 MONTHS -----
--*WATER*--

TUNING
2.0 /
/

WELOPEN
  'INJ-G' 'SHUT' /
  'INJ-W' 'OPEN' /
/

TSTEP
  91.25
/

-- SECOND 3 MONTHS -----
--*GAS*--

TUNING
2.0 /

```

```
/
/
WELOPEN
  'INJ-W' 'SHUT' /
  'INJ-G' 'OPEN' /
/

TSTEP
  91.25
/

--THIRD 3 MONTHS -----
--*WATER*--

TUNING
2.0 /
/
/

WELOPEN
  'INJ-G' 'SHUT' /
  'INJ-W' 'OPEN' /
/

TSTEP
  91.25
/

-- FOURTH 3 MONTHS -----
--*GAS*--

TUNING
2.0 /
/
/

WELOPEN
  'INJ-W' 'SHUT' /
  'INJ-G' 'OPEN' /
/

TSTEP
  91.25
/-----
```

CO₂ injection: geological storage, EOR and ECBM

	Eclipse 100
x	Eclipse 300

One of the major issues of CO₂ injection problems is the chemical reactivity between the supercritical fluid (referred to as the gas phase in this document) and the liquid phase. There are several facilities available in Eclipse 300, depending on the site and operational conditions that need to be modeled.

Geological storage

In Aquifers

Using the [CO2STORE](#) option, three phases are considered: a CO₂-rich phase (labeled gas phase), an H₂O-rich phase (labeled liquid phase) and a solid phase.

This option gives accurate mutual solubilities of CO₂ in water (x_{CO_2}) and water in the CO₂-rich phase (y_{H_2O}). Salts are described as components of the liquid/solid phase. To allow for a solid phase the [SOLID](#) option should be used.

The [GASWAT](#) option provides a gas/aqueous phase equilibrium method. The liquid mole fraction x_{CO_2} is accurately predicted; however, the gas phase mole fraction y_{H_2O} is less accurate. The gas composition is not restricted to CO₂/H₂O, as other gases and their solubility in water can be included.

In Oil Depleted Reservoirs and EOR (Enhanced oil recovery)

The option best suited for this case is [CO2SOL](#). Water is not considered in the oil or gas phase, but the solubility of CO₂ in the water phase is accounted for. The gas/oil composition is not restricted to CO₂ but only CO₂ is considered soluble in water.

In Gas Depleted Reservoirs

This case can be modeled using the [GASWAT](#) option by adding gas components, such as CH₄ and H₂S for example.

ECBM

The [COAL](#) option provides a diffusive flow and sorption model for Enhanced Coal Bed Methane recovery (ECBM). A full description of this option is provided in "[Coal bed methane model](#)".

Since the [CO2STORE](#), [CO2SOL](#) and [GASWAT](#) options estimate the fluid properties differently, they cannot be used together.

To model depleted reservoirs together with the [THERMAL](#) option, the [CO2STORE](#) keyword should be used with the required hydrocarbon components. A full description of this option is provided in "[Thermal CO2STORE](#)".

CO2STORE

The mutual solubilities of CO₂ and H₂O are calculated to match experimental data for typical CO₂ storage conditions: typically 12-250 degrees C and up to 600 bars. The lower limit of 12 °C can be extended with the low temperature and/or hydrate formation options, discussed below. This extension is only applicable to systems with these two components and does not apply to systems with additional hydrocarbon. Solubilities are calculated following the procedure given by [Spycher and Pruess \(2005\)](#) and [Spycher and Pruess \(2009\)](#), based on fugacity equilibration between water and a CO₂ phase. Water fugacity and CO₂ fugacity are calculated using a modified Redlich-Kwong equation of state.

See "[CO2 storage](#)".

- Salt precipitation is enabled by the use of the [SOLID](#) option. If the [SOLID](#) option is not used, salts are assumed to stay in the liquid phase.
- The gas density is obtained by an accurately tuned cubic equation of state.
- The brine density is first approximated by the pure water density and then corrected for salt and CO₂ effects by Ezrokhi's method ([Zaytsev and Aseyev, 1993](#)).
- The CO₂ gas viscosity is calculated from [Vesovic et al \(1990\)](#) and [Fenghour et al \(1998\)](#).

Using CO2STORE

RUNSPEC section

The option switched on using the [CO2STORE](#) keyword. The number of components is specified by the [COMPS](#) keyword. Five components are currently allowed: CO₂, H₂O, NaCl, CaCl₂ and CaCO₃.

PROPS section

Fluid component properties are assigned internally based on the names given in the [CNAMES](#) keyword (labeled 'CO2', 'H2O', 'NaCl', 'CaCl2', 'CaCO3').

The water relative permeability is input using the [WSF](#) keyword, while the gas relative permeability and the gas-water capillary pressure are entered using the [GSF](#) keyword. These keywords are aliases for the [SOF2](#) and [SGFN](#) keywords, which can also be used (with all references to oil treated as references to water).

Initial state constructed by equilibration

If an initial gas-water contact exists within the reservoir you may specify either the vapor or the aqueous phase composition at the contact; either [ZMFVD](#) or [COMPVD](#) can be used.

If only a single phase is present the composition may be entered using [ZI](#) or with respect to depth using [ZMFVD](#).

SOLUTION section

The [EQUIL](#) keyword is used for initial state equilibration.

- Initial state set by enumeration
- Liquid, gas and total compositions are specified explicitly with keywords [XMF](#), [YMF](#) and [ZMF](#) respectively.

SCHEDULE section

The composition of the injected gas can be specified using the keyword [WINJGAS](#).

Mole fractions xCO₂ and yH₂O are output by writing [XMF](#) and [YMF](#) in [RPTSCHED](#).

Note: See the note about salt precipitation in CO₂ Storage "[Compatibility](#)".

CO2SOL

[CO2SOL](#) is a three-phase compositional option for simulating CO₂ flooding in oil reservoirs (EOR, storage in oil depleted reservoirs). Water is not allowed to dissolve in the gas or oil phases, CO₂ can be present in the three phases, and other components in the oil and gas phases.

See ["CO₂/H₂S solution in the aqueous phase"](#).

CO₂ partitioning between the oil and gas phases is calculated by a fugacity equilibration method. A cubic equation of state is used to model oil and gas phase densities and fugacities.

The amount of CO₂ dissolved in water, and other aqueous phase properties, are computed using solubility data that you enter with either the [SOLUBILI](#), [SOLUBILS](#) or [SOLUBILT](#) keywords or defaulted following [Chang et al, 1996](#). When using [SOLUBILI](#) or [SOLUBILT](#) the salting-out effect is taken into account by correcting solubility in pure water with the Chang et al correlation for brine ([Chang et al, 1996](#)). The [SOLUBILS](#) keyword can be used to describe a more complex salt dependency using the [BRINE](#) tracking. The [SOLUBILT](#) keyword can be used to describe a more complex temperature dependency. To obtain details on the solubility data used in a simulation the SOLU mnemonic may be specified in the [RPTPROPS](#) keyword.

Molecular diffusion between gases can be modeled using the [DIFFUSE](#) option. See also ["Diffusion"](#).

It is possible to set initial dissolved CO₂ as a function of depth using the [RSWVD](#) or [RSW](#) keywords. This initial solution honors this value unless it exceeds the saturated CO₂ concentration in which case the saturated value is used.

Notes

- The initial CO₂ concentration may not be in exact equilibrium with the hydrocarbon phase. In such a case the initial flash slightly modifies the input values.
- The first line of the solubility table defined using the [SOLUBILI](#) keyword represents water at standard conditions, when the dissolved CO₂ amount is at a minimum. The simulator does not assume that the amount of CO₂ in water at the lowest pressure is zero; if you want to specify it as such then you should set it to zero in the first line of the table.
- Instantaneous equilibrium is assumed between the oil, gas and water phases.

Using CO2SOL

RUNSPEC section

This option is switched on using the [CO2SOL](#) keyword. Unlike in the [CO2STORE](#) option, water is not defined as a component in [COMPS](#) and [CNAMES](#), as it is absent in the oil and gas phase.

The [DIFFUSE](#) option allows for the molecular diffusion of gases. See also ["GASWAT Option"](#).

The brine tracking option is activated by the [BRINE](#) keyword.

PROPS section

The gas relative permeability and oil-gas capillary pressure are input using the [SGFN](#) keyword, while the water relative permeability and the gas-water capillary pressure are entered using the [SWFN](#) keyword. Oil relative permeabilities are defined with the [SOF3](#) keyword.

Fluid component properties can be generated from the PVT *i* database. The component 'CO₂' must be present in [CNAMES](#).

The initial dissolved CO₂ may be specified as a function of depth using the [RSWVD](#) keyword.

The [SOLUBILI](#) keyword allows you to enter a solubility data set as a function of reservoir pressure. The [SOLUBILT](#) keyword is similar to [SOLUBILI](#) but allows you to enter a temperature dependent table. For

both keywords, the table will be automatically corrected for the salinity effect following [Chang et al \(1996\)](#) if either the [SALINITY](#) or [SALINITYR](#) keyword has been used. In this case, the temperature used in that correction will be taken from the [RTEMP](#) keyword in the case of [SOLUBILI](#) and from table itself for [SOLUBILT](#). The density of water at surface condition can be adjusted using the [DENSITY](#) or [GRAVITY](#) keywords.

The [SOLUBILS](#) keyword is similar to [SOLUBILI](#) but you enter a salt concentration dependent table. If the Brine option is active, [SOLUBILS](#) supersedes [PVTWSALT](#) and [BDENSITY](#) and the salt concentration is taken from the brine tracer.

If none of the [SOLUBILI](#), [SOLUBILS](#) or [SOLUBILT](#) keywords are used the solubility data is defaulted as detailed in the description of [SOLUBILI](#).

The coefficients for molecular diffusion must be set with [DIFFCGAS](#) if the [DIFFUSE](#) option is used.

If CO₂ appears as liquid, it will be labelled as oil phase. In such a case, make sure that the [SOF3](#) keyword (relative permeability for CO₂ liquid) is appropriate.

Initial state constructed by equilibration

If an initial oil-gas contact exists within the reservoir you may specify either the vapor or the liquid phase composition at the contact, either [ZMFVD](#) or [COMPVD](#) can be used.

If only a single phase is present the composition may be entered using [ZI](#) or with respect to depth using [ZMFVD](#).

SOLUTION section

The [EQUIL](#) keyword is used for equilibration.

- Initial state set by enumeration
- The oil phase composition is set with [XMF](#), the gas phase composition with [YMF](#), water composition with [AMF](#) and the total oil and gas composition with [ZMF](#). Water content in CO₂ can be defined by [RSW](#) or [RSWVD](#).

SCHEDULE section

The composition of the injected gas can be specified using the keyword [WINJGAS](#).

The aqueous mole fractions a CO₂ and a H₂O can be output by adding the argument [AMF](#) in [RPTSCHED](#).

CO2SOL/H2SSOL/GASSOL

[CO2SOL](#) / [H2SSOL](#) is a three-phase compositional option for simulating CO₂ and H₂S flooding in oil reservoirs (EOR, storage in oil depleted reservoirs).

[GASSOL](#) is a three-phase compositional option for simulating multi-component gas flooding in oil reservoirs (EOR, storage in oil depleted reservoirs). Water is not allowed to dissolve in the gas or oil phases; any gas component can be present in all three phases; all other components can be present in the oil and gas phases.

See "[CO2/H2S solution in the aqueous phase](#)" and "[GASSOL solution in the aqueous phase](#)".

Multi-component gas partitioning between the oil and gas phases is calculated by a fugacity equilibration method. A cubic equation of state is used to model oil and gas phase densities and fugacities.

The amount of multi-component gas dissolved in water, and other aqueous phase properties, is computed using solubility data entered by the user using [SOLUCO2H](#), [SOLUH2SC](#) or [SOLUAQA](#). This solubility data may also be defaulted. To obtain details regarding the solubility data used in a simulation, the [SOLU](#) mnemonic may be specified using the [RPTPROPS](#) keyword.

Molecular diffusion between gases can be modeled using the [DIFFUSE](#) option. See also "[Diffusion](#)".

It is possible to set the initial concentrations of dissolved gases as a function of depth using the [RSWVD](#) or [RSW](#) keywords. The initial solution honors these values unless they exceed the saturated multi-component gas concentration, in which case the saturated value is used.

Notes

- The initial multi-compound gas concentration may not be in exact equilibrium with the hydrocarbon phase. In such a case, the initial flash slightly modifies the input values.
- Instantaneous equilibrium is assumed between the oil, gas and water phases.

Using GASSOL

RUNSPEC section

This option is switched on using the [GASSOL](#) keyword. Unlike in the [CO2STORE](#) option, water is not defined as a component in [COMPS](#) and [CNAMES](#), as it is absent in the oil and gas phases.

The [DIFFUSE](#) option allows for the molecular diffusion of gases. See also "[GASWAT option](#)".

PROPS section

The gas relative permeability and oil-gas capillary pressure are input using the [SGFN](#) keyword, while the water relative permeability and the gas-water capillary pressure are entered using the [SWFN](#) keyword. Oil relative permeabilities are defined with the [SOF3](#) keyword.

Fluid component properties can be generated from the PVT *i* database. The dissolved gas components must be present in [CNAMES](#).

The initial concentrations of dissolved gases may be specified as a function of depth using the [RSWVD](#) keyword.

The [SOLUAQA](#) keyword allows you to enter a solubility data set as a function of reservoir pressure. The reservoir temperature is taken from the [RTEMP](#) keyword. Note that the solubility data may be defaulted for any given soluble component. In such a case, a default set of pressures and corresponding solubilities is generated. If the solubilities for all soluble components are defaulted, then each table comprises twenty rows of data. If the solubilities for some (but not all) soluble components are defaulted, then the number of rows in the defaulted solubility table(s) equals the maximum number of rows from the remaining input solubility tables. Since the default solubilities depend strongly on the data entered in the [PROPS](#) section, this may lead to unrealistic values if the component data is not accurate enough. After computing the dissolving hydrocarbons' mole fraction in the aqueous phase using the phase equilibrium algorithm mentioned previously, the water density and the water viscosity μ_{sat} can be calculated using the Ezrokhi data specified using [DENAQA](#) and [VISCAQA](#), respectively.

The coefficients for molecular diffusion must be set with [DIFFCGAS](#) if the [DIFFUSE](#) option is used.

Initial state constructed by equilibration

If an initial oil-gas contact exists within the reservoir, you can specify either the vapor or the liquid phase composition at the contact, using [ZMFVD](#) or [COMPVD](#).

If only a single phase is present, the composition may be entered using [ZI](#) or (with respect to depth) using [ZMFVD](#).

SOLUTION section

The [EQUIL](#) keyword is used for equilibration.

- Initial state set by enumeration.
- The oil phase composition is set with [XMF](#), the gas phase composition with [YMF](#), water composition with [AMF](#) and the total oil and gas composition with [ZMF](#). Aqueous phase content can be defined by [RSW](#) or [RSWVD](#).

SCHEDULE section

The composition of the injected gas can be specified using the keyword [WINJGAS](#).

The aqueous mole fractions of dissolved gas components can be output by adding the mnemonic [AMF](#) in [RPTSCHED](#).

GASWAT

The [GASWAT](#) option uses a modified Peng-Robinson equation of state in order to model gas/aqueous phase equilibria.

GASWAT allows a multi-component gas phase. CO₂, N₂, and H₂S solubilities in water are treated with the Peng Robinson equation of state (EOS) modified by Soreide and Whitson. However, solubilities of other gases like methane are treated by the original Peng Robinson EOS by default. "[GASWAT option](#)" provides an exhaustive description of this feature.

Molecular diffusion between gases can be modeled using the [DIFFUSE](#) option. See "[Diffusion](#)".

Using GASWAT for CO₂ storage issues**RUNSPEC section**

This option is switched on using the [GASWAT](#) keyword. The number of components is specified by the [COMPS](#) keyword. Water has to be defined as a component; salts are not components. The [WATER](#) keyword should not be present, as H₂O is a component specified in [COMPS](#).

The Diffusion option allows for molecular diffusion of gases.

PROPS section

The water relative permeability is input using the [WSF](#) keyword, while the gas relative permeability and the gas-water capillary pressure are entered using the [GSF](#) keyword.

Salinity can be set up by the [SALINITY](#) keyword or on a regional basis with the [SALINITR](#) keyword

The coefficients for molecular diffusion must be set with [DIFFCGAS](#) if the Diffusion option is used.

Initial state constructed by equilibration

If an initial gas-water contact exists within the reservoir you may specify either the vapor or the aqueous phase composition at the contact, either [ZMFVD](#) or [COMPVD](#) can be used.

If only a single phase is present the composition may be entered using [ZI](#) or with respect to depth using [ZMFVD](#).

SOLUTION section

The [EQUIL](#) keyword is used for initial state equilibration.

- Initial state set by enumeration
- Liquid phase composition can be set with [XMF](#), the gas phase composition with [YMF](#) and the total composition with [ZMF](#).

SCHEDULE section

The composition of the gas injected can be specified using the keyword [WINJGAS](#).

Mole fractions x_{CO_2} and y_{H_2O} can be output by writing arguments [XMF](#) and [YMF](#) in [RPTSCHED](#).

Boundary conditions

Several facilities are available.

- Numerical aquifers contain only a water phase. For [CO2STORE](#) and [GASWAT](#) cases, make sure that no supercritical CO_2 (CO_2 rich phase being labeled as gas phase) reaches the boundary. Likewise for the liquid CO_2 for a [CO2SOL](#) run.
- For [CO2STORE](#) and [GASWAT](#) cases, analytic aquifers can be used to let the CO_2 rich phase leave the reservoir. For this, use [AQUCHWAT](#), [AQUCHGAS](#) and [AQSTREAM](#) and ensure that the composition of the gas aquifer specified by [AQUCHGAS](#) is as close as the composition of the gas reaching the boundary.

Note on conversions for salinity units

As the various CO_2 options use different frameworks within Eclipse, units for salinity input differ.

Salt concentrations in [XMF](#) and [YMF](#) must be entered in mole fraction. The [SALINITY](#) keyword unit is molality (gm-M/kg of solvent). Salinity in [SOLUBILS](#) must be expressed in kg/m³ of brine.

Conversions between these different units are described below.

Molality to mole fraction

Mole fraction of component i in the aqueous phase: $x_i = \text{molality}_i \times \frac{M_{\text{solvent}}}{1000} \times x_{\text{solvent}}$

where:

mole fraction of solvent in the aqueous phase:

$$x_{\text{solvent}} = \frac{n_{\text{solvent}}}{n_{\text{total}}} = \frac{n_{\text{solvent}}}{n_{\text{solvent}} + \sum_{i \neq \text{solvent}} n_i}$$

molality of component i : $molality_i = \frac{n_i}{m_{solvent}} [\text{gm-M/kg}]$,

$M_{solvent}$ is the molecular weight of solvent [gm/gm-M]

Thus:

$molality_{solvent} = 1000/M_{solvent} [\text{gm-M/kg}]$,

In most of the cases, the solvent is H_2O . As $M_{\text{H}_2\text{O}} = 18.014 \text{ gm/gm-M}$, $molality_{\text{H}_2\text{O}} = 55.5 \text{ gm-M/kg}$.

Concentration (mg/L) to mole fraction

Chemical analysis usually gives concentration in mg/L ($=10^{-3}\text{kg/m}^3$) and the initial composition of the brine/water in Eclipse must be entered in mole fraction.

Concentrations in mg/L can be converted to mole fraction through:

$$x_i = \frac{\left(\frac{c_i}{M_i}\right)}{\sum_j \left(\frac{c_j}{M_j}\right)} = \frac{[i]}{\sum_j [j]}$$

where

$[i]$ denotes the molarity of component i in gm-M/L,

c_i the concentration in mg/L and

M_i the molecular weights in mg/gm-M.

With a standard water analysis at hand, a few additional considerations are required to process the conversion by this formula, as highlighted in the following example:

Example

A brine sample has been analyzed and the results are the following:

Species	Concentration (mg/L)	Species	Concentration (mg/L)
Na^+	90110	HCO_3^-	45
Ca^{2+}	1760	I^-	1
Mg^{2+}	2230	HS^-	nb
K^+	470	CO_3^{2-}	0
NH_4^{2+}	0	OH^-	0
Fe^{2+}	0	TDS (Total Dissolved Solid)	250000
Cl^-	141470	pH	5.3
SO_4^{2-}	6600	Water Density at reservoir conditions (kg/m^3)	1164
Br^-	64		

Table 6.1: Brine sample analysis

$$c_{Na^+} = 90.11 \text{ gm/L}$$

$$c_{Ca^{2+}} = 1.76 \text{ gm/L}$$

$$c_{Cl^-} = 141.47 \text{ gm/L}$$

For one liter of brine:

$$m_{H_2O} = \text{reservoir brine density} - \text{TDS} = 1164 - 250 = 914 \text{ gm}$$

$$n_{H_2O} = \frac{m_{H_2O}}{M_{H_2O}} = \frac{914}{18.014} = 50.738 \text{ gm-M}$$

$$n_{Na^+} = \frac{90.11}{23} = 3.919 \text{ gm-M}$$

$$n_{Ca^{2+}} = \frac{1.76}{40} = 0.044 \text{ gm-M}$$

$$n_{Cl^-} = \frac{141.47}{35} = 4.042 \text{ gm-M}$$

For $Na^+ + Cl^- \rightleftharpoons NaCl$:

	Sodium (Na)	Chlorine (Cl)	Sodium Chloride (NaCl)
Initial	n_{Na^+0}	n_{Cl^-0}	0
At t time n_t	$n_{Na^+0} - \xi$	$n_{Cl^-0} - \xi$	ξ
Final n_f	0	$n_{Cl^-f} = n_{Cl^-0} - n_{Na^+0}$	n_{Na^+0}

where ξ is the reaction advancement.

Here $n_{NaCl(aq)} = n_{Na^+0}$ because Na^+ is the limiting factor.

For $Ca^{2+} + 2Cl^- \rightleftharpoons CaCl_2$:

	Calcium (Ca)	Chlorine (Cl)	Calcium Chloride ($CaCl_2$)
Initial	$n_{Ca^{2+}0}$	n_{Cl^-f}	0
At t time n_t	$n_{Ca^{2+}0} - \psi$	$n_{Cl^-f} - 2\psi$	ψ
Final n_f	0	$n_{Cl^-f} - 2n_{Ca^{2+}0}$	$n_{Ca^{2+}0}$

where ψ is the reaction advancement.

Here $n_{CaCl_2(aq)} = n_{Ca^{2+}0}$ because Ca^{2+} is the limiting factor.

The ionic species are reconciled into neutral component in Eclipse. The neutral components in this case are NaCl and $CaCl_2$. These are formed from the ionic species in order of increasing solubility. Starting with the salt with lowest solubility, in this case NaCl, the limiting ionic species is Na^+ , that is, 3.92 gm-M of NaCl can be formed, leaving $(4.00 - 3.92) = 0.08$ gm-M/L of Cl^- for the formation of 0.04 gm-M/L of $CaCl_2$. It should be noted that the water analysis in this case is consistent with the Ca^{2+} concentration.

$$n_{total} = n_{NaCl(aq)} + n_{CaCl_2(aq)} + n_{H_2O} = 54.701 \text{ gm-M}$$

$$x_{NaCl} = \frac{n_{NaCl(aq)}}{n_{total}} = \frac{3.919}{54.701} = 0.0716$$

$$x_{CaCl_2} = \frac{n_{CaCl_2(aq)}}{n_{total}} = \frac{0.044}{54.701} = 0.0008$$

$$x_{H_2O} = \frac{n_{H_2O}}{n_{total}} = \frac{46.408}{54.701} = 0.9276$$

Note: In this example, the molality of salt you have to input for the keyword **SALINITY** (as used in the CO2SOL and GASWAT options) is:

$$\text{for one liter of solution, } molality_{salt} = \frac{\sum_{salt} n_i}{mass_{H_2O}} = \frac{3.963}{0.916} = 4.34 \text{ gm-M/kg}$$

Molality to mass per volume unit (kg/m³)

$$\text{Mass per volume unit} = \frac{d}{B} \times \frac{\text{molality} \times 10^{-3} \times M}{1 + \text{molality} \times 10^{-3} \times M}$$

where

M = salt molecular weight (typically 58.44 kg/kg-M for NaCl)

B = brine volume factor

d = surface brine density

CO₂/H₂S solution in the aqueous phase

	Eclipse 100
x	Eclipse 300

The CO₂ and H₂S solution algorithm allows carbon dioxide and/or hydrogen sulfide to dissolve in the aqueous phase. This is switched on using respectively the [CO2SOL](#) and/or [H2SSOL](#) keyword in the RUNSPEC section. The basic model is a fugacity function for aqueous CO₂ and H₂S which is constructed to match solubility data and which takes the form:

$$f_{CO_2}^A = Pa_{CO_2} \varphi_{CO_2}(P) \quad \text{Eq. 6.64}$$

$$f_{H_2S}^A = Pa_{H_2S} \varphi_{H_2S}(P)$$

The phase equilibrium between the aqueous CO₂ or H₂S and the hydrocarbon phases is then defined by the conditions that the fugacity values are equal. It is also possible to construct a CO₂ and H₂S aqueous phase Gibbs energy contribution.

The water phase properties may be entered using the [SOLUBILI](#) keyword. Alternatively, a default set of data is provided for CO₂:

- The CO₂ solubility is calculated from the correlations of [Chang et al \(1998\)](#).
- The effects of salt and CO₂ on water density are calculated using Ezrokhi's method ([Zaytsev and Aseyev, 1993](#)). These densities are then used to calculate the saturation formation volume factor, unless [item 115](#) of [OPTIONS3](#) is set to 1, when the method from [Chang et al \(1998\)](#) is used.
- Water compressibility and viscosity are taken from the [PVTW](#) data, unless [item 164](#) of [OPTIONS3](#) is set to 1, when water viscosity is calculated using Ezrokhi's method.

For H₂S no default values exist.

To obtain details of the solubility data used in a simulation the SOLU mnemonic may be specified with the [RPTPROPS](#) keyword.

The function $\varphi(P)$ is constructed by considering a pure CO₂ or H₂S/aqueous mixture. The gas phase fugacity is obtained using the equation of state.

The initial dissolved CO₂ or H₂S solubility as a function of depth may be specified using the [RSWVD](#) keyword. This initial solution honors this value unless it exceeds the saturated CO₂ concentration in which case the saturated value is used.

Notes

- The initial CO₂ or H₂S concentration may not be in exact equilibrium with the hydrocarbon phases and initial flash modifies the input values slightly.
- The first line of the solubility table defined using the [SOLUBILI](#) keyword represents water at standard conditions, when the dissolved CO₂ or H₂S amount is at a minimum. The simulator does not assume that the amount of CO₂ or H₂S in water at the lowest pressure is zero; if you want to specify it as such then you should set it to zero in the first line of the table. The data of this first line can be set to obtain a satisfying density of water with a small amount of dissolved CO₂ or H₂S.

For further information see "[CO2SOL](#)".

Water solubilities of both CO₂ and H₂S

If both components are made soluble this should be entered by the keywords [SOLUCO2H](#) and [SOLUH2SC](#) specifying the solubility of CO₂ in the presence of H₂S and the solubility of H₂S in the presence of CO₂. Please note that no default values exist in this case. If only CO₂ or H₂S is made soluble the keyword [SOLUBILI](#) is used to specify either CO₂ or H₂S as the component reacting with the aqueous phase.

Note: The CO₂ component is always regarded as the first component followed by the H₂S component when entering composition dependent data with [RSW](#) or [AMF](#).

Water component properties

After calculating the CO₂ aqueous mole fraction using the phase equilibrium algorithm mentioned previously, the saturated formation volume factor FVF_{sat} , the water compressibility C_{sat} , the water viscosity μ_{sat} and the water saturation pressure P_{sat} can be calculated using the CO₂-saturated water properties tables given by either [SOLUBILI](#), [SOLUBILS](#) or [SOLUBILT](#) using linear interpolation.

The formation volume factor FW at pressure P is calculated by:

$$\frac{1}{FVF} = \frac{1}{FVF_{sat}} \cdot (1 + X_c + 0.5X_c^2)$$

where the compressibility term X_c is given by

$$X_c = c_{sat} \cdot (P - P_{sat})$$

Density

The reservoir density of water as given by the mnemonic DENW of [RPTREST](#) or [RPTSOL](#) is given by:

$$\frac{\rho_s}{MW} \cdot \frac{MW_r}{FVF} \quad \text{Eq. 6.65}$$

where

ρ_s is the surface density, specified with either keywords [DENSITY](#), [GRAVITY](#) or [SOLUBILS](#).

MW is the molecular weight of pure water (18.015 kg/kg-M).

MW_r is the molecular weight of water in the reservoir calculated using the aqueous mole fraction of CO₂ (AMF_{CO_2}) and water and their molecular weights ($AMF_{CO_2} \cdot MW_{CO_2} + (1 - AMF_{CO_2}) \cdot MW$).

The molecular weight of CO₂ is specified using the keyword [MW](#).

FVF is the formation volume factor calculated as explained earlier.

Viscosity

The viscosity of water at pressure P is given by

$$\frac{\mu_{sat}}{(1 + Y_c + 0.5Y_c^2)} \quad \text{Eq. 6.66}$$

where the term Y_c is given by

$$Y_c = -c_V \cdot (P - P_{sat})$$

where

c_V is the “viscosibility”, specified with keyword [PVTW](#)

Aquifer properties

An analytic water aquifer defined by [AQUCHWAT](#), [AQUCT](#), [AQUFETP](#) or [AQUFLUX](#) contains only the “WATER” component. Any water component is allowed to leave the reservoir. If [item 131](#) of the `OPTIONS3` keyword is set to 1, this reverts to the pre-2008.1 behavior where the water entering or leaving the reservoir is made up entirely of the “WATER” component. Additionally, the water properties of the analytic aquifers are calculated as described above. If the [item 119](#) of `OPTIONS3` is set to 1, this reverts to the pre-2008.1 behavior where the water PVT tables were used. See "[Aquifer modeling facilities](#)".

GASSOL solution in the aqueous phase

	Eclipse 100
x	Eclipse 300

The GASSOL solution algorithm allows a specified number of gas components to dissolve in the aqueous phase. This option is selected using the [GASSOL](#) keyword in the RUNSPEC section. The basic model consists of fugacity functions for dissolving hydrocarbons in the aqueous phase. These are constructed to match the supplied solubility data and take the form:

$$f_i^A = x_i^A P \phi_i^A(P, T) \quad \text{Eq. 6.67}$$

where x_i^A is the saturated mole fraction from the solubility data for component i and $\phi_i^A(P, T)$ is the fugacity coefficient for component i in the aqueous phase at pressure P and temperature T . The phase equilibrium between the aqueous hydrocarbons and the hydrocarbon phases is defined by the condition that the fugacity values are equal. Using the equation of state with a sample pure “gas” composition, the corresponding vapor fugacity f_i^V is computed. The aqueous fugacity coefficient for component i is then given by:

$$\phi_i^A(P, T) = \frac{f_i^A}{P x_i^A} \quad \text{Eq. 6.68}$$

It is also possible to construct the aqueous hydrocarbon component Gibbs energy contributions.

The solubility properties of the hydrocarbon components in the aqueous phase may be entered as a function of pressure using the [SOLUAQA](#) keyword. The data for one or more soluble components may be defaulted, whereupon a default set of solubilities is generated internally as a function of pressure. It is strongly recommended, however, that appropriate values are supplied via the [SOLUAQA](#) keyword to ensure sufficient accuracy in the modeling of the solubility behavior.

The solubility data used in the simulation, which may be a combination of input and default values, can be reported by via the SOLU mnemonic of the [RPTPROPS](#) keyword.

The initial dissolved hydrocarbons solubility may be specified as a function of depth using the [RSWVD](#) keyword.

Notes

- The initial dissolved hydrocarbon concentrations may not be in exact equilibrium with the hydrocarbon oil and gas phases. In addition, the initial flash may also modify the input values slightly.
- The [SOLUAQA](#) keyword employs two records per component to specify the solubility data. The first record consists simply of the component number, which should correspond with numerical index of the component in the sequence of components defined by the [CNAMES](#) keyword. The second record consists of a table of solubility as a function of pressure.
- When using the [SOLUAQA](#) keyword, the solubilities of one or more dissolving hydrocarbon components may optionally be defaulted. In such cases, a default solubility table is generated by applying the GASWAT framework with the component properties specified in the [PROPS](#) section. The calculations assume a feed composition comprising an equal split between water and the component whose properties are being calculated for a range of pressures that is defined internally.
- If the solubilities for all of the dissolving hydrocarbon components are defaulted, the default tables will consist of twenty pressure nodes per table. If the solubilities for only some of the dissolving hydrocarbon components are defaulted, the number of pressure nodes in the defaulted tables will be equal to the maximum number of pressure nodes found in the non-defaulted tables. It is strongly

recommended that appropriate input values are supplied rather than relying upon the internally generated default values in order to ensure sufficient accuracy in the modeling of the aqueous behavior of the hydrocarbons.

Aqueous phase properties

Once the aqueous hydrocarbon mole fractions have been determined as described above the water density and viscosity can be calculated using Ezrokhi's method ([Zaytsev and Aseyev, 1993](#)) together with the density and viscosity coefficients specified using the [DENAQA](#) and the [VISCAQA](#) keywords respectively.

Density

The water density ρ is calculated using Ezrokhi's method ([Zaytsev and Aseyev, 1993](#)) and a correlation of the form:

$$\log_{10}(\rho) = \log_{10}(\rho_0(P, T)) + \sum_i A_i(T)w_i \quad \text{Eq. 6.69}$$

where

w_i is the weight fraction of the non-water component i and

$\rho_0(P, T)$ is the density of pure water.

The coefficients $A_i(T)$ are given by:

$$A_i(T) = a_{0,i} + a_{1,i}T + a_{2,i}T^2 \quad \text{Eq. 6.70}$$

where the temperature T is specified in Celsius.

The three regression coefficients for each component $a_{0,i}$, $a_{1,i}$ and $a_{2,i}$ are specified by the [DENAQA](#) keyword.

The reservoir density of aqueous phase can be output using the mnemonic DENW with the [RPTRST](#) or [RPTSOL](#) keywords.

Viscosity

The water viscosity μ is calculated using Ezrokhi's method ([Zaytsev and Aseyev, 1993](#)) and a correlation of the form:

$$\log_{10}(\mu) = \log_{10}(\mu_0(P, T)) + \sum_i B_i(T)w_i \quad \text{Eq. 6.71}$$

where

w_i is the weight fraction of the non-water component i and

$\mu_0(P, T)$ is the viscosity of pure water.

The coefficients $B_i(T)$ are given by:

$$B_i(T) = b_{0,i} + b_{1,i}T + b_{2,i}T^2 \quad \text{Eq. 6.72}$$

where the temperature T is specified in Celsius.

The three regression coefficients for each component $b_{0,i}$, $b_{1,i}$ and $b_{2,i}$ are specified by the [VISCAQA](#) keyword.

Aquifer properties

An analytic water aquifer defined by [AQUCHWAT](#), [AQUCT](#), [AQUFETP](#) or [AQUFLUX](#) contains only the pure water component. Any water component is allowed to leave the reservoir. Additionally, the water properties of the analytic aquifers are calculated as described above.

Polymer flood model

x	Eclipse 100
x	Eclipse 300

The main objective of polymer injection during water flooding of oil reservoirs is to decrease the mobility of the injected water. This decrease results in a more favorable fractional flow curve for the injected water, leading to a more efficient sweep pattern and reduced viscous fingering. Some plugging effects within highly permeable layers may also occur and result in the diversion of the injected water into less permeable zones of the reservoir.

The mobility decrease of the injected water resulting from the addition of polymer is due to two effects. Firstly, the viscosity of the polymer solution is higher than that of pure water (the viscosity of the polymer solution increases with the polymer concentration). Secondly, the rock permeability to water is reduced after the passage of a polymer solution through the rock material (the permeability to oil is, however, largely unaffected). Both effects combine to reduce the water mobility while that for the oil is unaltered.

To achieve maximum efficiency, the polymer solution is often applied in the form of a tapered slug. At the front edge of the slug, the displacement is stable but the interface between the water and the polymer solution smears due to physical dispersion of the polymer. At the rear edge, the mobility ratio is unfavorable and is dominated by viscous fingering. Both effects cause deterioration of the slug. In Eclipse 100 these effects are modeled by means of a mixing parameter applied to the viscosity terms in the fluid flow equations. In Eclipse 300 these effects are currently not modeled; it is assumed that the polymer is fully mixed in the water phase.

Certain polymer solutions, such as those of the PA type (Polyamides), exhibit a strong sensitivity to the presence of certain salts. In particular, sodium chloride can influence the viscosity characteristics of PA solutions to such an extent that in highly saline reservoirs it is often necessary to pre-flush with fresh water to reduce the exposure of the polymer solution to the reservoir brine. Both Eclipse 100 and Eclipse 300 models allow you to investigate the effect of varying brine concentrations on the efficiency of the polymer flood. The brine effect is, however, optional.

When a polymer solution is injected into the reservoir some of the long chain molecules constituting the polymer are adsorbed onto the rock surfaces. Mechanical entrapment of some of the large molecules at the entrance to small pore throats may also occur and account for an apparent loss of polymer from the invading solution. Experimentally, the reservoir rock material is believed to retain a specific capacity of polymer. The main effects of polymer loss occur at the leading edges of the polymer slug where a stripped water bank is created and the slug width is gradually reduced in time. Some desorption effects can occur as the trailing edge of the slug passes but these effects are usually small compared with the adsorption losses. Eclipse 100 allows you to either neglect the desorption effect or to allow the adsorption isotherm to be retraced after passage of the polymer slug. Eclipse 300 allows you to model the effects of adsorption and desorption by combining the Polymer Flood Model with the Solid Model.

Eclipse 100 considerations

A further effect caused by the adsorption and entrapment processes is a reduction in the relative permeability of the polymer solution. The reduction results from an interaction between the aqueous solution and the polymer retained by the rock material. For modeling purposes it will be assumed that the reduction in permeability to the polymer solution is proportional to the quantity of polymer lost to the rock material. The permeability of the rock to water is thus permanently reduced after the passage of a polymer slug compared to its value before the passage. Experimentally, it is found that only a very small change occurs to the hydrocarbon relative permeability and the Eclipse 100 model assumes that the change is negligible.

The effect of temperature variations on the behavior of the polymer solution can be optionally modeled.

In core flooding experiments, it is often observed that injected polymer slugs break through to producers earlier than tracer slugs (for example, NaCl). The polymer fluid velocity is higher than that of the tracer fluid within the porous medium and this is due to the fact that only a fraction of the total pore space is available to the polymer fluid. As the inaccessible pore space to the polymer fluid increases, the effective polymer velocity through the rock increases and leads to a faster breakthrough of polymer. This chromatographic effect is modeled by assuming that the dead pore space is constant for each rock type:

- and either does not exceed the corresponding irreducible water saturation (pre 2008.2).
- or independent of the water saturation (post 2008.2 included)

The rheology of polymer solutions is not simple. At low flow rates the viscosity of the solution is approximately constant and depends only on the concentration of polymer in the solution. At higher flow rates the solution viscosity reduces in a reversible (elastic) manner. At even higher velocities the large polymer molecules begin to break up, and the viscosity reduction becomes irreversible (plastic). The effects tend to be greatest in the vicinity of injection wells where the fluid velocity is greatest, and so is the shear rate.

The Eclipse 100 polymer flood simulation model

The flow of the polymer solution through the porous medium is assumed to have no influence on the flow of the hydrocarbon phases. The standard black-oil equations are therefore used to describe the hydrocarbon phases in the model.

Modification is required to the standard aqueous (water) equation and additional equations are needed to describe the flow of polymer and brine within the finite difference grid. The water, polymer and brine equations used in the model are as follows:

$$\frac{d}{dt} \left(\frac{VS_w}{B_r B_w} \right) = \sum \left[\frac{Tk_{rw}}{B_w \mu_{w,eff} R_k} (\delta P_w - \rho_w g D_z) \right] + Q_w \quad \text{Eq. 6.73}$$

$$\begin{aligned} \frac{d}{dt} \left(\frac{V^* S_w C_p}{B_r B_w} \right) + \frac{d}{dt} \left(V \rho_r C_p^a \frac{1-\phi}{\phi} \right) &= \sum \left[\frac{Tk_{rw}}{B_w \mu_{p,eff} R_k} (\delta P_w - \rho_w g D_z) \right] C_p \\ &+ Q_w C_p \end{aligned} \quad \text{Eq. 6.74}$$

$$\frac{d}{dt} \left(\frac{VS_w C_n}{B_r B_w} \right) = \sum \left[\frac{Tk_{rw} C_n}{B_w \mu_{s,eff} R_k} (\delta P_w - \rho_w g D_z) \right] + Q_w C_n \quad \text{Eq. 6.75}$$

pre-2008.2

$$V^* = V \left(1 - \frac{S_{dpv}}{S_w} \right) \quad \text{Eq. 6.76}$$

post-2008.2

$$V^* = V (1 - S_{dpv}) \quad \text{Eq. 6.77}$$

where

S_{dpv} denotes the dead pore space within each grid cell

C_p^a denotes the polymer adsorption concentration

ρ_r	denotes the mass density of the rock formation
φ	denotes the porosity
ρ_w	denotes the water density
Σ	denotes the sum over neighboring cells
R_k	denotes the relative permeability reduction factor for the aqueous phase due to polymer retention
C_p, C_n	denote the polymer and salt concentrations respectively in the aqueous phase
$\mu_{a\text{eff}}$	denotes the effective viscosity of the water ($a=w$), polymer ($a=p$) and salt ($a=s$).
D_z	is the cell center depth.
B_r, B_w	are the rock and water formation volumes
T	is the transmissibility
k_{rw}	is the water relative permeability
S_w	is the water saturation
V	is the block pore volume
Q_w	is the water production rate
P_w	is the water pressure
g	is the acceleration due to gravity

The model makes the assumption that the density and formation volume factor of the aqueous phase are independent of the polymer and salt concentrations. The polymer solution, reservoir brine and the injected water are represented in the model as miscible components in the aqueous phase, where the degree of mixing is specified through the viscosity terms in the conservation equations.

The principal effects of polymer and salt on the flow of the aqueous phase are represented by equations 6.73 to 6.76 above. The fluid viscosities ($\mu_{w\text{ eff}}, \mu_{p\text{ eff}}, \mu_{s\text{ eff}}$) are dependent on the local concentrations of salt and polymer in the solution. Polymer adsorption is represented by the additional mass accumulation term on the left hand side of equation 6.74. The adsorption term requires that you specify the adsorption isotherm, C_p^a for each rock type. The effect of pore blocking and adsorption on the aqueous phase relative permeability is treated through the term, R_k , which requires the input of a residual resistance factor for each rock type.

The equations solved for the polymer model are a discretized form of the differential equations 6.73 - 6.76. In order to avoid numerical stability problems that could be triggered by strong changes in the aqueous phase properties over a timestep (resulting from large changes in the local polymer/salt concentrations) a fully implicit time discretization is used, which is therefore free from this type of instability.

Treatment of fluid viscosities

The viscosity terms used in the fluid flow equations contain the effects of a change in the viscosity of the aqueous phase due to the presence of polymer and salt in the solution. However, to incorporate the effects of physical dispersion at the leading edge of the slug and also the fingering effects at the rear edge of the slug the fluid components are allocated effective viscosity values that are calculated using the Todd-Longstaff technique.

Consider the effective polymer viscosity term. The viscosity of a fully mixed polymer solution as an increasing function of the polymer concentration in solution ($\mu_m(C_p)$) must be input. The viscosity of the solution at the maximum polymer concentration also needs to be specified and denotes the injected polymer concentration in solution (μ_p). The effective polymer viscosity is taken to be:

$$\mu_{p, \text{ eff}} = (\mu_m(C_p))^\omega \cdot \mu_p^{1-\omega} \quad \text{Eq. 6.78}$$

where

ω is the Todd-Longstaff mixing parameter.

The mixing parameter is useful in modeling the degree of segregation between the water and the injected polymer solution. If $\omega = 1$ then the polymer solution and water are fully mixed in each.

If $\omega = 0$, the polymer solution is completely segregated from the water.

The partially mixed water viscosity is calculated in an analogous manner using the fully mixed polymer viscosity and the pure water viscosity (μ_w):

$$\mu_{w, e} = (\mu_m(C_p))^\omega \cdot \mu_w^{1-\omega} \quad \text{Eq. 6.79}$$

In order to calculate the effective water viscosity to be inserted into equation 6.73, the total water equation is written as the sum of contributions from the polymer solution and the pure water. The following expression then gives the effective water viscosity to be inserted into equation 6.73:

$$\frac{1}{\mu_{w, \text{ eff}}} = \frac{1 - \bar{C}}{\mu_{w, e}} + \frac{\bar{C}}{\mu_{p, \text{ eff}}} \quad \text{Eq. 6.80}$$

$$\bar{C} = \frac{C_p}{C_{p, \text{ max}}} \quad \text{Eq. 6.81}$$

where

$\bar{C}$ is the effective saturation for the injected polymer solution within the total aqueous phase in the cell.

If the brine option is active, the above expressions still hold for the effective polymer and water viscosity terms. The injected salt concentration needs to be specified in order to evaluate the maximum polymer solution viscosity, μ_p . The effective salt component viscosity to be used in equation 6.75 is set equal to the effective water viscosity.

Note that the effective viscosities are computed at the pressure of the oil phase.

Treatment of fluid viscosities in the well

The multisegment well and friction well models both include frictional pressure drops which depend upon fluid viscosity. Polymer bearing water will by default have a viscosity as defined by equation 6.81, where the Todd-Longstaff mixing parameter ω is assigned to the well using the [WELSPECS](#) or [WELSPECL](#)

keyword. There is a further option to control this viscosity; which is using the keyword **WPOLYRED** to set the polymer reduction factor α . The water phase viscosity in the wellbore is then calculated as

$$\mu_w = \alpha\mu_{w\text{ eff}} + (1-\alpha)\mu_w \quad \text{Eq. 6.82}$$

Polymer and water are assumed to travel with the same velocity in the wellbore.

Treatment of polymer adsorption

Adsorption is treated as an instantaneous effect in the model. The effect of polymer adsorption is to create a stripped water bank at the leading edge of the slug. Desorption effects may occur as the slug passes. The isotherm adsorption can be specified as:

- either a look-up table of adsorbed polymer as a function of polymer concentration using the **PLYADS** keyword if there is no salinity dependence of the adsorption, or the **PLYADSS** keyword if there is. For salinity dependence, the **ADSALNOD** keyword is used for the salinity values, not the **SALTNODE** keyword.
- or by a generic analytical adsorption model using the **ADSORP** keyword with **POLYMER** as an argument. This model enables the adsorption to depend on the salinity and rock permeability.

Note that the multi-component brine model (**ECLMC** keyword in the **RUNSPEC** section) cannot currently be used with the polymer/brine flood model.

If desorption is prevented (item 4 of the **PLYROCK** keyword) then the adsorbed polymer concentration may not decrease with time. If desorption is allowed then each grid block retraces the adsorption isotherm as the polymer concentration rises and falls in the cell.

Treatment of permeability reductions and dead pore volume

The adsorption process causes a reduction in the permeability of the rock to the passage of the aqueous phase and is directly correlated to the adsorbed polymer concentration. In order to compute the reduction in rock permeability, you are required to specify the residual resistance factor (RRF) for each rock type (item 2 of the **PLYROCK** keyword). The actual resistance factor can then be calculated:

$$R_k = 1.0 + (\text{RRF} - 1.0) \frac{C_p^a}{C_p^{\text{amax}}} \quad \text{Eq. 6.83}$$

The value of the maximum adsorbed concentration, C_p^{amax} , depends on the rock type and needs to be specified as item 5 of the **PLYROCK** keyword. This value must be non-zero.

The dead pore volume (item 1 of the **PLYROCK** keyword) must also be specified for each rock type. It represents the amount of total pore volume in each grid cell that is inaccessible to the polymer solution. The effect of the dead pore volume within each cell is to cause the polymer solution to travel at a greater velocity than inactive tracers embedded in the water. This chromatographic effect is modeled by assuming that the dead pore space is constant for each rock type:

- and either does not exceed the corresponding irreducible water saturation (pre 2008.2)
- or is independent of the water saturation (post 2008.2 included).

If required, the RRF can be set on a cell-by-cell basis using the **PLYKRRF** keyword. Any cells for which it is not set will use the value for their rock type given by item 2 of the **PLYROCK** keyword.

If the temperature option is specified, then a temperature dependence can be provided for the RRF using the **PLYTRRF** keyword.

Similarly, the value of the maximum adsorbed concentration can be set on a cell-by-cell basis using the [PLYCAMAX](#) keyword. Any cells for which it is not set will use the value for their rock type given by item 5 of the [PLYROCK](#) keyword.

Treatment of the non-Newtonian rheology

You can choose from three models to take into account some of the non-Newtonian rheological behavior reported for polymer solutions. Non-Newtonian rheology of the polymer solution is represented by modifying the solution's viscosity (for models 1 and 3) or effective transmissibility (model 2).

Note: These modified properties are applied to the flows between cells, and are based on the flow magnitude (velocity, flow rate or shear rate) through the face and cell properties based on both cells, or on the upstream cell. In contrast, the unmodified fluid viscosity is calculated based on the properties of one cell and applied at the cell center. For this reason, although the modified properties are always displayed in the cell with the lower cell address relative to the face (that is it is always an I+, J+ or K+ face property displayed at the cell (I,J,K)), the actual fluid properties influencing the modified viscosity or transmissibility may come from the cell on the other side of the face if that cell is the upstream cell.

Model 1

This model targets the shear thinning or thickening of polymer that has the effect of changing the polymer viscosity at higher flow rates. It assumes that shear rate is proportional to the flow velocity. This assumption is not valid in general, as for example, a given flow in a low permeability rock will have to pass through smaller pore throats than the same flow in a high permeability rock, and consequently the shear rate will be higher in the low permeability rock. However, under the consideration of a single type of reservoir rock, this assumption can be reasonable.

This model is activated by the [PLYSHEAR](#) keyword. Note that shear thinning can cause runaway positive feedback due to the loop: increased speed causes decreased viscosity causes increased speed. It may be easier to control the shear-thinning effect by using Model 3 below. Shear thickening may exacerbate convergence problems in wells where crossflow is disallowed. Please refer to the keyword description for more details. Note that crossflow in wells is now allowed with polymer.

Model 2

The second model is the Herschel-Bulkley model that can be used to model shear thinning and thickening as well as yield stress, dependent on polymer concentration. Refer to "[Non-Newtonian fluid flows](#)" for more details.

Model 3

The third model is similar to Model 1 but instead uses a logarithmic look-up calculation in order to update the water phase and polymer flow properties in the presence of shear thinning/thickening. This model is activated by the [PLYSHLOG](#) keyword. The shear multipliers can be specified either as a function of water phase/polymer flow velocity or as function of water phase/polymer flow shear rate; keyword [SHRATE](#) is required for the latter.

Model 1 description

In Model 1, the velocity corresponding to the given water/polymer flow is calculated as:

$$V_i = B_w \cdot \frac{F_i}{\phi A(S_w - S_{wcr})} \quad \text{Eq. 6.84}$$

where

F_i is the water/polymer ($i = w, p$ respectively) flow rate in surface units

B_w is the water formation volume factor

ϕ is the average porosity of the two cells

A is the flow area between the two cells

S_w is the water saturation

S_{wcr} is the critical water saturation.

The shear effect is skipped in those cases where the mobile water saturation is less than or equal to zero.

Note:

Note that the pre-2014.1 velocity calculation is given by:

$$V_i = B_w \cdot \frac{F_i}{\phi A} \quad \text{Eq. 6.85}$$

and may be activated by setting OPTIONS keyword [item 228](#) to 1.

The reduction in each of the effective component viscosities is assumed to be reversible as a function of the relevant velocity. The resulting shear water and shear polymer viscosities are calculated, respectively, as:

$$\mu_{w,sh} = \mu_{w,eff} Z_w^*, \quad \mu_{p,sh} = \mu_{p,eff} Z_p^* \quad \text{Eq. 6.86}$$

where

$\mu_{w,sh}$ is the shear water viscosity following shear thinning/thickening

$\mu_{p,sh}$ is the shear polymer viscosity following shear thinning/thickening

$\mu_{w,eff}$ is the effective water viscosity (see equation [6.80](#))

$\mu_{p,eff}$ is the effective polymer viscosity (see equation [6.78](#))

Z_w^* is the shear factor applied to the effective water viscosity and is calculated based on the water phase flow velocity prior to shear thinning/thickening

Z_p^* is the shear factor applied to the effective polymer viscosity and is calculated based on the polymer flow velocity prior to shear thinning/thickening

and each shear factor is constructed in the form

$$Z = \frac{1 + (P - 1)M}{P} \quad \text{Eq. 6.87}$$

where

P is the viscosity multiplier assuming no shear effect (entered using the [PLYVISC](#), [PLYVISCS](#), [PLYVISCT](#) or [PLYVSCST](#) keywords)

M is the shear thinning/thickening multiplier supplied in the `PLYSHEAR` keyword.

Note: For $M = 1$, or no shear effect, we recover the effective component viscosities, and for $M = 0$, or maximum shear thinning, the shear viscosities are

$$\mu_{w,sh} = \frac{\mu_{w,eff}}{P}, \quad \mu_{p,sh} = \frac{\mu_{p,eff}}{P} \quad \text{Eq. 6.88}$$

which corresponds to the minimum viscosity that can be obtained. If the polymer concentration is zero ($P=1$) we recover the effective component viscosities which, in that particular case, equal the pure water viscosity.

With the above definitions, there is a local implicit relationship between the water phase velocity through a grid block face and its associated velocity-dependent water phase viscosity in the presence of polymer. The equation for this relationship is

$$V_{i,sh} = V_{i,0} \left(\frac{\mu_{i,eff}}{\mu_{i,sh}(V_{i,sh})} \right) \quad \text{Eq. 6.89}$$

where

$V_{i,sh}$ is the shear-modified velocity associated with the water phase/polymer ($i = w, p$ respectively) flow,

$V_{i,0}$ is the Darcy velocity without shear calculated from equation 6.84 (or from equation 6.85 if `OPTIONS` keyword `item 228` is set to 1),

and the velocity dependence of the shear-modified viscosity is explicitly shown. This equation is solved by an appropriate root finding method. Once the solution of this equation is available for a given grid block face, the sheared velocity at that location is known along with the additional derivatives that it gives rise to; these modified values and derivatives then enter the global implicit calculation for reservoir flows for the current Newton iteration.

The well inflows are treated in a manner analogous to the treatment of block to block flows. The viscosity of the polymer solution flowing into the well is calculated assuming a velocity at a representative radius from the well. The representative radius is taken to be:

$$R_r = \sqrt{R_w R_a} \quad \text{Eq. 6.90}$$

where

R_w is the well bore radius (taken from diameter input in `COMPDAT`)

R_a is the area equivalent radius of the grid block in which the well is completed. This is the radius of a circle with the same area as the cross section of the connected cell perpendicular to the well

penetration direction, so for a connection penetrating in the Z direction $R_a = \sqrt{\frac{D_x D_y}{\pi}}$.

In the present version of Eclipse 100, the radial inflow equation is not integrated over distance from the well to account for the local viscosity reduction due to the local velocity.

Model 3 description

Model 3 is implemented via two approaches, as detailed below. Note that the representative radius for the well inflows described above for Model 1 also applies in each approach below.

Approach I: Based on velocity

In this first approach, for a specified set of reference conditions comprising polymer concentration, salinity and temperature, the shear thinning/thickening multipliers, Z^{ref} , are specified as a function of water phase/polymer flow velocity, V , this time using the `PLYSHLOG` keyword.

As in Model 1, the velocities corresponding to the water and polymer flows are calculated according to equation 6.84 (or according to equation 6.85 if `OPTIONS` keyword `item 228` is set to 1).

The resulting shear viscosities are again obtained using equation 6.86 where the calculation of each shear thinning/thickening multiplier (multiplier Z_w^* for the water phase flows and multiplier Z_p^* for the polymer flows) comprises the following series of steps:

- For each velocity in `PLYSHLOG`, calculate a corresponding viscosity multiplier using

$$Z(V) = \frac{1 + (P-1)M^{ref}(V)}{P} \quad \text{Eq. 6.91}$$

where

M^{ref} is calculated by applying the reverse of equation 6.91 and using the reference data provided in `PLYSHLOG`, and

P is the viscosity multiplier interpolated at the given conditions using the relevant polymer viscosity keyword (one of `PLYVISC`, `PLYVISCS`, `PLYVISCT`, `PLYVSCST`).

- Take the natural logarithm of each viscosity multiplier calculated above and the natural logarithm of each corresponding velocity and construct the piecewise linear function through the resulting set of data points:

$$f: \ln V \mapsto \ln Z \quad \text{Eq. 6.92}$$

- Find where the relationship between the unsheared and sheared velocities satisfies the input shear data obtained experimentally, this done by locating the point of intersection of the above piecewise linear function and the straight line represented by:

$$\ln V + \ln Z = \ln V_0 \quad \text{Eq. 6.93}$$

in which V_0 is the Darcy velocity corresponding to the water phase/polymer flow calculated from equation 6.84 (or equation 6.85 if `OPTIONS` keyword `item 228` is set to 1) without any shear effects. This intersection provides the shear-modified velocity for the given water phase/polymer flow and the relevant shear thinning/thickening multiplier.

Note that the shear data specified via the `PLYSHLOG` keyword must satisfy the constraint

$$\frac{d \ln M^{ref}}{d \ln V} > -1 \quad \text{Eq. 6.94}$$

which henceforth ensures a unique point of intersection. In view of equation 6.86, the reservoir flow rates in equation 6.73 and equation 6.74 are modified accordingly together with the associated derivatives, that is, the unsheared flows are multiplied by the reciprocal of the relevant shear multiplier calculated as per above. A similar transformation is likewise applied to the unsheared well component mobilities.

Approach II: Based on shear rate

In this second approach, the shear multiplier is specified instead as a function of water phase/polymer flow shear rate. Once again the shear data is specified using the `PLYSHLOG` keyword, for a specified set of reference conditions. Keyword `SHRATE` must also be specified in order to activate this option. In this case, the first column in each `PLYSHLOG` table corresponds to the shear rate, γ .

This approach follows Approach I very closely, but with the following modifications:

- The calculation of shear multiplier in equation 6.91 is done for each shear rate (as opposed to velocity) entered in the `PLYSHLOG` table.
- Take the natural logarithm of each shear rate in `PLYSHLOG` and take the natural logarithm of the corresponding viscosity multiplier calculated via equation 6.91. Construct the piecewise linear function through the resulting set of data points:

$$f: \ln \gamma \mapsto \ln Z \quad \text{Eq. 6.95}$$

- Locate the point of intersection of the above piecewise linear function and the straight line represented by:

$$\ln \gamma + \ln Z = \ln \gamma_0 \quad \text{Eq. 6.96}$$

in which γ_0 is the shear rate corresponding to the given inter-block water phase/polymer flow prior to shear thinning/thickening and is calculated as follows,

$$\gamma_0 = C_\gamma V_0 \sqrt{\frac{\varphi S_w}{K k_{rw}}} \quad \text{Eq. 6.97}$$

where:

V_0 is the unsheared inter-block velocity calculated from equation 6.84 (or from equation 6.85 if `OPTIONS` keyword `item 228` is set to 1)

K is the absolute permeability

k_{rw} is the water relative permeability associated with the flow from the upwind cell

S_w is the water saturation in the upwind cell

φ is the average porosity of the two cells and

C_γ is the shear rate constant specified via the `SHRATE` keyword.

At a well connection, the transformation 6.97 is applied once again to convert, this time, from the velocity in the connection to the corresponding shear rate. Corresponding grid block quantities are used in order to carry out this default conversion which therefore replicates that applied in the reservoir flow calculation.

Note:

The pre-2014.1 conversion from velocity to shear rate used for reservoir flows is given by

$$\gamma_0 = C_\gamma V_0 \sqrt{\frac{\varphi}{K S_w k_{rw}}} \quad \text{Eq. 6.98}$$

and the pre-2014.1 conversion from velocity to shear rate for well connections is given by

$$\gamma_0 = \frac{C_\gamma V_0^{wc}}{D^{wc}} \quad \text{Eq. 6.99}$$

in which V_0^{wc} and D^{wc} represent, respectively, the unsheared velocity and the diameter in the given wellbore connection. Note that conversions 6.98 and 6.99 may be activated in conjunction with the new default velocity calculation (by setting `OPTIONS` keyword item 228 to 2) or activated in conjunction with the pre-2014.1 velocity calculation (by setting `OPTIONS` keyword item 228 to 3). The possible options and corresponding conversions are summarized in the following table.

	Default	OPTIONS item 228 set to 1	OPTIONS item 228 set to 2	OPTIONS item 228 set to 3
Conversions applied	Equations 6.84 and 6.97	Equations 6.85 and 6.97	Equations 6.84, 6.98 and 6.99	Equations 6.85, 6.98 and 6.99

Table 6.2: Options available for converting between (i) flow rate and velocity, and (ii) velocity and shear rate for models using the `SHRATE` keyword

The shear effect is skipped in those cases where a conversion from velocity to shear rate is not possible due to zero denominator terms associated with 6.97 and 6.98.

As with Approach I, the input `PLYSHLOG` data must satisfy the constraint

$$\frac{d \ln M^{ref}}{d \ln \gamma} > -1 \quad \text{Eq. 6.100}$$

which yields a unique point of intersection. Again, the unsheared reservoir flow rates and unsheared well component mobilities are modified by multiplying through by the reciprocal of the relevant shear thinning/thickening factor. The associated derivatives are modified in line with this shear effect.

Cases where the shear effect is omitted

Shear thinning/thickening is not applied in those cases where:

- The value of the unsheared velocity (or shear rate) is less than the minimum velocity (or shear rate) specified in `PLYSHLOG`.
- There is no significant effect due to the polymer concentration (case where $P = 1$)

In such cases, the shear viscosities are equal to the effective component viscosities, that is, $Z_w^* = Z_p^* = 1$.

Temperature effects

Various effects of temperature on the polymer flood model are considered. First, the viscosity of the polymer solution is now allowed to vary with temperature. This is analogous to the variation with salinity specified by `PLYVISC`, and two new keywords implement this: `PLYVISCT`, which provides tables of temperature variation only as a function of polymer concentration, and `PLYVSCST`, which includes both temperature and salinity variations. All these keywords (and `PLYVISC`, which just considers the effect of polymer concentration) provide a multiplier for the given conditions which modifies the pure water viscosity specified in `PVTW`.

Note that a temperature effect on the pure water viscosity can be specified with `WATVISCT`, which is not a multiplier but a replacement viscosity value. The data input in `PVTW` is used only to provide a multiplier for

the pressure variation (viscosity) in this case. A warning is given if the new polymer viscosity keywords for temperature variation are used with WATVISCT, but the cumulative effect of both specifications is allowed. You should decide which combination of inputs best fits your model.

Second, the polymer can degrade due to thermal effects. This is modeled as a half-life of the reduction of polymer concentration, and is input as a table of half-life against temperature by the keyword `PLYDHFLF`. The equation 6.74 is modified by the addition of the term

$$Q_{HL} = -\alpha \left(V^* S_w C_p + \frac{1-\phi}{\phi} V \rho_r C_p^a \right) \quad \text{Eq. 6.101}$$

on the right-hand side, where α is related to the half-life T_H by $\alpha = \ln(2)/T_H$.

Thirdly, the adsorption of the polymer can also be temperature dependent. In this case, multiple occurrences of the `PLYADS` or `PLYADSS` keywords specifying the adsorbed concentration can be used. Each occurrence refers to a different temperature, which is specified by the use of the `PLYATEMP` keyword.

Finally, the residual resistance factor of the adsorbed polymer (keywords `PLYROCK` and `PLYKRRF`) can be made temperature dependent using the `PLYTRRF` keyword.

The temperature model must be activated with the keyword `TEMP` for these keywords to function.

See also *Clifford and Sorbie (1985)*.

Temperature-sensitive popping polymer microgels

Various technologies (such as Bright WaterTM; see *Pritchett et al (2003)* and *Garmeh et al (2011)*) now exist to inject polymer microgels which have low impact on viscosity and residual resistance factor at low injection temperatures, but, when heated to reservoir temperature, “pop” (expand their molecular size considerably) to produce much higher solution viscosities and adsorbed material residual resistance factors. This allows them to block high permeability streaks (“thief zones”) and ensure better sweep efficiency.

In Eclipse 100, these two effects are modeled by separate keywords. Temperature-dependent polymer solution viscosity is represented by the two keywords `PLYVISCT`, which provides tables of temperature variation only as a function of polymer concentration, and `PLYVSCST`, which includes both temperature and salinity variations. To model the temperature sensitivity of these microgels, the multipliers in either keyword should increase markedly between injection and reservoir temperatures. The temperature dependence of the residual resistance factor is represented by the `PLYTRRF` keyword, which multiplies the residual resistance factor of the adsorbed polymer (keywords `PLYROCK` and `PLYKRRF`) by a factor dependent on the temperature. Again, the multiplier value should increase markedly between injection and reservoir temperatures.

The temperature model must be activated with the keyword `TEMP` for these keywords to function. This model will also allow rock conduction of heat from the reservoir rock to the cooler injected fluid and it is important that the conductivity (`THCONR`) is large enough to heat the polymer solution as it passes through the formation, for the gel-popping representation to work.

It should be noted that, while the microgel “popping” behavior is expected to be irreversible, the current modeling described here has more limited irreversibility. The adsorbed polymer should be prevented from returning to solution by setting item 4 of keyword `PLYROCK` to the value 2. The viscosity of the “popped” microgel cannot be prevented from returning to its low-temperature value if the solution is later cooled - which may happen by mixing with water or other polymer solution injected later in the process. It is hoped that the blocking effect of the initial viscosity increase will act to decrease such mixing and retain the blocking effect to later times.

For the temperature dependence of the residual resistance factor, there are additional factors governing how the irreversibility of the popping effect is represented. In principle, the cell containing the adsorbed polymer can be heated and cooled multiple times. As the cell heats, existing adsorbed polymer can expand further according to the temperature, as represented by the `PLYTRRF` data. If the cell is re-cooled by further injectant, and more polymer is adsorbed, then there will be at least two populations of adsorbate in the cell with two different RRF values. Eclipse 100 provides two approximations to this effect. In the first, the re-cooling effect is ignored, and the polymer is assumed to respond to the maximum temperature experienced by the cell while polymer is present. This allows for initial cooling by the injectant, while minimizing the memory usage required to represent adsorbate at multiple RRF values. This option uses only the `PLYTRRF` keyword.

The second approximation allows the existence of more than one adsorbate RRF state. As an example, in a given grid cell, polymer adsorbed at the start of injection, and subsequently heated by the reservoir, is assigned the RRF value of the reservoir temperature. Polymer injected later and adsorbed in the same cell at a cooler temperature is assigned the RRF value of the injection temperature. The total RRF value for that cell is a weighted average of the two different RRFs, with the amounts of adsorbed polymer providing the weights. The temperatures are specified by ranges between the values of the `PLYTRRF` table, and in this case the RRF multipliers are assumed constant in a given temperature range rather than being piece-wise linear.

This approximation is activated by the `PLYTRRFA` keyword, which also specifies the number of temperature intervals used. In this case, a separate adsorbed polymer concentration is stored for each cell and each interval. More intervals can be specified, to represent partial expansion of the microgel, but this will increase the amount of memory used. There is a trade-off between any performance degradation due to this extra memory compared to the improved representation of the polymer history, just as there is a trade-off between using piece-wise constant multipliers for the intervals in this option, compared with the piecewise linear representation in the previous option.

In this case, the part of the adsorbed polymer whose current RRF value corresponds to the nnn^{th} temperature interval of the `PLYTRRF` table can be viewed using the mnemonic `CABINnnn`, where `nnn` has a format between 001 and 999 but must not exceed the `PLYTRRFA` argument value.

The polymer flood model and wells

Prior to the 2012.1 release, the polymer flood option was not compatible with crossflowing wells. From 2012.1, the polymer concentration is solved as an implicit variable in the well model. Brine concentration is also solved this way for cases with polymer and brine. This allows the standard well model to calculate polymer and brine crossflows. Pre-2012.1 behavior may be recovered using `OPTIONS` keyword [item 176](#). Polymer and brine are solved as tracers within the water phase rather than as a separate phase and travel with the same velocity as the water phase.

Multisegment and friction wells

If the wellbore friction Model or multisegment well model are used, then water viscosity may be calculated taking polymer concentration into account. The effective viscosity of the water/polymer mixture is calculated as described above in equation [6.82](#). The Todd-Longstaff mixing parameter ω and maximum polymer and brine concentrations are determined by the polymer mixing table number defined for the well. See the `WELSP ECS` and `WELSP ECL` keywords for details of how to specify the table number. The water viscosity may be further tuned using the `WPOLYRED` keyword.

Note: The effect of non-Newtonian rheology is not included in the wellbore.

Using the polymer flood model

The option is activated by the keyword **POLYMER** in the RUNSPEC section. If salt-sensitivity for polymer is required, this can be set by adding the keyword **BRINE** to the RUNSPEC section. By default the polymer model is not salt-sensitive.

The mixing parameter data is obligatory and should be provided using the keyword **PLMIXPAR**. The maximum number of mixing parameter regions is set using the parameter **NPLMIX** in the keyword **REGDIMS**. The associated mixing region for each grid cell is specified using the **PLMIXNUM** keyword. The mixing region for each well is specified using **WELSPECS** or **WELSPECL** as appropriate. The maximum polymer and salt concentrations to be used in calculating the effective fluid component viscosities are entered under the keyword **PLYMAX**.

The viscosity of a fully mixed polymer solution needs to be defined as function of the solution concentration. This should be done using the keyword **PLYVISC**. If the salt-sensitive option is activated (**BRINE** specified), then the keywords **SALTNODE** and **PLYVISC** should be used to enter the two-dimensional table for the viscosity of the solution.

The polymer adsorption data should be entered using either with the keyword **PLYADS**, with the **PLYADSS** keyword accompanied by the keyword **ADSALNOD**, or with the **ADSORP** keyword with **POLYMER** as argument in the PROPS section. When the **ADSORP** keyword is used, the polymer adsorption can be set to depend on the salinity and rock permeability. In this case, with the standard brine option, the salt's mass concentration is directly used in the analytic model, and therefore the units of the coefficients need to be interpreted accordingly.

Note that the multi-component brine model (**ECLMC** keyword in the RUNSPEC section) cannot currently be used with the polymer/brine flood model.

Other polymer-rock parameters such as the rock mass density used in the adsorption calculation, the dead pore volume and the residual resistance factor are input using the keyword **PLYROCK**.

If required, individual cell values can be set for the rock mass density, the residual resistance factor, and the maximum adsorbed polymer using **PLYRMDEN**, **PLYKRRF** and **PLYCAMAX** respectively.

Non-Newtonian rheology can be activated by setting one of the following:

- The **PLYSHEAR** keyword in the PROPS section for model 1.
- Opting for the Herschel-Bulkley model as described in "Non-Newtonian fluid flows".
- The **PLYSHLOG** keyword in the PROPS section for model 3 (approach I).
- The **PLYSHLOG** and **SHRATE** keywords in the PROPS section for model 3 (approach II).

If Model 1 is activated, the shear thinning/thickening data consists of tables of viscosity reduction as a function of local velocity. The values of $1/(\varphi \cdot A)$ used to calculate the velocity can be printed out using the mnemonic **POLYMER** in the **RPTGRID** keyword. The representative radius used in the well inflow calculation is output with the well connection data.

If model 3 (approach I) is activated, the shear thinning/thickening data consists of tables of shear multipliers as a function of local velocity, and these correspond to a specified set of reference conditions. If model 3 (approach II) is activated, the multipliers are specified as a function of local shear rate instead.

Note however that if non-Newtonian rheology is not an important effect, it should be omitted as the calculation of the velocity dependent viscosity can significantly increase the CPU requirements for the run.

When the brine option is activated in conjunction with polymer, the initial salt concentrations are specified with either the **SALT** keyword (if the enumeration method is used for initialization) or the **SALTV**

keyword (if the [EQUIL](#) keyword is used). The initial polymer concentration and the initial adsorbed polymer concentration are both automatically set to zero by default; however the [SPOLY](#) keyword can be used to override this.

If the model contains analytic aquifers, the initial salt concentration in each aquifer should be specified using the appropriate keywords [AQUCT](#), [AQUFET](#) or [AQUFETP](#). Since the aquifer volume is large, the polymer concentration is assumed to be always zero - any polymer entering the aquifer due to backflow is adsorbed by the aquifer rock.

If the standard brine is used, the polymer and salt well injection concentration should be set using the keyword [WPOLYMER](#) in the [SCHEDULE](#) section.

Note that the multi-component brine model ([ECLMC](#) keyword in the [RUNSPEC](#) section) cannot currently be used with the Polymer/Brine Flood Model (that is, the combination of polymer and brine using the [SALTNODE](#) keyword; [ECLMC](#) can be used with the basic Polymer Flood Model).

A bisection algorithm has been added in 2015.1 to counter a particular type of convergence problem sometimes encountered during a timestep solve. This type of problem occurs when the solution polymer concentration oscillates between two distinct intervals of the adsorption data supplied by either [PLYADS](#) or [PLYADSS](#). Note that, if the latter keyword is used, then potential oscillations involving the salt concentration are also considered. For those cases where an oscillation is detected, the proposed solution change is bisected in order to break the non-converging cycle of Newton iteration updates. The bisection is only applied after a minimum number of Newton iterations, the latter set using Item 11 of Record 3 of the [TUNING](#) keyword. This bisection algorithm can be deactivated by setting [OPTIONS](#) keyword [item 261](#) to 1.

Note: When both the polymer and surfactant options are active, the surfactant option is applied first to modify the properties of the water used by the polymer calculations. Surfactant property reporting therefore includes only the effect of the surfactant, while polymer property reporting includes both the effect of the surfactant and the polymer.

Restrictions

The IMPES solution algorithm cannot currently be used with the Polymer option.

If a cell is defined with zero pore volume in the [GRID](#) section, then making it active by changing the pore volume in the [EDIT](#) section using [PORV](#), is not allowed with the polymer option.

When using [RESTART](#) with the polymer flood option the base case must have neither polymer nor brine, or it must have the same polymer and brine choices as the restart. This is in part due to the fact that brine is solved as a tracer in the absence of polymer, but as a fully implicit variable in the presence of polymer. If a base case with brine alone is intended for use with a restart using polymer, then the [POLYMER](#) keyword must be added to the base case as well. The following table shows the compatible base cases and restart cases.

	No POLYMER or BRINE (Base case)	POLYMER (Base case)	BRINE (Base case)	POLYMER and BRINE (Base case)
No POLYMER or BRINE (Restart case)	Compatible	Not compatible	Not compatible	Not compatible

	No POLYMER or BRINE (Base case)	POLYMER (Base case)	BRINE (Base case)	POLYMER and BRINE (Base case)
POLYMER (Restart case)	Compatible	Compatible	Not compatible	Not compatible
BRINE (Restart case)	Compatible	Not compatible	Compatible	Not compatible
POLYMER and BRINE (Restart case)	Not compatible	Not compatible	Not compatible	Compatible

Table 6.3: Polymer and Brine restart compatibility

The Eclipse 300 polymer flood model

In Eclipse 300 polymer is modeled as a water component as described in "Water components". At least one water component must be identified as a polymer (CWTYPPE) and can be input into the reservoir by defining the injection water composition (WINJW).

In contrast with the Eclipse 100 polymer flood model, in Eclipse 300 the polymer is assumed to be fully mixed with the water and to move at the same velocity as the water phase.

Polymer can be adsorbed onto the rock surface. For Eclipse 300 a solid component can be used to represent adsorbed polymer, and adsorption and polymer can be modeled using a chemical reaction. See "Chemical Reactions" and "Solid Phase".

Treatment of fluid viscosities

The viscosity terms used in the fluid flow equations contain the effects of the viscosity change of the water phase due to the presence of polymer and any other water component defined in the model as described in "Water components". This is, the water phase viscosity is calculated as a linear combination of the viscosity of each water component as specified in equation 4.157.

The variation of the water phase viscosity with polymer concentration can be specified using the keyword **PLYVISC**. In this case, the water phase viscosity is given by $m(C_p)\mu_w$ where the multiplier m is a function of polymer concentration C_p as specified in **PLYVISC** and μ_w is calculated using equation 4.157.

In salt sensitive studies, the variation of the viscosity of the water phase with polymer concentration can be modeled using the keyword **PLYVISCs**. The water phase viscosity is given by $m(C_p, E_s)\mu_w$ where the multiplier m is a function of polymer concentration C_p and effective salinity E_s as specified by the keywords **PLYVISCs** and **ESPNode**. The effective salinity is calculated as a function of the water component concentrations as defined by the keyword **SALTEFF**.

The viscosity of each water component at their reference pressure can be defined with the keywords **VREFW** and **PREFW**. However, for the polymer model it is advised that these parameters are set to those of water so that the polymer concentration effect of the water phase viscosity is only defined by the keywords **PLYVISC** or **PLYVISCs**.

Treatment of polymer adsorption

Adsorbed polymer can be modeled using the **SOLID** model. The **COMPS** or **COMPSOL** keyword should be used to set the number of non-water components (including solids). The **CVTYPE** keyword should be used

to identify the solid component. The properties of adsorbed polymer are defined using the solid keywords, see "[Solid PROPS keywords](#)".

The polymer absorption and desorption processes can be modeled using chemical reactions, see "[Chemical Reactions](#)". The [REACTION](#) keyword should be used to specify the number of chemical reactions to be simulated. Different chemical reactions are available to model the absorption of polymer. In particular, dissolution processes can be modeled using "[Equilibrium deviation reaction rates](#)", see "[Adsorption of dissolved water components](#)".

In Eclipse 100 absorption is treated as an instantaneous process. In Eclipse 300 this effect could be modeled by setting a large constant reaction rate with the keyword [REACRATE](#). This might have a significant impact in convergence.

Restrictions

The following features of the Eclipse 100 polymer flood model are currently not available in the Eclipse 300 polymer flood model:

- The effect of Todd-Longstaff mixing parameter to model the physical dispersion at the leading edge of the slug and the fingering effects at the rear edge of the slug.
- The effect of temperature variations on the behavior of the polymer solution.
- The effect of reduced relative permeability of the polymer solution.
- The effect of dead pore volume on increasing the velocity of the polymer relative to the water phase velocity.
- The effect of non-Newtonian rheology.

Summary of Eclipse 100 keywords

RUNSPEC

Keyword	Description
BRINE	Enables the Polymer Salt-sensitivity option.
OPTIONS	If item 261 is set to 1, this triggers the pre-2015.1 behavior for the Newton iteration solution update in POLYMER flood models with adsorption. This is done by deactivating a bisection algorithm (the algorithm is otherwise applied to deal with a particular type of convergence problem - the latter caused by an oscillation between two distinct intervals in the adsorption data supplied by either PLYADS or PLYADSS).
REGDIMS	Item 10 sets the maximum number of mixing parameter regions.
POLYMER	Enables the Polymer Flood Model.

GRID

Keyword	Description
RPTGRID	Controls output from the GRID section. The POLYMER argument outputs a map of the $1/(\text{poro} \times \text{area})$ values used for shear thinning calculations.

PROPS

Keyword	Description
ADSALNOD	Salt concentration nodes for polymer solution adsorption (by saturation region).
PLYADS	Polymer adsorption isotherms without salinity dependence.
PLYADSS	Polymer adsorption isotherms with salinity dependence.
ADSORP	Analytical adsorption isotherms with salinity and permeability dependence.
PLYMAX	Polymer/salt concentrations for mixing calculations.
PLYROCK	Specifies the polymer-rock properties.
PLYSHEAR	Polymer shear thinning/thickening data.
PLYSHLOG	Polymer shear thinning/thickening data (alternative logarithmic model).
SHRATE	Activate the shear rate form of the PLYSHLOG shear thinning/thickening option.
PLYVISC	Polymer solution viscosity function.
PLYVISCS	Polymer/salt solution viscosity function.
PLYVISCT	Polymer solution viscosity function with temperature
PLYVSCST	Polymer/salt solution viscosity function with temperature.
PLYDHFLF	Polymer half life for thermal degradation.
RPTPROPS	Controls output from the PROPS section, including arguments for controlling the properties output for the Polymer Flood Model.
SALTNODE	Salt concentration nodes for polymer solution viscosity (by pressure region).
TEMPNODE	Temperature nodes for polymer solution viscosity.
PLMIXPAR	Todd-Longstaff mixing parameter.
PLYKRRF	Residual resistance factor (RRF) of individual grid blocks.
PLYTRRF	Temperature dependence of the RRF by region.
PLYTRRFA	Number of temperature intervals activating the second mode of PLYTRRF.
PLYCAMAX	Maximum polymer adsorption value of individual grid blocks.
PLYRMDEN	Mass density of individual grid blocks.
PLYATEMP	Temperature specifications for temperature dependence of polymer adsorption tables PLYADS or PLYADSS

REGIONS

The keywords are:

- PLMIXNUM Polymer mixing region numbers.

SOLUTION

Keyword	Description
RPTSOL	Controls output from the SOLUTION section. You can set the following arguments: • FIPPLY outputs polymer/salt fluid in place reports. • PBLK outputs initial grid block polymer concentration values. • PLYADS outputs the current adsorbed polymer concentration on the formation rock. • RK outputs the water relative permeability reduction factor. • SALT outputs the initial grid block salt concentration values • SALTVD outputs the tables of initial salt concentration versus depth.
SALT	Initial salt concentrations.
SALTVD	Salt concentration versus depth for equilibration.
SPOLY	Initial polymer concentrations.

SUMMARY

Refer to [Eclipse 100 Polymer flood SUMMARY output controls](#) in the *Eclipse Reference Manual*.

SCHEDULE

Keyword	Description
WELSPECS	Polymer mixing region number for wells.
WELSPECL	Polymer mixing region number for wells.
WSEGINIT	Initial segment polymer and brine concentrations for multisegment wells.
RPTSCHED	Controls output from the SCHEDULE section. You can set the following arguments. • FIPSALT: polymer/salt fluid in place reports. • PBLK: polymer concentration values. • PLYADS: adsorbed polymer concentration on the formation rock. • RK: water relative permeability reduction factor. • SALT: salt concentration values.

Keyword	Description
RPTRST	Argument VISC allows to output extra viscosities when Polymer is active. You can obtain the following output: • EPVIS: effective polymer viscosity (Polymer option). • EMVIS: effective mixture (water/polymer) viscosity (Polymer option) within a cell. Note that this is a cell-based quantity; shear-viscosity quantities (see below) which depend on the upstream cell properties may not correspond to EMVIS if the EMVIS cell is not the upstream cell. • EWV_POL: effective water viscosity (Polymer option). • EWV_SAL: effective water viscosity due to salt concentration (Brine option). • PLADALK: polymer adsorption multiplier as a function of alkaline concentration. • SFIPPLY/SFIPPOL: polymer in place (mass) • RRFFAC: residual resistance factor. • TRXBRRF: I-direction transmissibility divided by the residual resistance factor. • TRYBRRF: J-direction transmissibility divided by the residual resistance factor. • TRZBRRF: K-direction transmissibility divided by the residual resistance factor. • SHWVISI, SHWVISJ, SHWVISK: shear viscosity of the water/polymer solution due to shear thinning/thickening at the positive I-, J- and K-faces respectively of the relevant cell, (PLYSHEAR and PLYSHLOG keywords). Note that these properties are upwinded (or upstreamed, that is take their formative values from the cell that the flow is “from”) and consequently if the flow is in the negative direction through these faces, then the corresponding formative values and the unsheared viscosity (EMVIS) value will be found in the cell on the other side of the face from the display cell. • POLYMAX: maximum polymer concentration. • RRFTMAX: RRF maximum temperature. • POLYMER: polymer concentration.
TUNING	Item 11 of Record 3 sets the minimum number of Newton iterations prior to applying a bisection algorithm that counters a particular type of convergence problem in polymer flood models. The convergence issue in question is caused by an oscillation between two distinct intervals in the adsorption data supplied by either PLYADS or PLYADSS.
WPOLYMER	Sets polymer/salt concentrations for injection wells. The arguments can be used as User-Defined Arguments (UDAs).
WPOLYRED	Sets the polymer/water mixture viscosity reduction parameter.

Summary of Eclipse 300 keywords

The polymer component and water properties must be defined by the water component keywords. See ["Water component keywords"](#).

A solid component can be used to represent adsorbed polymer. See ["Solid Phase"](#).

Adsorption and desorption could then be modeled using a pair of chemical reactions. See "[Chemical Reactions](#)".

PROPS

The keywords are:

Keyword	Description
ESPNode	Effective salinity nodes.
PLYOPTS	Polymer model options.
PLYVISC	Polymer solution viscosity function.
PLYVISCs	Polymer/salt solution viscosity function.
SALTEFF	Set of coefficients to compute the effective salinity.
RPTPROPS	Controls output from the PROPS section. Arguments: • PLYVISC: echoes the polymer viscosity table specified by the PLYVISC keyword. • PLYVISCs: echoes the polymer viscosity table specified by the PLYVISCs keyword.

SOLUTION

Keyword	Description
RPTSOL	Controls output from the SOLUTION section. Arguments: • POLY: polymer concentration values. • POLYVM: polymer solution viscosity multiplier. • ESALTP: effective salinity values.
RPTRST	Arguments: • POLY: polymer concentration values. • POLYVM: polymer solution viscosity multiplier. • ESALTP: effective salinity values.

SUMMARY

Refer to [Eclipse 300 Polymer flood SUMMARY output controls](#) in the *Eclipse Reference Manual*.

Note that "[Water component](#)", "[Solid Phase](#)" and "[Chemical Reactions](#)" summary vectors may be used to provide additional output.

SCHEDULE

Keyword	Description
PLYVISC	Polymer solution viscosity function

Keyword	Description
PLYVISC	Polymer solution viscosity function.
RPTSCHED	Controls output from the SOLUTION section. Arguments: - POLY: polymer concentration values. - POLYVM: polymer solution viscosity multiplier. - ESALTP: effective salinity values.
RPTRST	Arguments: - POLY: polymer concentration values. - POLYVM: polymer solution viscosity multiplier. - ESALTP: effective salinity values.

Eclipse 100 examples

Example 1

This is a 1-D example modeling water injection into oil at adverse mobility ratios. To aid recovery, a polymer slug is injected into the reservoir some time after the primary water drive. Water injection is resumed after the polymer injection period. The performance of the polymer drive is assumed not be salt sensitive in this study.

RUNSPEC

```

RUNSPEC =====
TITLE
Polymer flood model test problem - oil/water/polymer

DIMENS
15      1      1  /

OIL
WATER
POLYMER
FIELD

TABDIMS
1      1      50      2      2      20      1      20  /

REGDIMS
2      1      0      0      0      1  /

WELLDIMS
2      1      1      2  /

START
1 'JAN' 1983  /

FMTOUT
FMTIN

```

GRID

```

GRID =====
EQUALS

```

```

'DX'      100    /
'DY'      100    /
'DZ'      20     /
'PERMX'   100    /
'PERMY'   100    /
'TOPS'    3990.0 /
'PORO'     0.5    /
/
RPTGRID
'PORO'    'PORV'  /

```

PROPS

```

PROPS =====
SWFN
    .20    .0    4.0
    .7     0.7   2.0
    1.0    1.0   0.0
/
SOF2
    .3000  .0000
    .8000  1.0000
/
PVTW
    .0    1.0    3.03E-06   .5   0.0 /
PVDO
    .0     1.0     5.0
    8000.0 .92     5.0
/
ROCK
    4000.0    .30E-05 /
DENSITY
    52.0000  64.0000   .04400 /

```

Define the viscosity of the polymer solution as a function of the polymer concentration. Note that the function values refer to the multiplication factor for the pure water viscosity to obtain the true polymer solution viscosity.

```

PLYVISC
    0.0    1.0
    7.0   20.0 /

```

Define the polymer-rock properties. A series of entries is required for each rock region. These values represent the inaccessible pore volume, the residual resistance factor, the rock mass density, the adsorption index and the maximum polymer concentration, to be used in the calculation of the actual resistance factor.

```

PLYROCK
    0.15  2.67  1000.0  2  0.0035 /

```

Define the polymer adsorption properties. A table of values is required for each rock region. Each table represents the equilibrium concentration of polymer adsorbed on the rock and the concentration of polymer in the surrounding aqueous phase.

```

PLYADS
    0.0  0.000
    2.0  0.0015
    8.0  0.0025 /

```

Mixing parameter to indicate the degree of segregation between the polymer solution and the water at the trailing edge of the slug. In this case the polymer and water in each grid cell are fully mixed. A value of the mixing parameter should be entered for each mixing region.

```
PLMIXPAR
1.0 /
```

Maximum value for the polymer and salt concentration values to be used in the mixing calculation of fluid viscosities. Since the salt option has been disabled in this run, the second value is ignored. A pair of entries is required for each mixing region.

```
PLYMAX
5.0 0.0 /
```

The polymer properties can be echoed using keyword [RPTPROPS](#).

```
RPTPROPS
'PLYVISC' /
```

REGIONS

```
REGIONS =====
FIPNUM
1 1 1 1 1 1 2 2 2 2 2 2 2 /
```

SOLUTION

```
SOLUTION =====
PRESSURE
15*4000.0 /

SWAT
15*0.2
```

Note that the mixing regions have not been specified in this run since the region numbers are all unity in this problem.

```
RPTSOL
'PRES' 'SOIL' 'SWAT' 'RESTART=2' 'PBLK'
'SALT' 'PLYADS' 'FIPPLY=2' /
```

SUMMARY

```
SUMMARY =====
RUNSUM
```

SUMMARY file keywords are used to indicate which data are to be written to the Eclipse SUMMARY files at each timestep. In this case, we are writing field water-cuts, polymer production rates, the cumulative polymer total and the cumulative oil production total at each timestep.

```
FWCT
FCPR
FCPT
FOPT
RPTSMRY
1 /
```

SCHEDULE

```

=====
SCHEDULE
RPTSCHED
'PRES' 'SOIL' 'SWAT' 'RESTART=2' 'NEWTON=2'
'PBLK' 'SALT' 'PLYADS' 'RK' /

WELSPESCS
'I' 'G' 1 1 4000 'WAT' 0.0 'STD' 'SHUT' 'NO' /
'P' 'G' 15 1 4000 'WAT' 0.0 'STD' 'SHUT' 'NO' /
/

COMPDAT
'I' ' ' 1 1 1 1 'OPEN' 0 .0 1.0 /
'P' ' ' 15 1 1 1 'OPEN' 0 .0 1.0 /
/

WCONPROD
'P' 'OPEN' 'LRAT' 3* 1000.0 /
/

WCONINJE
'I' 'WAT' 'OPEN' 'RATE' 1000.0 /
/

```

Initial water flood

Specify polymer concentration in the injection stream of well 'I'. The second figure refers to the injection salt concentration and is ignored in this run.

```

WPOLYMER
'I' 0.0 0.0 /
/
TSTEP
1.0 99.0 2*100.0 /

```

Polymer slug in fresh water

Specify the polymer concentration in the injection stream of well 'I'.

```

WPOLYMER
'I' 5.0 0.0 /
/
TSTEP
5*100.0 /

```

Chase water period

Specify the polymer concentration in the injection stream of well 'I'.

```

WPOLYMER
'I' 0.0 0.0 /
/
TSTEP
10*100.0 /
END

```

Example 2

This is a version of the previous example with the salt option activated to study the effect of the interaction between the injected polymer and the salt. The polymer is strongly salt-sensitive and the efficiency of the drive is expected to be lower than in the previous example.

RUNSPEC

```
RUNSPEC =====
TITLE
Polymer flood model test problem - oil/water/polymer/salt

DIMENS
15      1      1  /

OIL
WATER
POLYMER
BRINE
FIELD

TABDIMS
1      1      50      2      2      20      1      20  /

REGDIMS
2      1      0      0      0      1  /

WELLDIMS
2      1      1      2  /

START
1 'JAN' 1983  /

FMTOUT
FMTIN
```

GRID

```
GRID =====
EQUALS
'DX'      100      /
'DY'      100      /
'DZ'      20       /
'PERMX'   100      /
'PERMY'   100      /
'TOPS'    3990.0   /
'PORO'    0.5      /
/

RPTGRID
'PORO'    'PORV'   /
```

PROPS

```
PROPS =====
SWFN
.20      .0      4.0
.7       0.7     2.0
1.0      1.0     0.0
/

SOF2
.3000    .0000
.8000    1.0000
/

PVTW
.0      1.0      3.03E-06   .5      0.0  /

PVDO
.0       1.0      5.0
8000.0   .92     5.0
/

ROCK
4000.0    .30E-05  /
```

```
DENSITY
52.0000 64.0000 .04400 /
```

Define the viscosity of the polymer solution as a function of both the polymer and the salt concentrations. A 2-dimensional table input structure is required. The salt concentration nodal values are defined using the [SALTNODE](#) keyword. The polymer concentration nodal values and the function values are entered using the [PLYVISC](#) keyword. Note that the function values refer to the multiplication factor which is to be applied to the water viscosity to give the true solution viscosity.

```
SALTNODE
0.0
0.35 /

PLYVISC
0.0 1.0
      1.0 /
7.0 20.0
      2.0 /
/
```

Define the polymer-rock properties. A series of entries is required for each rock region. These values represent the inaccessible pore space, the residual resistance factor, the rock mass density, the adsorption index and the maximum polymer concentration, to be used in the calculation of the actual resistance factor.

```
PLYROCK
0.15 2.67 1000.0 2 0.0035 /
```

Define the polymer adsorption properties. A table of values is required for each rock region. Each table represents the equilibrium concentration of polymer adsorbed on the rock and the concentration of polymer in the surrounding aqueous phase.

```
PLYADS
0.0 0.000
2.0 0.0015
8.0 0.0025 /
```

Mixing parameter to indicate the degree of segregation between the polymer solution and the water at the trailing edge of the slug. In this case, the polymer and water in each grid cell are fully mixed. A value of the mixing parameter should be entered for each mixing region.

```
PLMIXPAR
1.0 /
```

Maximum value for the polymer and salt concentration values to be used in the mixing calculation of fluid viscosities. A pair of entries is required for each mixing region.

```
PLYMAX
5.0 0.0 /
```

The polymer properties can be echoed using keyword [RPTPROPS](#).

```
RPTPROPS
'PLYVISC' /
```


REGIONS

```
REGIONS =====
FIPNUM
1 1 1 1 1 1 1 2 2 2 2 2 2 2 /
```

Define the mixing region for each grid cell

```
PLMIXNUM
15*1 /
```

SOLUTION

```
SOLUTION =====
PRESSURE
15*4000.0 /
SWAT
15*0.2 /
```

Define the initial salt concentration by enumeration.

```
SALT
15*0.3 /
RPTSOL
'PRES' 'SOIL' 'SWAT' 'RESTART=2' 'PBLK'
'SALT' 'PLYADS' 'FIPPLY=2' /
```

SUMMARY

```
SUMMARY =====
RUNSUM
```

SUMMARY file keywords are used to indicate which data is to be written to the Eclipse summary files at each timestep. In this case, we are writing field water-cuts, polymer production rates, salt production rates and cumulative oil production totals, at each timestep.

```
FWCT
FCPR
FSPR
FOPT
RPTSMRY
1 /
```

SCHEDULE

```
SCHEDULE =====
RPTSCHED
'PRES' 'SOIL' 'SWAT' 'RESTART=2' 'NEWTON=2'
'PBLK' 'SALT' 'PLYADS' 'RK' /
WELSPEDS
'I' 'G' 1 1 4000 'WAT' 0.0 'STD' 'SHUT' 'NO' /
'P' 'G' 15 1 4000 'WAT' 0.0 'STD' 'SHUT' 'NO' /
/
COMPDAT
'I' ' ' 1 1 1 1 'OPEN' 0 .0 1.0 /
'P' ' ' 15 1 1 1 'OPEN' 0 .0 1.0 /
/
WCONPROD
'P' 'OPEN' 'LRAT' 3* 1000.0 /
/
WCONINJE
```

```
'I' 'WAT' 'OPEN' 'RATE' 1000.0 /  
/
```

Initial water flood

Specify polymer and salt concentrations in the injection stream of well 'I'.

```
WPOLYMER  
'I' 0.0 0.3 /  
/  
TSTEP  
1.0 99.0 2*100.0 /
```

Polymer slug in fresh water

Specify polymer and salt concentrations in the injection stream of well 'I'.

```
WPOLYMER  
'I' 5.0 0.0 /  
/  
TSTEP  
5*100.0 /
```

Chase water period

Specify polymer and salt concentrations in the injection stream of well 'I'.

```
WPOLYMER  
'I' 0.0 0.3 /  
/  
TSTEP  
10*100.0 /  
END
```

Eclipse 300 examples

Example 1

A polymer slug is injected into the reservoir and water injection is resumed after the polymer injection period. The performance of the polymer drive is assumed not to be salt sensitive in this study.

RUNSPEC

```
RUNSPEC =====  
  
TITLE  
Test for Polymer in E300  
  
FIELD  
  
DIMENS  
6 6 2 /  
  
WATER  
  
OIL  
  
GAS  
  
COMPW  
2/  
  
COMPS  
6 /
```

```
TABDIMS
1 1 20 50 1 20 /

WELLDIMS
2 10 1 /

START
21 'DEC' 2013 /
```

GRID

```
GRID      =====

INIT

GRIDFILE
1 1 /

TOPS
36*2000/

DX
72*100/

DY
72*100/

DZ
72*100/

PORO
72*0.2/

PERMX
72*100/

PERMY
72*100/

PERMZ
72*100/
```

PROPS

```
PROPS      =====

EOS
PR/

PRCORR

RTEMP
160/

STCOND
60 14.7/

CNames
C1 C3 C6 C10 C15 C20 /

TCRIT
343.0
665.7
913.4
1111.8
1270.0
1380.0 /

PCRIT
667.8
616.3
```

```

436.9
304.0
200.0
162.0 /

ZCRIT
0.290
0.277
0.264
0.257
0.245
0.235 /

MW
16.04
44.10
86.18
149.29
206.00
282.00 /

ACF
0.013
0.1524
0.3007
0.4885
0.6500
0.8500 /

BIC
0.0
0.0 0.0
0.0 0.0 0.0
0.05 0.005 0.0 0.0
0.05 0.005 0.0 0.0 0.0 /

-- Rock fluid properties
SWFN
0.2 0 45.0
0.2899 0.0022 19.03
0.3778 0.0180 10.07
0.4667 0.0607 4.90
0.5556 0.1438 1.8
0.6444 0.2809 0.5
0.7000 0.4089 0.05
0.7333 0.4855 0.01
0.8222 0.7709 0.0
0.9111 1.0000 0.0
1.00 1.0000 0.0 /

SGFN
0.00 0.000 0.0
0.05 0.000 0.0
0.0889 0.001 0.0
0.1778 0.010 0.0
0.2667 0.030 0.001
0.3556 0.05 0.001
0.4444 0.10 0.03
0.5333 0.20 0.8
0.6222 0.35 3.0
0.65 0.39 4.0
0.7111 0.56 8.0
0.80 1.0 30.0 /

SOF3
0.00 0.0 0.0
0.0889 0.0 0.0
0.1500 0.0 0.0
0.1778 0.0 0.0110
0.2667 0.0 0.0370
0.3 0.0 0.0560
0.3556 0.0123 0.0878
0.4444 0.0835 0.1715
0.5333 0.2178 0.2963
0.6222 0.4153 0.4705

```

```

0.7111 0.6769 0.7023
0.80 1.0 1.0 /

ZMFVD
1000.0 0.5 0.03 0.07 0.2 0.15 0.05
10000.0 0.5 0.03 0.07 0.2 0.15 0.05 /

DENSITY
1* 62.4 1* /

ROCK
75.0 5.0E-04 /

```

Define the water component properties. Here there are two water components: fresh water and polymer.

```

-- Water and polymer properties
WNAMEs
PURE POLYMER /

CWTYPE
1* POLY /

CREFW
3.E-8 3.E-8 /

MWW
18 1E5 /

PREFW
2000 2000 /

VREFW
1.2 1.2 /

CREFWS
3.E-8 3.E-8 /

MWWS
18 1E5 /

PREFWS
14.6959 14.6959 /

```

Define the viscosity of the polymer solution as a function of the polymer concentration. Note that the function values refer to the multiplication factor for the pure water viscosity to obtain the true polymer solution viscosity.

```

PLYVISC
0 1
1 2.25
2 5.5
3 7.75
4 10 /

PLYOPTS
2.E-6 /

```

The polymer properties can be echoed using keyword [RPTPROPS](#).

```

RPTPROPS
'PLYVISC' /

```

SOLUTION

```

SOLUTION =====
PRESSURE
72*2000/

```

```

SWAT
72*0.3/

SGAS
72*0/

WMF
72*1 72*0 /

XMF
72*0.5 72*0.03 72*0.07 72*0.2 72*0.15 72*0.05/

YMF
72*0 72*0 72*0 72*0 72*0 72*0/

```

The polymer properties can be reported using the [RPTRST](#) and the [RPTSOL](#) keywords.

```

RPTRST
'BASIC=2' POLY POLYVM /

RPTSOL
POLY POLYVM /

```

SUMMARY

```

SUMMARY =====

FWMIR
/

FWMIT
/

FWMPR
/

FWMPT
/

FAMF
/

WWMPR
PROD /
/

WWMIR
INJ/
/

BAMF
1 1 2 /
3 3 2 /
6 6 2 /
/

BMLST
1 1 2 /
3 3 2 /
6 6 2 /
/

BMLSC
1 1 2 /
3 3 2 /
6 6 2 /
/

BVWAT
1 1 2 /

```

```

3 3 2 /
6 6 2 /
/

BPOLY
1 1 2 /
3 3 2 /
6 6 2 /
/

BPOLYVM
1 1 2 /
3 3 2 /
6 6 2 /
/

FOE

ALL

RUNSUM

```

SCHEDULE

```

SCHEDULE =====

RPTSCHED
MLSC MWAT POLY POLYVM /

RPTRST
BASIC=3 FREQ=1 MLSC MWAT POLY POLYVM /

WELLSPEC
INJ FIELD 1 1 3500 1* 0 /
PROD FIELD 6 6 3500 1* 0 /
/

COMPDAT
INJ 1* 1* 1 2 OPEN /
PROD 1* 1* 1 2 OPEN /
/

WCONINJE
INJ WATER OPEN RATE 10000 /
/

-- Definition of the injection streams
WELLSTRW
FRESH 1.0 0.0 /
POLYM 0.9999998 2E-7 /
/

-- Injection of polymer slug
WINJW
'I*' STREAM POLYM /
/

DATES
25 DEC 2013 /
/

WCONPROD
PROD OPEN BHP 1* 1* 1* 1* 1* 200 /
/

DATES
1 FEB 2014 /
/

-- Update the properties of the polymer
PLYVISC
0 1
1 2.0

```

```

2    5.2
3    7.4
4    9.5 /

DATES
1 MAR 2014 /
1 APR 2014 /
1 MAY 2014 /
1 JUN 2014 /
/

WCONINJE
INJ WATER OPEN RATE 10000 /
/

-- Resume fresh water injection
WINJW
'I*' STREAM FRESH /
/

DATES
1 JUL 2014 /
1 AUG 2014 /
1 SEP 2014 /
1 OCT 2014 /
1 NOV 2014 /
1 DEC 2014 /
/

END

```

Example 2

A polymer slug is injected into the reservoir and water injection is resumed after the polymer injection period. In this study the injected polymer is salt sensitive.

RUNSPEC

```

RUNSPEC =====
TITLE
Test for Polymer in E300 with salinity effects

FIELD

DIMENS
6 6 2 /

WATER

OIL

GAS

COMPW
4 /

COMPS
6 /

TABDIMS
1 1 20 50 1 20 /

WELLDIMS
2 10 1 /

START
21 'DEC' 2013 /

FULLIMP

```



```
CPR
/

FMTOUT
UNIFOUT
```

GRID

```
GRID      =====

INIT

GRIDFILE
1 1 /

TOPS
36*2000/

DX
72*100/

DY
72*100/

DZ
72*100/

PORO
72*0.2/

PERMX
72*100/

PERMY
72*100/

PERMZ
72*100/
```

PROPS

```
PROPS      =====

EOS
PR/

PRCORR

RTEMP
160/

STCOND
60 14.7/

CNAMES
C1 C3 C6 C10 C15 C20 /

TCRIT
343.0
665.7
913.4
1111.8
1270.0
1380.0 /

PCRIT
667.8
616.3
436.9
304.0
200.0
```

```

162.0 /
ZCRIT
0.290
0.277
0.264
0.257
0.245
0.235 /
MW
16.04
44.10
86.18
149.29
206.00
282.00 /
ACF
0.013
0.1524
0.3007
0.4885
0.6500
0.8500 /
BIC
0.0
0.0 0.0
0.0 0.0 0.0
0.05 0.005 0.0 0.0
0.05 0.005 0.0 0.0 0.0 /
-- Rock fluid properties
SWFN
0.2 0 45.0
0.2899 0.0022 19.03
0.3778 0.0180 10.07
0.4667 0.0607 4.90
0.5556 0.1438 1.8
0.6444 0.2809 0.5
0.7000 0.4089 0.05
0.7333 0.4855 0.01
0.8222 0.7709 0.0
0.9111 1.0000 0.0
1.00 1.0000 0.0 /
SGFN
0.00 0.000 0.0
0.05 0.000 0.0
0.0889 0.001 0.0
0.1778 0.010 0.0
0.2667 0.030 0.001
0.3556 0.05 0.001
0.4444 0.10 0.03
0.5333 0.20 0.8
0.6222 0.35 3.0
0.65 0.39 4.0
0.7111 0.56 8.0
0.80 1.0 30.0 /
SOF3
0.00 0.0 0.0
0.0889 0.0 0.0
0.1500 0.0 0.0
0.1778 0.0 0.0110
0.2667 0.0 0.0370
0.3 0.0 0.0560
0.3556 0.0123 0.0878
0.4444 0.0835 0.1715
0.5333 0.2178 0.2963
0.6222 0.4153 0.4705
0.7111 0.6769 0.7023
0.80 1.0 1.0 /

```

```

ZMFVD
1000.0 0.5 0.03 0.07 0.2 0.15 0.05
10000.0 0.5 0.03 0.07 0.2 0.15 0.05 /

DENSITY
1* 62.4 1* /

ROCK
75.0 5.0E-04 /

```

Define the water component properties. Here there are four water components: fresh water, two salts and polymer.

```

-- Water and polymer properties
WNAMEs
PURE NaCl CaCO3 POLYMER /

CWTYPE
1* 1* 1* POLY /

CREFW
3.E-8 3.E-8 3.E-8 3.E-8 /

MWW
18 58.44 100.0869 1E5 /

PREFW
2000 2000 2000 2000 /

VREFW
1.2 1.2 1.2 1.2 /

CREFWS
3.E-8 3.E-8 3.E-8 3.E-8 /

MWWS
18 58.44 100.0869 1E5 /

PREFWS
14.6959 14.6959 14.6959 14.6959 /

```

Define the effective salinity as the concentration of water components two and three:

```

-- Effective salinity
SALTEFF
-- model      effective      a_0 a_1 a_2 a_3 a_4 b_0 b_1 b_2 b_3 b_4
-- type      salinity type
POLY          1          0.0 0.0 1.0 1.0 0.0 1.0 0.0 0.0 0.0 0.0 /

```

Define the viscosity of the polymer solution as a function of the polymer concentration and effective salinity:

```

ESPNode
0
10
20/

PLYVISCS
0 1
  1.05
  1.1 /
1 2.25
  2
  1.75 /
2 5.5
  5
  4.5 /
3 7.75
  7
  6.3 /

```

```
4      10
      9.2
      8.1  /
/

PLYOPTS
2.E-6  /
```

The polymer properties can be echoed using keyword [RPTPROPS](#).

```
RPTPROPS
'PLYVISCS'  /
```

SOLUTION

```
SOLUTION  =====

PRESSURE
72*2000/

SWAT
72*0.3/

SGAS
72*0/

WMF
72*0.995
72*0.005
72*0
72*0
/

XMF
72*0.5 72*0.03 72*0.07 72*0.2 72*0.15 72*0.05/

YMF
72*0 72*0 72*0 72*0 72*0 72*0/
```

The polymer properties can be reported using the [RPTRST](#) and the [RPTSOL](#) keywords.

```
RPTRST
'BASIC=2' POLY POLYVM ESALTP /

RPTSOL
POLY POLYVM ESALTP /
```

SUMMARY

```
SUMMARY  =====

FWMIR
/
FWMIT
/
FWMPR
/
FWMPT
/
FAMF
/
WWMPR
PROD /
/
WWMIR
INJ/
/
```

```

BAMF
1 1 2 /
3 3 2 /
6 6 2 /
/

BMLST
1 1 2 /
3 3 2 /
6 6 2 /
/

BMLSC
1 1 2 /
3 3 2 /
6 6 2 /
/

BVWAT
1 1 2 /
3 3 2 /
6 6 2 /
/

BPOLY
1 1 2 /
3 3 2 /
6 6 2 /
/

BPOLYVM
1 1 2 /
3 3 2 /
6 6 2 /
/

BESALTP
1 1 2 /
3 3 2 /
6 6 2 /
/

FOE

ALL
RUNSUM

```

SCHEDULE

```

SCHEDULE =====
RPTSCHED
MLSC MWAT POLY POLYVM ESALTP/

RPTRST
BASIC=3 FREQ=1 MLSC MWAT POLY POLYVM ESALTP/

WELLSPEC
INJ FIELD 1 1 3500 1* 0 /
PROD FIELD 6 6 3500 1* 0 /
/

COMPDAT
INJ 1* 1* 1 2 OPEN /
PROD 1* 1* 1 2 OPEN /
/

WCONINJE
INJ WATER OPEN RATE 10000 /
/

-- Definition of the injection streams
WELLSTRW

```

```

FRESH 1.0      0.0  0.0  0.0  /
BRINE  0.99    0.005 0.005 0.0  /
POLYM  0.989998 0.005 0.005 2E-6 /
POLYM2 0.989999 0.005 0.005 1E-6 /
/

-- Injection of Brine
WINJW
'I*' STREAM BRINE /
/

DATES
25 DEC 2013 /
/

-- Injection of Polymer Slug
WINJW
'I*' STREAM POLYM /
/

WCONPROD
PROD OPEN BHP 1* 1* 1* 1* 1* 200 /
/

DATES
1 FEB 2014 /
1 MAR 2014 /
1 APR 2014 /
1 MAY 2014 /
1 JUN 2014 /
/

-- Injection of Polymer 2 Slug
WINJW
'I*' STREAM POLYM2 /
/

DATES
1 JUL 2014 /
1 AUG 2014 /
/

WCONINJE
INJ WATER OPEN RATE 10000 /
/

-- Resume fresh water injection
WINJW
'I*' STREAM FRESH /
/

DATES
1 SEP 2014 /
/

WCONPROD
PROD OPEN BHP 1* 1* 1* 1* 1* 250 /
/

DATES
1 OCT 2014 /
1 NOV 2014 /
1 DEC 2014 /
/

END

```

Example 3

A salt-sensitive polymer slug is injected into the reservoir and water injection is resumed after the polymer injection period. Polymer adsorption is also included in the model.

RUNSPEC

```

RUNSPEC  =====

TITLE
Test for Polymer in E300 with salinity and adsorption effects

FIELD

DIMENS
6 6 2 /

WATER

OIL

GAS

SOLID

COMPW
4 /

COMPS
7 /

REACTION
1 1 1* 1* 1 /

TABDIMS
1 1 20 50 1 20 /

WELLDIMS
2 10 1 /

START
21 'DEC' 2013 /

FULLIMP

CPR
/

FMTOUT
UNIFOUT

```

GRID

```

GRID  =====

INIT

GRIDFILE
1 1 /

TOPS
36*2000/

DX
72*100/

DY
72*100/

DZ
72*100/

PORO
72*0.2/

PERMX
72*100/

```

```
PERMY
72*100/

PERMZ
72*100/

ROCKDEN
72*165.43 /
```

PROPS

```
PROPS  =====
EOS
PR/

PRCORR

RTEMP
160/

STCOND
60 14.7/

CNAMES
C1 C3 C6 C10 C15 C20 APOLY/

CVTYPE
6* SOLID /

TCRIT
343.0
665.7
913.4
1111.8
1270.0
1380.0
1380.0 /

PCRIT
667.8
616.3
436.9
304.0
200.0
162.0
162.0 /

ZCRIT
0.290
0.277
0.264
0.257
0.245
0.235
0.235 /

MW
16.04
44.10
86.18
149.29
206.00
282.00
1E5 /

ACF
0.013
0.1524
0.3007
0.4885
0.6500
```



```

0.8500
0.8500 /

ZI
0.5 0.03 0.07 0.2 0.15 0.05 0/

BIC
0.0
0.0 0.0
0.0 0.0 0.0
0.05 0.005 0.0 0.0
0.05 0.005 0.0 0.0 0.0
0.0 0.0 0.0 0.0 0.0 0.0 /

-- Rock fluid properties
SWFN
0.2 0 45.0
0.2899 0.0022 19.03
0.3778 0.0180 10.07
0.4667 0.0607 4.90
0.5556 0.1438 1.8
0.6444 0.2809 0.5
0.7000 0.4089 0.05
0.7333 0.4855 0.01
0.8222 0.7709 0.0
0.9111 1.0000 0.0
1.00 1.0000 0.0 /

SGFN
0.00 0.000 0.0
0.05 0.000 0.0
0.0889 0.001 0.0
0.1778 0.010 0.0
0.2667 0.030 0.001
0.3556 0.05 0.001
0.4444 0.10 0.03
0.5333 0.20 0.8
0.6222 0.35 3.0
0.65 0.39 4.0
0.7111 0.56 8.0
0.80 1.0 30.0 /

SOF3
0.00 0.0 0.0
0.0889 0.0 0.0
0.1500 0.0 0.0
0.1778 0.0 0.0110
0.2667 0.0 0.0370
0.3 0.0 0.0560
0.3556 0.0123 0.0878
0.4444 0.0835 0.1715
0.5333 0.2178 0.2963
0.6222 0.4153 0.4705
0.7111 0.6769 0.7023
0.80 1.0 1.0 /

ZMFVD
1000.0 0.5 0.03 0.07 0.2 0.15 0.05 0
10000.0 0.5 0.03 0.07 0.2 0.15 0.05 0 /

DENSITY
1* 62.4 1* /

SDREF
6* 970.0 /

ROCK
75.0 5.0E-04 /

```

Define the water component properties. Here there are four water components: fresh water, two salts and polymer.

```

-- Water and polymer properties
WNAMEs
PURE NaCl CaCO3 POLYMER /

CWTYPE
1* 1* 1* POLY /

CREFW
3.E-8 3.E-8 3.E-8 3.E-8 /

MWW
18 58.44 100.0869 1E5 /

PREFW
2000 2000 2000 2000 /

VREFW
1.2 1.2 1.2 1.2 /

CREFWs
3.E-8 3.E-8 3.E-8 3.E-8 /

MWWS
18 58.44 100.0869 1E5 /

PREFWs
14.6959 14.6959 14.6959 14.6959 /

```

Define the effective salinity as the concentration of water components two and three:

```

-- Effective salinity
SALTEFF
-- model      effective      a_0 a_1 a_2 a_3 a_4 b_0 b_1 b_2 b_3 b_4
-- type      salinity type
POLY          1          0.0 0.0 1.0 1.0 0.0 1.0 0.0 0.0 0.0 0.0 /

```

Define the viscosity of the polymer solution as a function of the polymer concentration and effective salinity:

```

ESPNode
0
10
20/

PLYVISCS
0 1
  1.05
  1.1 /
1 2.25
  2
  1.75 /
2 5.5
  5
  4.5 /
3 7.75
  7
  6.3 /
4 10
  9.2
  8.1 /
/

PLYOPTS
2.E-6 /

```

The polymer properties can be echoed using keyword [RPTPROPS](#).

```

RPTPROPS
'PLYVISCS' /

```

The polymer adsorption properties are defined using [EQLDREAC](#) keyword model 4. This provides a reversible reaction between the solid component (component 7 in this example) and the POLYMER water component (component 11 in this example).

```
--Adsorption reactions
STOREAC
-- C1 C3 C6 C10 C15 C20 APOLY W NaCl CaCO3 POLY
   0 0 0 0 0 0 0 0 0 0 1 / Adsorbed Polymer <-> Dissolved
Polymer
STOPROD
-- C1 C3 C6 C10 C15 C20 APOLY W NaCl CaCO3 POLY
   0 0 0 0 0 0 1 0 0 0 0 / Adsorbed Polymer <-> Dissolved
Polymer

REACRATE
1 /

EQLDREAC
1 4 11 0 1 BOTH 1 /

EQLDTAB
--WMF_Poly Ca
   0.0      0.0
   5E-8     0.0001
   1E-7     0.0002
   2E-6     0.0002
/
```

SOLUTION

```
SOLUTION =====
PRESSURE
72*2000/

SWAT
72*0.3/

SGAS
72*0/

WMF
72*0.995
72*0.005
72*0
72*0
/

XMF
72*0.5 72*0.03 72*0.07 72*0.2 72*0.15 72*0.05 72*0/

YMF
72*0 72*0 72*0 72*0 72*0 72*0 72*0/
RPTRST
'BASIC=2' POLY POLYVM ESALTP /

RPTSOL
POLY POLYVM ESALTP /
```

SUMMARY

```
SUMMARY =====
FWMIR
/
FWMIT
/
FWMPR
/
```

```
FWMPT
/
FAMF
/
WWMPR
PROD /
/
WWMIR
INJ/
/
BAMF
1 1 2 /
3 3 2 /
6 6 2 /
/

BMLST
1 1 2 /
3 3 2 /
6 6 2 /
/

BMLSC
1 1 2 /
3 3 2 /
6 6 2 /
/

BVWAT
1 1 2 /
3 3 2 /
6 6 2 /
/

BPOLY
1 1 2 /
3 3 2 /
6 6 2 /
/

BPOLYVM
1 1 2 /
3 3 2 /
6 6 2 /
/

BESALTP
1 1 2 /
3 3 2 /
6 6 2 /
/

BREAC
1 1 2 /
3 3 2 /
6 6 2 /
/

FOE

ALL
RUNSUM
```

SCHEDULE

```
SCHEDULE =====
RPTSCHED
MLSC MWAT POLY POLYVM ESALTP/

RPTRST
BASIC=3  FREQ=1  MLSC MWAT POLY POLYVM ESALTP/
```

```

WELLSPEC
INJ FIELD 1 1 3500 1* 0 /
PROD FIELD 6 6 3500 1* 0 /
/

COMPDAT
INJ 1* 1* 1 2 OPEN /
PROD 1* 1* 1 2 OPEN /
/

WCONINJE
INJ WATER OPEN RATE 10000 /
/

-- Definition of the injection streams
WELLSTRW
FRESH 1.0 0.0 0.0 0.0 /
BRINE 0.99 0.005 0.005 0.0 /
POLYM 0.989998 0.005 0.005 2E-6 /
POLYM2 0.989999 0.005 0.005 1E-6 /
/

-- Injection of Brine
WINJW
'I*' STREAM BRINE /
/

DATES
25 DEC 2013 /
/

-- Injection of Polymer Slug
WINJW
'I*' STREAM POLYM /
/

WCONPROD
PROD OPEN BHP 1* 1* 1* 1* 1* 200 /
/

DATES
1 FEB 2014 /
1 MAR 2014 /
1 APR 2014 /
1 MAY 2014 /
1 JUN 2014 /
/

-- Injection of Polymer 2 Slug
WINJW
'I*' STREAM POLYM2 /
/

DATES
1 JUL 2014 /
1 AUG 2014 /
/

WCONINJE
INJ WATER OPEN RATE 10000 /
/

-- Resume fresh water injection
WINJW
'I*' STREAM FRESH /
/

DATES
1 SEP 2014 /
/

WCONPROD
PROD OPEN BHP 1* 1* 1* 1* 1* 250 /
/

```

```
DATES
1 OCT 2014 /
1 NOV 2014 /
1 DEC 2014 /
/
END
```

Alkaline model

x	Eclipse 100
	Eclipse 300

Alkaline flooding requires the injection of alkaline chemicals (lye or caustic solutions, in other words, high pH solutions) into a reservoir that react with petroleum acids to form in-situ surfactants that help release the oil from the rock by reducing interfacial tension, changing the rock surface wettability, and spontaneous emulsification. The oil can then be moved more easily from the reservoir towards the production wells.

Alkaline flooding is usually more efficient if the acid content of the reservoir oil is relatively high.

Note: In Eclipse 300, it is possible to model the effect of alkaline on water-oil surface tension when Surfactant option is activated. This can be achieved by specifying ALK as a water component type, see "Water component keywords", and specifying keyword `ALSURFST` in the PROPS section.

When used in conjunction with surfactant and polymer to perform an Alkaline-Surfactant-Polymer (ASP) flooding, the alkaline can reduce the adsorption of both surfactant and polymer on the rock surface, therefore enhancing the effectiveness of the surfactant and polymer drive.

The alkaline model described in the following sections provides a simplified model that does not take into account the in-situ surfactant creation and the phase behavior. This simplified model provides an easy way to take into account some effects of the alkaline on ASP flooding performance. With it you can analyze the effect of the alkaline on the water-oil surface tension and adsorption reduction of surfactant and polymer.

Alkaline model for ASP flooding

This section describes the simplified alkaline model for ASP flooding.

The alkaline conservation equation

The alkaline is assumed to exist only in the water phase. To specify the concentration of alkaline in a water injector, refer to the `WALKALIN` keyword. The distribution of the injected alkaline is modeled by solving a conservation equation:

$$\frac{d}{dt} \left(\frac{VS_w C_a}{B_r B_w} \right) + \frac{d}{dt} \left(V \rho_r C_a^a \frac{1 - \varphi}{\varphi} \right) = \sum \left[\frac{Tk_{rw}}{B_w \mu_{s\ eff}} (\delta P_w - \rho_w g D_z) \right] C_a + Q_w C_a \quad \text{Eq. 6.102}$$

where

V is the block pore volume

S_w is the water saturation

C_a denotes the alkaline concentration

B_w, B_r are the water and rock formation volume respectively

ρ_w, ρ_r denote the water and rock density respectively

C_a^a denotes the adsorbed alkaline concentration

φ denotes the porosity

Σ	denotes the sum over neighboring cells
T	is the transmissibility
k_{rw}	is the water relative permeability
$\mu_{s\text{ eff}}$	denotes the effective viscosity of the salt
P_w	is the water pressure
g	is the gravity acceleration
D_z	is the cell center depth
Q_w	is the water production rate.

By default, the alkaline concentrations are updated at the end of a timestep after the inter-block phase flows have been determined.

If the alkaline effects of the flow properties are very strong, the alkaline concentrations can be solved at each Newton iteration so that the tracer concentrations are updated simultaneously with the phase equations. To do this, set [item 126](#) of the `OPTIONS` keyword to 1.

If the `ECLMC` keyword is found in the `RUNSPEC` section, the default solution is to update the alkaline concentration at each Newton iteration. The new alkaline concentrations are then used to update the flow properties for the subsequent iterations.

Treatment of adsorption

The adsorption of alkaline is assumed to be instantaneous. The isotherm adsorption can be specified as:

- either a lookup table of adsorbed alkaline as a function of alkaline concentration using the `ALKADS` keyword
- or by a generic analytical adsorption model using the `ADSORP` keyword. This model enables the adsorption to depend on rock permeability as well.

If desorption is prevented (refer to the item 1 of the `ALKROCK` keyword), then the adsorbed alkaline concentration may not decrease with time. If desorption is allowed, then each grid block retraces the adsorption isotherm as the alkaline concentration falls in the cell.

Note: Because this simplified model is targeted at looking at some of the effects of the alkaline on Alkaline-Surfactant-Polymer flooding (ASP), the rock mass density is taken from either the `PLYROCK` keyword or the `SURFROCK` keyword.

Alkaline effect on water-oil surface tension

The effect of alkaline on the water-oil surface tension is modeled as a combined effect with surfactant by modifying the water-oil surface tension as follows:

$$\sigma_{wo} = \sigma_{wo}(C_{surf})A_{st}(C_{alk}) \quad \text{Eq. 6.103}$$

where

$\sigma_{wo}(C_{\text{surf}})$ is the surface tension at surfactant concentration and zero alkaline concentration ([SURFST](#) keyword).

$A_{st}(C_{alk})$ is the surface tension multiplier at alkaline concentration ([ALSURFST](#) keyword).

Alkaline effect on surfactant/polymer adsorption

The alkaline can reduce the adsorption of both surfactant and polymer on the rock surface. This is modeled by modifying the mass of adsorbed surfactant or polymer as follows:

$$V \rho_r C_{s,p}^a \frac{1-\varphi}{\varphi} A_{ad}(C_{alk}) \quad \text{Eq. 6.104}$$

where

V is the pore volume of the cell

ρ_r is the mass density of the rock (see the [SURFROCK](#) / [PLYROCK](#) keywords)

$C_{s,p}^a$ is the surfactant/polymer adsorbed concentration obtained from the [SURFADS](#) / [PLYADS](#) / [PLYADSS](#) / [ADSORP](#) keywords

φ is the porosity

$A_{ad}(C_{alk})$ is the adsorption multiplier at alkaline concentration (refer to the [ALSURFAD](#) / [ALPOLADS](#) keywords).

When the alkaline is assumed to be adsorbed irreversibly, by preventing desorption in the [ALKROCK](#) keyword, its effect on the surfactant/polymer adsorption is assumed to be irreversible as well. Using the previous notations, the mass of adsorbed surfactant or polymer then becomes:

$$V \rho_r C_{s,p}^a \frac{1-\varphi}{\varphi} A_{ad}(C_{alk \max}) \quad \text{Eq. 6.105}$$

where

$C_{alk \max}$ is the maximum alkaline concentration reached in the block, corresponding to the actual effect on surfactant/polymer adsorption.

Note: If surfactant and polymer are used in the same run the [PLYROCK](#) keyword will supersede the [SURFROCK](#) keyword.

Using the alkaline model

This model is activated by specifying the [ALKALINE](#) keyword in the [RUNSPEC](#) section. The polymer flood model or the surfactant flood model should be active as well.

The presence of at least one of the following keyword in the [PROPS](#) section is obligatory.

Restriction	Keyword	Description
Obligatory if the Surfactant Flood Model is active	ALSURFST	Table of oil/water surface tension as a function of alkaline concentration.

Restriction	Keyword	Description
Obligatory if the SURFADS keyword is used	ALSURFAD	Table of surfactant adsorption as a function of alkaline concentration
Obligatory if the Polymer Flood Model is active.	ALPOLADS	Table of polymer adsorption as a function of alkaline concentration.
Optional	ALKADS	Tables of adsorption functions.
Obligatory if ALKADS is used	ALKROCK	Specifies alkaline-rock properties.

The concentration of the injected alkaline in a water injector is specified using the [WALKALIN](#) keyword.

For the list of output associated with the alkaline model, refer to the [SUMMARY](#) and [SCHEDULE](#) sections below.

The [SUMMARY](#) outputs follow the passive tracer format with a keyword concatenated with the tracer name. Here, alkaline is referred to as [ALK](#). For example, the keyword for the field total alkaline production would be [FTPRALK](#).

Summary of keywords

RUNSPEC section

[ALKALINE](#) activates the Alkaline model.

PROPS section

Restriction	Alkaline model keyword	Description
Only if the surfactant option is activated.	ALSURFST	tables of oil/water surface tension as a function of alkaline concentration.
Only if the surfactant option is activated.	ALSURFAD	tables of surfactant adsorption as a function of alkaline concentration.
Only if the polymer option is activated.	ALPOLADS	table for polymer adsorption as a function of alkaline concentration.
	ALKADS	table for alkaline adsorption function as a function of alkaline concentration
	ADSORP	analytical adsorption isotherm (alternative to ALKADS)
	ALKROCK	alkaline-rock adsorption/desorption properties

SUMMARY section

Refer to [Alkaline SUMMARY output controls](#) in the *Eclipse Reference Manual*.

SCHEDULE section

Keyword	Description
RPTSCHED	Controls output from the SCHEDULE section to the print file. Mnemonic ALKALINE outputs the alkaline concentrations and the alkaline adsorption when active. Outputs are: • ALKALINE: alkaline concentration • ALKADS: alkaline concentration adsorbed • ALKMAX: maximum alkaline concentration adsorbed
WALKALIN	Specifies the alkaline concentration for a water injector. This concentration can be set as a UDA (user-defined argument).

Example problem

This example mentions only those sections of an Eclipse data file that apply to alkaline data input and output.

RUNSPEC

```

RUNSPEC
TITLE
Example Alkaline and Surfactant Case.

DIMENS
  5    5    1  /

NONNC
OIL
WATER
SURFACT
ALKALINE
FIELD

TABDIMS
  2    2   20    5    6   20  /

REGDIMS
  6    1    0    0    0    1  /

WELLDIMS
  2    1    1    2  /

START
  1 'JAN' 1983  /

NSTACK
  8  /

```

PROPS

```

PROPS

SWFN
-- Table 1 - Immiscible curves - SATNUM
  .25    .0    4.0

```

```

      .7      1.0      .0
/
-- Table 2 - Miscible curves      - SURFNUM
      .25      .0      4.0
      .99      1.0      .0
/

SOF2
-- Table 1 - Immiscible curves - SATNUM
      .3000      .0000
      .7600      1.0000
/
-- Table 2 - Miscible curves      - SURFNUM
      .0100      .0000
      .7600      1.0000
/

PVTW
4000.0      1.0      3.03E-06      .5      0.0 /
4000.0      1.0      3.03E-06      .5      0.0 /

PVDO
      .0      1.0      2.0
8000.0      .92      2.0
/
-- Table 2 defaulted
/

ROCK
4000.0      .30E-05 /
4000.0      .30E-05 /

DENSITY
52.0000      64.0000      .04400 /
52.0000      64.0000      .04400 /

-- Water Viscosity vs surfactant concentration tables.
SURFVISC
-- Surfactant      Water viscosity at 4000 psi
-- concentration
--      LB/STB      CENTIPOISE
--      0.0      1.0
--      0.1      1.1
--      0.2      1.2
--      0.8      1.3 /
--      0.0      0.5
--      0.1      0.6
--      0.2      0.7
--      0.8      0.8 /

-- Surfactant Adsorption tables
SURFADS
-- Surfactant      Adsorbed saturation
-- concentration
--      LB/STB      LB/LB
--      0.0      0.00000
--      0.1      0.00005
--      0.2      0.00005
--      1.0      0.00005
/
-- Table 2 defaulted
/
-- Alkaline adsorption
ALKADS
-- Alkaline      Adsorbed
-- concentration      saturation
--      LB/STB
--      0.0      0.00000
--      0.2      0.00002
--      1.0      0.00005 /
/
-- Alkaline adsorption properties
ALKROCK
-- De-adsorption model
--      1      /

```

```

/
-- Alkaline multipliers for Surfactant Adsorption
ALSURFAD
-- alkaline          adsorption
-- concentration     multiplier
--   LB/STB
--   0.0              1.00000
--   0.1              0.99995
--   0.2              0.99500
--   1.0              0.50000 /
/

-- Water/Oil surface tension vs Surfactant concentration
SUREFST
-- Surfactant        Water/Oil Surface Tension
-- concentration
--   LB/STB          LB/IN
--   0.0             100.0
--   0.1             50.0
--   0.2             10.0
--   0.5             0.0 /
/

-- Water/Oil surface tension multipliers vs alkaline concentration
ALSURFST
-- Alkaline          Water/Oil Surface Tension
-- concentration     multiplier
--   LB/STB
--   0.0             1.0
--   0.1             0.9
--   0.2             0.9
--   0.5             0.9 /
/

-- Capillary de-saturation curve
SURFCAPD
-- LOG10(Capillary Number)  Miscibility
--                          Function
--   -9                     0.0
--   2                      0.0
--   5                      1.0
--   10                     1.0 /
/

-- Extra rock properties.
SURFROCK
-- De-adsorption model      Rock mass
-- indicator                 Density
--   1 - reversible
--   2 - irreversible        LB/RB
--   1                      1000 /
--   1                      1000 /

```

SUMMARY

```

SUMMARY
CTFRALK
'P' 5 5 1 /
/
FTPRALK
FTPTALK

FTIRALK
FTITALK

WTPALK
'P' /

WTITALK
'I' /

BTCNFALK

```

```
1 1 1 /
2 2 1 /
5 5 1 /
/
BTADSALK
1 1 1 /
2 2 1 /
5 5 1 /
/

FTPRSUR

WTPRSUR
'P' /

WTPTSUR
'P' /
WTITSUR
'I' /

BTCNFSUR
1 1 1 /
2 2 1 /
5 5 1 /
/

FTIPTSUR
FTIPFSUR

RTIPTSUR
1 /

RTIPFSUR
1 /

BTIPTSUR
1 1 1 /
/

FTADSUR

RTADSUR
1 /

BTADSUR
1 1 1 /
/

BTCASUR
1 1 1 /
/
```

SCHEDULE

```
SCHEDULE
RPTSCHED

                                / ALKALINE

WELSPECS
'I' 'G' 1 1 4000 'WAT' /
'P' 'G' 5 5 4000 'OIL' /
/

COMPDAT
'I'      ' 1 1 1 1 'OPEN' 0 .0 1.0 /
'P'      ' 5 5 1 1 'OPEN' 0 .0 1.0 /
/

WCONPROD
'P' 'OPEN' 'BHP' 5* 3999.0 /
```

```
/
WCONINJE
'I' 'WAT' 'OPEN' 'RATE' 200.0 /
/

-- Start injecting the Surfactant Slug
WSURFACT
'I' 0.8 /
/

-- Start injecting the Alkaline
WALKALIN
'I' 0.8 /
/

TSTEP
2*100
/

-- Water chase
WSURFACT
'I' 0.0 /
/

WALKALIN
'I' 0.0 /
/

TSTEP
5*100 /

END
```

Surfactant model

x	Eclipse 100
x	Eclipse 300

Most large oil fields are now produced with some type of secondary pressure maintenance scheme, such as water flooding. Water flooding can increase recovery from around 1% to the 20-40% range. The remaining oil can be divided into two classes, firstly residual oil to the water flood, and secondly oil bypassed by the water flood. A surfactant flood is a recovery mechanism aimed at reducing the residual oil saturation in water swept zones.

Typically a water flood that contacts 100% of a given oil zone will leave a residual oil saturation of, say, 30%. This is the saturation at which the oil phase relative permeability value is zero. The oil is immobile at this saturation because of the surface tension between oil and water; the water pressure alone is unable to overcome the high capillary pressure required to move oil out of very small pore volumes.

A surfactant offers a way of recovering the residual oil by reducing the surface tension between the oil and water phases. A very low oil-water surface tension reduces the capillary pressure and hence allows water to displace extra oil. If it is possible to reduce the surface tension to zero, then theoretically the residual oil can be reduced to zero. In practice the residual oil to even high concentrations is unlikely to lead to 100% recovery of swept zones.

One of the effects that will influence the success or failure of a surfactant flood is the tendency of the surfactant being used to be adsorbed by the rock. If the adsorption is too high, then large quantities of surfactant will be required to produce a small quantity of additional oil.

The Eclipse surfactant model does not aim to model the detailed chemistry of a surfactant process, but rather to model the important features of a surfactant flood on a full field basis.

The simulation model

Surfactant conservation equation

Eclipse 100

In Eclipse 100 the distribution of injected surfactant is modeled by solving a conservation equation for surfactant within the water phase. The surfactant concentrations are updated fully-implicitly at the end of each timestep after the oil, water and gas flows have been computed. The surfactant is assumed to exist only in the water phase, and the input to the reservoir is specified as a concentration at a water injector (see the [WSURFACT](#) keyword).

Eclipse 300

In Eclipse 300 surfactant is modeled as a water component as described in "[Water components](#)". A water component must be identified as a surfactant (see the [CWTYPE](#) keyword) and can be input into the reservoir by defining the injection water composition (see the [WINJW](#) keyword).

Surfactant can be adsorbed on to the rock. The Eclipse 100 keywords are described in "[Treatment of adsorption](#)". For Eclipse 300 a solid component can be used to represent adsorbed surfactant, and adsorption and desorption can be modeled using a chemical reaction. See "[Chemical reactions](#)" and "[Solid phase](#)".

Surfactant effects

The presence of surfactant can affect reservoir performance in three different ways:

- The surfactant modifies the oil-water surface tension (see the [SURFST](#) keyword or [SURFSTES](#) if the [BRINE](#) option is activated). This affects the capillary pressure directly as described in "[Capillary pressure](#)" and the oil and water relative permeabilities via the capillary number as described in the "[Relative permeability model](#)".

- The surfactant can modify the water properties. This is described in "[Water PVT properties](#)" for Eclipse 100, or in "[Water component properties](#)" for Eclipse 300.
- The adsorbed surfactant can affect the wettability of the rock. This is described in "[Modeling the change to wettability](#)" for Eclipse 100, or in "[Solid wettability affects](#)" for Eclipse 300.

Calculation of the capillary number

The capillary number is a dimensionless group that measures the ratio of viscous forces to capillary forces. The capillary number is given by:

$$N_c = \frac{C_N}{C_D} \left\| \frac{T \Delta P_o}{A} \right\| \frac{1}{\sigma_{ow}} = C_N \frac{\|K \cdot \nabla P_o\|}{\sigma_{ow}} \quad \text{Eq. 6.106}$$

where

T is the transmissibility number

A is the flow cross-sectional area

K is the permeability

P_o is the potential

σ_{ow} is the interfacial tension (see [SURFST](#) or [SURFSTES](#))

C_D is the Darcy Constant

C_N is a conversion factor depending on the units used.

Relative permeability model

The Surfactant Relative Permeability Model allows a transition from immiscible relative permeability curves at low capillary number to miscible relative permeability curves at high capillary number. The [SURFCAPD](#) keyword is used to define an interpolation parameter F_{kr} as a tabulated function of the logarithm (base 10) of the capillary number, such that the weighting function F is given by $F = F_{kr}(\log N_c)$.

The relative permeability used at a given value of the miscibility function is calculated in two steps. Figure [6.3](#) illustrates the calculation for a look-up saturation of S_t . Firstly, the saturation end points of the curves are interpolated and both the immiscible and the miscible curves are scaled using F to honor these points. Secondly, the relative permeability values are looked up on both curves as functions of the current grid block saturation, and the final relative permeability is taken as an interpolation between these two values using F . This procedure is used to calculate the water relative permeability k_{rw} and the oil-to-water relative permeability k_{row} .

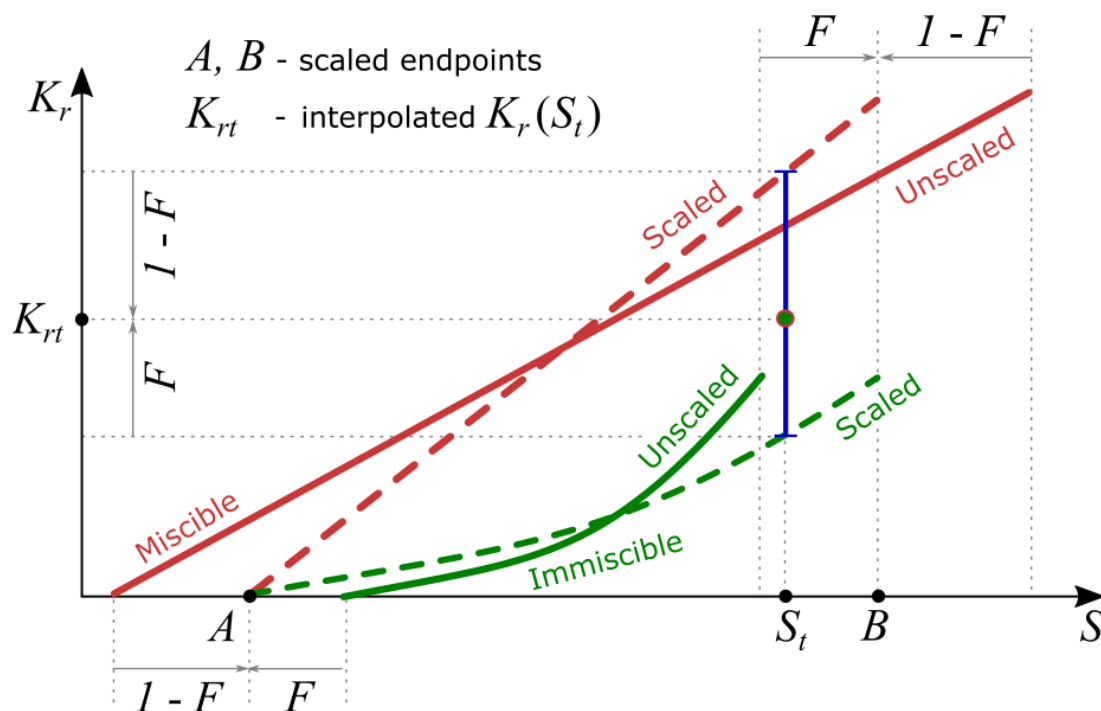


Figure 6.3. Calculation of the relative permeability

In the three-phase case, the oil relative permeability to water k_{row} is then used with either of the two STONE's three-phase relative permeability models or the default model to calculate the oil relative permeability k_{ro} .

Eclipse 100

For the default three-phase oil relative permeability model, both the oil relative permeability to water and the oil relative permeability to gas are interpolated in the manner described above before being combined to form the three-phase oil relative permeability.

The well relative permeability model will be taken as the block relative permeability following the above procedure, if the connection relative permeability tables are defaulted. However, if the connection relative permeability table number is set in the `COMPDAT` keyword, or if the `COMPRP` keyword is used, then that relative permeability table will be used with no modification to account for the surfactant present in the grid block.

The `LOWSALT` keyword can be used in conjunction with the Surfactant Option to model additionally the effects of salinity dependence. See "Brine tracking" for more information upon the `LOWSALT` option.

Eclipse 300

For the default three-phase oil relative permeability model, only the oil-to-water relative permeability is interpolated in the manner described above. For compositional cases, there is an extra step in calculating the oil relative permeability. The oil-to-water relative permeability k_{row} value is determined as above from the miscible and immiscible values. It is then combined with the gas-to-water relative permeability k_{rgw} to determine a hydrocarbon-to-water relative permeability k_{rhw} . See "Near critical oil and gas relative permeabilities". The hydrocarbon-to-water relative permeability k_{rhw} is then used in one of the three-phase relative permeability models to calculate k_{ro} .

The Surfactant Model does not currently support the modeling of relative permeability hysteresis in the water and oil phases.

Surfactant model in cells with no surfactant

The capillary number depends on both the velocity and the oil-water surface tension. Therefore, the capillary number, and hence F_{kr} , can vary even when there is no surfactant present. This means that when the surfactant model is used, relative permeability interpolation may occur even if surfactant is not injected.

Eclipse 100

This behavior is avoided in Eclipse 100. The interpolation of relative permeability is only performed for blocks with a non-zero surfactant concentration. For blocks with zero surfactant concentration the immiscible curves are used. The pre-2010.1 behavior, which is the same as Eclipse 300 behavior, of interpolating relative permeability for all blocks, regardless of surfactant concentration, can be restored by setting [item 147](#) of the `OPTIONS` keyword to a non-zero value.

Capillary pressure

The water oil capillary pressure will reduce as the concentration of surfactant increases; indeed it is the reduction in the oil water capillary pressure that gives rise to the reduction in the residual oil saturation. The oil water capillary pressure is taken as:

$$P_{cow} = F_{cp} \cdot P_{cow}(S_w) \quad \text{Eq. 6.107}$$

where

$F_{cp} = \frac{\sigma_{ow}(C_{\text{surf}})}{\sigma_{ow}(0)}$ is the capillary pressure multiplier

$\sigma_{ow}(C_{\text{surf}})$ is the surface tension at the present surfactant concentration,
determined from the table supplied by `SURFST` keyword,

$\sigma_{ow}(0)$ is the surface tension at zero concentration
determined from the table supplied by `SURFST` keyword, and

$P_{cow}(S_w)$ is the capillary pressure from the immiscible curves initially scaled to the interpolated end-points calculated in the relative permeability model.

Note: The scaling of the immiscible capillary pressure to honor the interpolated end-points calculated in the relative permeability model, was not done in versions before 2009.1. [Item 119](#) of the `OPTIONS` keyword can be used to restore the pre-2009.1 calculations.

Note: If a Leverett J-function is used (`JFUNC` or `JFUNC`), the immiscible capillary pressure is proportional to a user-defined constant surface tension σ_{Jow} . If this surface tension is set equal to the zero concentration surface tension, $\sigma_{Jow} = \sigma_{ow}(0)$ then when the immiscible capillary pressure is multiplied by F_{cp} , the resulting capillary pressure will be equal to that specified by the J-function, but with the surface tension $\sigma_{ow}(C_{\text{surf}})$.

If the salt-sensitivity option has been activated (in Eclipse 100 Brine option is active and the Polymer option is inactive, in Eclipse 300 the use of `SURFSTES` keyword activates the option), the capillary pressure multiplier becomes a function of salt concentration c_s as well:

$$F_{cp} = \frac{\sigma_{ow}(C_{\text{surf}}, c_s)}{\sigma_{ow}(0, c_s)}$$

where

$\sigma_{ow}(C_{surf}, c_s)$ is the surface tension at the present surfactant and salt concentrations

$\sigma_{ow}(0, c_s)$ is the surface tension at zero surfactant concentration and present salt concentration.

The surface tension values are determined from the two dimensional table supplied by [SURFSTES](#) and [ESSNODE](#).

Water PVT properties (Eclipse 100)

The surfactant modifies the viscosity of the pure or salted water input using the [PVTW](#) or the [PVTWSALT](#) keyword, respectively. The surfactant viscosity input as a function of surfactant concentration using the [SURFVISC](#) keyword is used to calculate the water-surfactant solution viscosity as follows:

$$\mu_{ws}(C_{surf}, P) = \mu_w(P) \frac{\mu_s(C_{surf})}{\mu_w(P_{ref})} \quad \text{Eq. 6.108}$$

If the brine option is active (and the polymer option is inactive), the previous equation becomes a function of salt concentration c_s as well:

$$\mu_{ws}(C_{surf}, P, c_s) = \mu_w(P, c_s) \cdot \frac{\mu_s(C_{surf})}{\mu_w(P_{ref}, c_{sref})} \quad \text{Eq. 6.109}$$

where

μ_s is the viscosity from the [SURFVISC](#) keyword(s)

μ_w is the viscosity from the [PVTW](#) or [PVTWSALT](#) keyword

μ_{ws} is the viscosity of the water-surfactant mixture

P_{ref} is the reference pressure in the [PVTW](#) or [PVTWSALT](#) keyword

c_{sref} is the reference salt concentration in the [PVTWSALT](#) keyword.

Note that the [SUMMARY](#) keyword [BEWV_SUR](#) should be used to output the effective water viscosity due to the surfactant concentration. The effects of any polymer concentration are not included in this output.

Treatment of adsorption (Eclipse 100)

The adsorption of surfactant is assumed to be instantaneous, and the quantity adsorbed is a function of the surrounding surfactant concentration. You have to supply an adsorption isotherm defined as a function of surfactant concentration (see the [SURFADS](#) keyword). As an alternative to [SURFADS](#), the surfactant adsorption can be specified using keyword [ADSORP](#) with [SURFACT](#) as an argument. [ADSORP](#) specifies coefficients for a generic, analytical adsorption isotherm which allows for dependencies of adsorption on rock permeability and effective brine salinity. When used in conjunction with the Multi-Component Brine Model, an effective salinity for surfactant can be calculated which can then be used in the adsorption isotherm. See "[Brine Tracking](#)" for information on the Multi-Component Brine Model and the calculation of effective salinity for surfactant. See the [ADSORP](#) keyword for information on the functional dependence of adsorption on salinity.

The quantity of surfactant adsorbed on to the rock is given by:

$$\text{Mass of adsorbed surfactant} = \text{PORV} \cdot \frac{1-\varphi}{\varphi} \cdot \text{MD} \cdot \text{CA}(C_{\text{surf}}) \quad \text{Eq. 6.110}$$

where

PORV is the pore volume of the cell

φ is the porosity

MD is the mass density of the rock (see [SURFROCK](#) keyword)

$\text{CA}(C_{\text{surf}})$ is the adsorption isotherm as a function of local surfactant concentration in solution.

There are two adsorption models that can be selected, using the first argument of [SURFROCK](#). The first model ensures that each grid block retraces the adsorption isotherm as the surfactant concentration falls in the cell. The second model assumes that the adsorbed surfactant concentration on the rock may not decrease with time and hence does not allow for any de-adsorption.

In the present version of Eclipse the adsorption concentrations are updated explicitly for the surfactant concentration. In the future it may be necessary to implement the adsorption implicitly.

Modeling changes in wettability due to surfactant adsorption (Eclipse 100)

This option models changes in the rock wettability due to the accumulation of surfactant by adsorption. It is activated by specifying the [SURFACTW](#) keyword, which also activates the Surfactant Model ([SURFACT](#) keyword).

The [SATNUM](#) and [SURFNUM](#) keywords are used to define the oil-wet and water-wet immiscible saturation regions respectively and hence the associated saturation tables. (The [HWSNUM](#) keyword may be used as an abbreviated alias for [SURFNUM](#).) The keyword [SURFNUM](#) is used to define the miscible saturation regions. Given two sets of saturation functions, one for immiscible oil-wet and one for immiscible water-wet, the immiscible water and oil table saturation end-points are interpolated firstly according to:

$$\begin{aligned} S_{wco}^{imm} &= F_4 S_{wco}^{ow} + (1-F_4) S_{wco}^{ww} \\ S_{wcr}^{imm} &= F_4 S_{wcr}^{ow} + (1-F_4) S_{wcr}^{ww} \\ S_{wmax}^{imm} &= F_4 S_{wmax}^{ow} + (1-F_4) S_{wmax}^{ww} \\ S_{owcr}^{imm} &= F_4 S_{owcr}^{ow} + (1-F_4) S_{owcr}^{ww} \\ S_{ogcr}^{imm} &= F_4 S_{ogcr}^{ow} + (1-F_4) S_{ogcr}^{ww} \end{aligned} \quad \text{Eq. 6.111}$$

and these are then combined with the miscible table saturation end-points according to:

$$\begin{aligned} S_{wco}^i &= F_3 S_{wco}^{mis} + (1-F_3) S_{wco}^{imm} \\ S_{wcr}^i &= F_3 S_{wcr}^{mis} + (1-F_3) S_{wcr}^{imm} \\ S_{wmax}^i &= F_3 S_{wmax}^{mis} + (1-F_3) S_{wmax}^{imm} \\ S_{owcr}^i &= F_3 S_{owcr}^{mis} + (1-F_3) S_{owcr}^{imm} \\ S_{ogcr}^i &= F_3 S_{ogcr}^{mis} + (1-F_3) S_{ogcr}^{imm} \end{aligned} \quad \text{Eq. 6.112}$$

where

F_3	is tabulated as a function of the capillary number (expressed in terms of its logarithm base ten) and corresponds to the second column of the SURFCAPD keyword.
F_4	is tabulated as a function of the adsorbed surfactant concentration and corresponds to the second column of the SURFADDW keyword.
S_{wco}	is the saturation table derived connate water saturation.
S_{wcr}	is the saturation table derived critical water saturation.
S_{wmax}	is the saturation table derived maximum water saturation.
S_{owcr}	is the saturation table derived critical oil saturation in water.
S_{ogcr}	is the saturation table derived critical oil saturation in gas.
superscripts ow , ww	denote the immiscible oil-wet and water-wet immiscible table derived saturation end-points respectively.
superscripts imm , mis	denote the immiscible and miscible table derived saturation end-points respectively.
superscript i	denotes the interpolation between the immiscible and miscible table derived saturation end-points.

The immiscible oil-wet and water-wet capillary pressures and relative permeabilities are looked up in the immiscible oil-wet and water-wet saturation tables by applying two-point saturation (horizontal) end-point scaling using the interpolated saturation end-points:

$$\begin{aligned}
 P_{cow}^{ow} &= f(S_w, S_{wco}^i, S_{wmax}^i, P_{cowmax}^{ow}) \\
 P_{cow}^{ww} &= f(S_w, S_{wco}^i, S_{wmax}^i, P_{cowmax}^{ww}) \\
 k_{rw}^{ow} &= f(S_w, S_{wcr}^i, S_{wmax}^i, k_{rwmax}^{ow}) \\
 k_{rw}^{ww} &= f(S_w, S_{wcr}^i, S_{wmax}^i, k_{rwmax}^{ww}) \\
 k_{rog}^{ow} &= f(S_{og}, S_{ogcr}^i, S_{omax}, k_{romax}^{ow}) \\
 k_{rog}^{ww} &= f(S_{og}, S_{ogcr}^i, S_{omax}, k_{romax}^{ww}) \\
 k_{row}^{ow} &= f(S_{ow}, S_{owcr}^i, S_{omax}, k_{romax}^{ow}) \\
 k_{row}^{ww} &= f(S_{ow}, S_{owcr}^i, S_{omax}, k_{romax}^{ww})
 \end{aligned}
 \tag{Eq. 6.113}$$

The miscible relative permeabilities are looked up in the miscible tables by applying two-point saturation (horizontal) end-point scaling using the interpolated saturation end-points:

$$\begin{aligned}
 k_{rw}^{mis} &= f(S_w, S_{wcr}^i, S_{wmax}^i, k_{rwmax}^{mis}) \\
 k_{rog}^{mis} &= f(S_{og}, S_{ogcr}^i, S_{omax}, k_{romax}^{mis}) \\
 k_{row}^{mis} &= f(S_{ow}, S_{owcr}^i, S_{omax}, k_{romax}^{mis})
 \end{aligned}
 \tag{Eq. 6.114}$$

where

f signifies the end-point scaling operation.

S_w is the grid block water saturation.

S_{ow} is the oil saturation for oil relative permeability to water.

S_{og} is the oil saturation for oil relative permeability to gas.

S_{omax} is the maximum oil saturation, $S_{omax} = (1 - S_{wco}^i - S_{gco}^{ow})$.

S_{gco} is the saturation table derived connate gas saturation.

k_{rwmax} is the saturation table derived maximum water relative permeability.

k_{romax} is the saturation table derived maximum oil relative permeability.

P_{cowmax} is the saturation table derived maximum oil-water capillary pressure.

For the default three-phase oil relative permeability model, the oil saturations used to look up the oil relative permeabilities are:

$$\begin{aligned} S_{ow} &= (1 - S_w - S_g) \\ S_{og} &= (1 - S_w - S_g) \end{aligned} \quad \text{Eq. 6.115}$$

For the Stone 1 and Stone 2 three-phase oil relative permeability models, the oil saturations used to look up the oil relative permeabilities are:

$$\begin{aligned} S_{ow} &= (1 - S_w) \\ S_{og} &= (1 - S_g - S_{wco}^i) \end{aligned} \quad \text{Eq. 6.116}$$

The immiscible oil-wet and water-wet relative permeabilities and capillary pressure are interpolated according to:

$$\begin{aligned} k_{rw}^{imm} &= F_4 k_{rw}^{ow} + (1 - F_4) k_{rw}^{ww} \\ k_{row}^{imm} &= F_4 k_{row}^{ow} + (1 - F_4) k_{row}^{ww} \\ k_{rog}^{imm} &= F_4 k_{rog}^{ow} + (1 - F_4) k_{rog}^{ww} \\ P_{cow}^{imm} &= F_4 P_{cow}^{ow} + (1 - F_4) P_{cow}^{ww} \end{aligned} \quad \text{Eq. 6.117}$$

The immiscible and miscible relative permeabilities are interpolated according to:

$$\begin{aligned} k_{rw}^i &= F_3 k_{rw}^{mis} + (1 - F_3) k_{rw}^{imm} \\ k_{row}^i &= F_3 k_{row}^{mis} + (1 - F_3) k_{row}^{imm} \\ P_{cow}^i &= P_{cow}^{imm} \end{aligned} \quad \text{Eq. 6.118}$$

Default three-phase oil relative permeability model:

$$k_{rog}^i = F_3 k_{rog}^{mis} + (1 - F_3) k_{rog}^{imm} \quad \text{Eq. 6.119}$$

Stone 1 and 2 three-phase oil relative permeability models:

$$k_{rog}^i = k_{rog}^{imm} \quad \text{Eq. 6.120}$$

The water-oil capillary pressure is only interpolated between the immiscible oil-wet and water-wet values. The gas saturation functions are looked up in the immiscible oil-wet table; that is, the gas relative permeability and gas-oil capillary pressure are assumed to exhibit no dependence upon surfactant concentration or surfactant adsorption.

When end-point scaling is specified via the **ENDSCALE** keyword the interpolations follow a similar pattern to that described above in the absence of end-point scaling. Hence given two sets of immiscible water and oil saturation functions, one for oil-wet and one for water-wet, the scaled saturation end points are interpolated firstly according to:

$$\begin{aligned} S_{WL}^{IMM} &= F_4 S_{WL}^{OW} + (1 - F_4) S_{WL}^{WW} \\ S_{WLPC}^{IMM} &= F_4 S_{WLPC}^{OW} + (1 - F_4) S_{WLPC}^{WW} \\ S_{WCR}^{IMM} &= F_4 S_{WCR}^{OW} + (1 - F_4) S_{WCR}^{WW} \\ S_{WU}^{IMM} &= F_4 S_{WU}^{OW} + (1 - F_4) S_{WU}^{WW} \\ S_{OWCR}^{IMM} &= F_4 S_{OWCR}^{OW} + (1 - F_4) S_{OWCR}^{WW} \\ S_{OGCR}^{IMM} &= F_4 S_{OGCR}^{OW} + (1 - F_4) S_{OGCR}^{WW} \end{aligned} \quad \text{Eq. 6.121}$$

and these are then combined with the miscible scaled end-points according to:

$$\begin{aligned} S_{WL}^I &= F_3 S_{WCO}^{MIS} + (1 - F_3) S_{WL}^{IMM} \\ S_{WCR}^I &= F_3 S_{WCR}^{MIS} + (1 - F_3) S_{WCR}^{IMM} \\ S_{WU}^I &= F_3 S_{WMAX}^{MIS} + (1 - F_3) S_{WU}^{IMM} \\ S_{OWCR}^I &= F_3 S_{OWCR}^{MIS} + (1 - F_3) S_{OWCR}^{IMM} \\ S_{OGCR}^I &= F_3 S_{OGCR}^{MIS} + (1 - F_3) S_{OGCR}^{IMM} \end{aligned} \quad \text{Eq. 6.122}$$

where

S_{WL}	is the scaled minimum water saturation for relative permeability.
S_{WLPC}	is the scaled minimum water saturation for oil-water capillary pressure.
S_{WCR}	is the scaled critical water saturation.
S_{WU}	is the scaled maximum water saturation.
S_{OWCR}	is the scaled critical oil saturation in water.
S_{OGCR}	is the scaled critical oil saturation in gas.
superscripts IMM , MIS	denote the immiscible interpolated and the miscible scaled saturation end-points respectively.
superscript I	denotes the interpolation between the immiscible and miscible scaled saturation end-points.

Lower and upper case subscripts and superscripts are used here to signify the table derived and the scaled end-points respectively.

The oil-wet immiscible saturation tables are scaled using the set of end-point scaling keywords consisting of: **SWL**, **SWLPC**, **SWCR**, **SWU**, **SGL**, **SGLPC**, **SGCR**, **SGU**, **SOWCR**, **SOGCR**, **KRW**, **KRWR**, **KRG**, **KRGR**, **KRO**, **KRORW**, **KRORG**, **PCW** and **PCG**. For versions prior to 2014.1, this set of end-points was also used for the water-wet immiscible saturation tables. Pre-2014.1 behavior may be restored by setting [item 225](#) of the **OPTIONS** keyword to 1.

The water-wet immiscible saturation tables are scaled using the reduced set of end-point scaling keywords consisting of **HWSWL**, **HWSWLPC**, **HWSWCR**, **HWSWU**, **HWSOWCR**, **HWSOGCR**, **HWKRW**, **HWKRWR**, **HWKRO**, **HWKRORW**, **HWKRORG** and **HWPCW**. This set does not include any gas end-points because only the oil-wet immiscible saturation functions are used for gas relative permeability and for oil-gas capillary pressure.

The miscible saturation tables are scaled using the reduced set of end-point scaling keywords consisting of **SSWL**, **SSWCR**, **SSWU**, **SSGL**, **SSGCR**, **SSOWCR**, **SSOGCR**, **SKRW**, **SKRWR**, **SKRO**, **SKRORW** and **SKRORG**.

The capillary pressure and relative permeabilities are looked up in the immiscible oil-wet and water-wet saturation tables using the sets of scaled end-points shown below. For capillary pressure, only two-point scaling is employed. For the relative permeabilities, two-point or three-point end-point scaling is employed according to whether item 1 of the **SCALECRS** keyword has been set to NO or YES respectively. (Two-point scaling is selected by default if this keyword is omitted.) For the immiscible oil relative permeability to water and for three-point scaling, the critical water saturation end-point is taken from the interpolated value or the respective oil-wet and water-wet values according to whether the default or the Stone 1 or 2 three-phase oil relative permeability models has been selected.

The immiscible oil-wet and water-wet oil-water capillary pressures are calculated according to:

$$\begin{aligned} P_{COW}^{OW} &= f(S_W, S_{WLPC}^I, S_{WU}^I, P_{CW}^{OW}) \\ P_{COW}^{WW} &= f(S_W, S_{WLPC}^I, S_{WU}^I, P_{CW}^{WW}) \end{aligned} \quad \text{Eq. 6.123}$$

For two-point scaling the immiscible oil-wet and water-wet relative permeabilities are calculated according to:

$$\begin{aligned} K_{RW}^{OW} &= f(S_W, S_{WCR}^I, S_{owcr}^{ow}, S_{WU}^I, K_{RW}^{OW}, K_{RWR}^{OW}) \\ K_{RW}^{WW} &= f(S_W, S_{WCR}^I, S_{owcr}^{ww}, S_{WU}^I, K_{RW}^{WW}, K_{RWR}^{WW}) \\ K_{ROG}^{OW} &= f(S_{OG}, S_{OGCR}^I, S_{grcr}^{ow}, S_{OMAX}, K_{RO}^{OW}, K_{RORG}^{OW}) \\ K_{ROG}^{WW} &= f(S_{OG}, S_{OGCR}^I, S_{grcr}^{ow}, S_{OMAX}, K_{RO}^{WW}, K_{RORG}^{WW}) \\ K_{ROW}^{OW} &= f(S_{OW}, S_{OWCR}^I, S_{wcr}^{ow}, S_{OMAX}, K_{RO}^{OW}, K_{RORW}^{OW}) \\ K_{ROW}^{WW} &= f(S_{OW}, S_{OWCR}^I, S_{wcr}^{ww}, S_{OMAX}, K_{RO}^{WW}, K_{RORW}^{WW}) \end{aligned} \quad \text{Eq. 6.124}$$

For three-point scaling the immiscible oil-wet and water-wet relative permeabilities are calculated according to:

$$\begin{aligned} K_{RW}^{OW} &= f(S_W, S_{GL}^H, S_{WCR}^I, S_{OWCR}^I, S_{WU}^I, K_{RW}^{OW}, K_{RWR}^{OW}) \\ K_{RW}^{WW} &= f(S_W, S_{GL}^H, S_{WCR}^I, S_{OWCR}^I, S_{WU}^I, K_{RW}^{WW}, K_{RWR}^{WW}) \\ K_{ROG}^{OW} &= f(S_{OG}, S_{WL}^I, S_{OGCR}^I, S_{GCR}^{OW}, S_{OMAX}, K_{RO}^{OW}, K_{RORG}^{OW}) \\ K_{ROG}^{WW} &= f(S_{OG}, S_{WL}^I, S_{OGCR}^I, S_{GCR}^{OW}, S_{OMAX}, K_{RO}^{WW}, K_{RORG}^{WW}) \end{aligned} \quad \text{Eq. 6.125}$$

Default three-phase oil relative permeability model:

$$\begin{aligned} K_{ROW}^{OW} &= f(S_{OW}, S_{GL}^H, S_{OWCR}^I, S_{WCR}^I, S_{OMAX}, K_{RO}^{OW}, K_{RORW}^{OW}) \\ K_{ROW}^{WW} &= f(S_{OW}, S_{GL}^H, S_{OWCR}^I, S_{WCR}^I, S_{OMAX}, K_{RO}^{WW}, K_{RORW}^{WW}) \end{aligned} \quad \text{Eq. 6.126}$$

Stone 1 and 2 three-phase oil relative permeability models:

$$\begin{aligned} K_{ROW}^{OW} &= f(S_{OW}, S_{GL}^H, S_{OWCR}^I, S_{WCR}^{OW}, S_{OMAX}, K_{RO}^{OW}, K_{RORW}^{OW}) \\ K_{ROW}^{WW} &= f(S_{OW}, S_{GL}^H, S_{OWCR}^I, S_{WCR}^{WW}, S_{OMAX}, K_{RO}^{WW}, K_{RORW}^{WW}) \end{aligned} \quad \text{Eq. 6.127}$$

where

S_W is the grid block water saturation.

S_{OW} is the oil saturation for oil relative permeability to water.

S_{OG} is the oil saturation for oil relative permeability to gas.

S_{wcr} is the saturation table derived critical water saturation.

S_{owcr} is the saturation table derived critical oil in water saturation.

S_{OMAX} is the scaled maximum oil saturation, $S_{OMAX} = (1 - S_{WL}^I - S_{GL}^{OW})$.

S_{GL} is the scaled minimum gas saturation.

S_{GCR} is the scaled critical gas saturation.

S_{gcr} is the saturation table derived critical gas saturation.

K_{RW} is the scaled maximum water relative permeability.

K_{RWR} is the scaled water relative permeability at residual oil saturation.

K_{RO} is the scaled maximum oil relative permeability.

K_{RORG} is the scaled oil relative permeability at residual gas saturation.

K_{RORW} is the scaled oil relative permeability at residual water saturation.

P_{CW} is the scaled maximum oil-water capillary pressure.

Noting that:

1. For the scaling of both the oil-wet and water-wet immiscible saturation tables, specifically, for the scaling of the relative permeabilities of oil to gas and water, only the oil-wet immiscible minimum and critical scaled gas saturations are used.
2. For two-point scaling, the positioning of the relative permeability at residual saturation of the displacing phase is determined by the two-point scaled table value of the displacing critical phase

saturation whereas, for three-point scaling, this is determined by the scaled end-point of the displacing phase critical saturation.

3. For the default three-phase oil relative permeability model, the critical water saturation used in the scaling of the oil-wet and water-wet oil relative permeabilities to water is based upon the interpolated value, whereas for the Stone 1 and 2 three-phase oil relative permeability models, the scaling is based upon the respective immiscible oil-wet and water-wet values.

For the miscible relative permeabilities, either two-point or three-point end-point scaling is employed according to whether item 1 of the [SCALECRS](#) keyword has been set to NO or YES respectively. For two-point scaling the miscible saturation functions are looked up in the miscible saturation tables applying the set of scaled end-points shown below:

$$\begin{aligned} K_{RW}^{MIS} &= f(S_W, S_{WCR}^I, S_{owcr}^{mis}, S_{WU}^I, K_{RW}^{MIS}, K_{RWR}^{MIS}) \\ K_{ROG}^{MIS} &= f(S_{OG}, S_{OGCR}^I, S_{gcr}^{mis}, S_{OMAX}, K_{RO}^{MIS}, K_{ROG}^{MIS}) \\ K_{ROW}^{MIS} &= f(S_{OW}, S_{OWCR}^I, S_{wcr}^{mis}, S_{OMAX}, K_{RO}^{MIS}, K_{RORW}^{MIS}) \end{aligned} \quad \text{Eq. 6.128}$$

For three-point scaling the miscible water relative permeability and the oil relative permeability to gas are looked up in the miscible tables applying the set of scaled end-points shown below:

$$\begin{aligned} K_{RW}^{MIS} &= f(S_W, S_{GL}^{MIS}, S_{WCR}^I, S_{OWCR}^{MIS}, S_{WU}^I, K_{RW}^{MIS}, K_{RWR}^{MIS}) \\ K_{ROG}^{MIS} &= f(S_{OG}, S_{WL}^I, S_{OGCR}^I, S_{GCR}^{MIS}, S_{OMAX}, K_{RO}^{MIS}, K_{ROG}^{MIS}) \end{aligned} \quad \text{Eq. 6.129}$$

For the miscible oil relative permeability to water and for three-point scaling, the critical water saturation end-point is taken from the interpolated value for the default three-phase oil relative permeability model:

$$K_{ROW}^{MIS} = f(S_{OW}, S_{GL}^{MIS}, S_{OWCR}^I, S_{WCR}^I, S_{OMAX}, K_{RO}^{MIS}, K_{RORW}^{MIS}) \quad \text{Eq. 6.130}$$

but from the miscible value for the Stone 1 and 2 three-phase oil relative permeability models:

$$K_{ROW}^{MIS} = f(S_{OW}, S_{GL}^{MIS}, S_{OWCR}^I, S_{WCR}^{MIS}, S_{OMAX}, K_{RO}^{MIS}, K_{RORW}^{MIS}) \quad \text{Eq. 6.131}$$

The immiscible oil-wet and water-wet relative permeabilities and capillary pressure are interpolated according to:

$$\begin{aligned} K_{RW}^{IMM} &= F_4 K_{RW}^{OW} + (1-F_4) K_{RW}^{WW} \\ K_{ROW}^{IMM} &= F_4 K_{ROW}^{OW} + (1-F_4) K_{ROW}^{WW} \\ K_{ROG}^{IMM} &= F_4 K_{ROG}^{OW} + (1-F_4) K_{ROG}^{WW} \\ P_{COW}^{IMM} &= F_4 P_{COW}^{OW} + (1-F_4) P_{COW}^{WW} \end{aligned} \quad \text{Eq. 6.132}$$

The miscible and immiscible relative permeabilities and capillary pressure are interpolated according to:

$$\begin{aligned} K_{RW}^I &= F_3 K_{RW}^{MIS} + (1-F_3) K_{RW}^{IMM} \\ K_{ROW}^I &= F_3 K_{ROW}^{MIS} + (1-F_3) K_{ROW}^{IMM} \\ P_{COW}^I &= P_{COW}^{IMM} \end{aligned} \quad \text{Eq. 6.133}$$

Default three-phase oil relative permeability model:

$$K_{ROG}^I = F_3 K_{ROG}^{MIS} + (1-F_3) K_{ROG}^{IMM} \quad \text{Eq. 6.134}$$

Stone 1 and 2 three-phase oil relative permeability models:

$$K_{ROG}^I = K_{ROG}^{IMM} \quad \text{Eq. 6.135}$$

The gas-oil capillary pressure and gas relative permeability are assumed to be independent of surfactant concentration and surfactant adsorption. Hence the gas-oil capillary pressure is calculated according to:

$$P_{COG} = f(S_G, S_{GLPC}^{OW}, S_{GU}^{OW}, P_{CG}^{OW}) \quad \text{Eq. 6.136}$$

For two-point scaling the gas relative permeability is calculated according to:

$$K_{RG} = f(S_G, S_{GCR}^{OW}, S_{ogcr}^{ow}, S_{GU}^{OW}, K_{RG}^{OW}, K_{RGR}^{OW}) \quad \text{Eq. 6.137}$$

For three-point scaling the gas relative permeability is calculated according to:

$$K_{RG} = f(S_G, S_{GCR}^{OW}, S_{OGCR}^{OW}, S_{GU}^{OW}, K_{RG}^{OW}, K_{RGR}^{OW}) \quad \text{Eq. 6.138}$$

where

S_G is the grid block gas saturation.

S_{GLPC} is the scaled minimum gas saturation for capillary pressure.

S_{GU} is the scaled maximum gas saturation.

S_{ogcr} is the saturation table derived critical oil saturation in gas.

P_{CG} is the scaled maximum oil-gas capillary pressure.

K_{RG} is the scaled maximum gas relative permeability.

K_{RGR} is the scaled gas relative permeability at residual oil saturation.

Hence only the oil-wet table and oil-wet scaled end-points are used for modeling the immiscible gas saturation functions. If the current grid block gas saturation exceeds the maximum oil-wet scaled gas saturation, the values of gas relative permeability and gas-oil capillary pressure are assumed to remain constant at their corresponding maximum values.

The [LOWSALT](#) keyword can be used in conjunction with the Surfactant Option ([SURFACT](#) keyword) or the Surfactant Option with Wettability Dependence ([SURFACTW](#) keyword) to model additionally the effects of salinity dependence. See ["Brine tracking"](#) for more information upon the [LOWSALT](#) option.

Saturation function interpolation (Eclipse 300)

Relative permeabilities

The interpolation method described in ["Relative permeability model"](#) applies to two relative permeability tables. The calculation described here is more complicated, particularly in compositional runs, although essentially it follows the same pattern.

Initially the saturation end-points are determined by interpolating between the miscible and immiscible endpoints. If solid wettability modeling has been selected, the immiscible end-points must first be interpolated between the two solid wettability tables corresponding to low and high concentrations of solid adsorption. The [SATNUM](#) and [SOLWNUM](#) keywords are used to define the low and high concentration

immiscible saturation regions for solid adsorption respectively. The keyword **SURFNUM** is used to define the miscible saturation regions.

If end-point scaling has not been specified, the immiscible table saturation end-points corresponding to the low and high concentrations of solid adsorption are interpolated firstly according to:

$$\begin{aligned}
 S_{wco}^{imm} &= F_w S_{wco}^h + (1-F_w) S_{wco}^l \\
 S_{wcr}^{imm} &= F_w S_{wcr}^h + (1-F_w) S_{wcr}^l \\
 S_{wmax}^{imm} &= F_w S_{wmax}^h + (1-F_w) S_{wmax}^l \\
 S_{owcr}^{imm} &= F_w S_{owcr}^h + (1-F_w) S_{owcr}^l \\
 S_{gco}^{imm} &= F_w S_{gco}^h + (1-F_w) S_{gco}^l
 \end{aligned}
 \tag{Eq. 6.139}$$

and these are then combined with the miscible table saturation end-points according to:

$$\begin{aligned}
 S_{wco}^i &= F_m S_{wco}^{mis} + (1-F_m) S_{wco}^{imm} \\
 S_{wcr}^i &= F_m S_{wcr}^{mis} + (1-F_m) S_{wcr}^{imm} \\
 S_{wmax}^i &= F_m S_{wmax}^{mis} + (1-F_m) S_{wmax}^{imm} \\
 S_{owcr}^i &= F_m S_{owcr}^{mis} + (1-F_m) S_{owcr}^{imm} \\
 S_{gco}^i &= F_m S_{gco}^{mis} + (1-F_m) S_{gco}^{imm}
 \end{aligned}
 \tag{Eq. 6.140}$$

where

F_w	is an interpolation factor dependent upon the concentration of adsorbed solid and is specified via the SOLWTAB keyword.
F_m	is an interpolation factor dependent upon the concentration of surfactant and is specified via the SURFCAPD keyword.
S_{wco}	is the saturation table derived connate water saturation.
S_{wcr}	is the saturation table derived critical water saturation.
S_{wmax}	is the saturation table derived maximum water saturation.
S_{owcr}	is the saturation table derived critical oil saturation in water.
S_{gco}	is the saturation table derived connate gas saturation.
superscripts l, h	denote the immiscible low and high concentrations of solid adsorption respectively.
superscripts imm, mis	denote the immiscible and miscible values respectively.
superscript i	denotes the interpolation between the immiscible and miscible end-points.

The immiscible relative permeabilities for the low and high concentrations of solid adsorption are looked up in the respective immiscible saturation tables by applying two-point saturation (horizontal) end-point scaling using the interpolated saturation end-points and the respective vertical end-points:

$$\begin{aligned}
k_{rw}^l &= f(S_w, S_{wcr}^i, S_{wmax}^i, k_{rwmax}^l) \\
k_{rw}^h &= f(S_w, S_{wcr}^i, S_{wmax}^i, k_{rwmax}^h) \\
k_{row}^l &= f(S_{ow}, S_{owcr}^i, S_{omax}^i, k_{romax}^l) \\
k_{row}^h &= f(S_{ow}, S_{owcr}^i, S_{omax}^i, k_{romax}^h)
\end{aligned}
\tag{Eq. 6.141}$$

The miscible relative permeabilities are looked up in the miscible tables by applying two-point saturation (horizontal) end-point scaling using the interpolated saturation end-points and the miscible vertical end-points:

$$\begin{aligned}
k_{rw}^{mis} &= f(S_w, S_{wcr}^i, S_{wmax}^i, k_{rwmax}^{mis}) \\
k_{row}^{mis} &= f(S_{ow}, S_{owcr}^i, S_{omax}^i, k_{romax}^{mis})
\end{aligned}
\tag{Eq. 6.142}$$

where:

f signifies the end-point scaling operation.

S_w is the grid block water saturation.

S_{ow} is the oil saturation for oil relative permeability to water.

S_{omax} is the maximum oil saturation, $S_{omax} = (1 - S_{wco}^i - S_{gco}^i)$.

S_{gco} is the saturation table derived connate gas saturation.

k_{rwmax} is the saturation table derived maximum water relative permeability.

k_{romax} is the saturation table derived maximum oil relative permeability.

The relative permeabilities for the low and high concentrations of solid adsorption are interpolated according to:

$$\begin{aligned}
k_{rw}^{imm} &= F_w k_{rw}^h + (1 - F_w) k_{rw}^l \\
k_{row}^{imm} &= F_w k_{row}^h + (1 - F_w) k_{row}^l
\end{aligned}
\tag{Eq. 6.143}$$

If solid wettability effects are not being modeled ($F_w = 0$), the immiscible relative permeabilities are taken directly from the saturation tables defined by [SATNUM](#) and no interpolation of the immiscible saturation end-points is required.

The immiscible and miscible relative permeabilities are interpolated according to:

$$\begin{aligned}
k_{rw}^i &= F_m k_{rw}^{mis} + (1 - F_m) k_{rw}^{imm} \\
k_{row}^i &= F_m k_{row}^{mis} + (1 - F_m) k_{row}^{imm}
\end{aligned}
\tag{Eq. 6.144}$$

In compositional cases, a hydrocarbon in water relative permeability is then calculated to ensure continuous behavior in super-critical regions. See ["Near critical oil and gas relative permeabilities"](#):

$$k_{rhw} = E k_{row} + (1 - E) k_{rgw}$$

The final step is to calculate three-phase relative permeabilities using the default method or Stone's method. See ["Three phase oil relative permeability models"](#) and ["Three phase gas relative permeability models"](#):

$$k_{ro} = k_{ro}(k_{rhw}, k_{rog})$$

$$k_{rg} = k_{rg}(k_{rhw}, k_{rgo})$$

Note: In compositional cases, the default extra scaling of the hydrocarbon relative permeabilities is turned off, see keyword [FORMOPTS](#).

Capillary pressures

If solid wettability effects are being modeled, the oil-water capillary pressure is interpolated for the two wettability regions, and the result is then multiplied by the miscibility factor:

$$P_{cow}^l = f(S_w, S_{wco}^i, S_{wmax}^i, P_{cowmax}^l)$$

$$P_{cow}^h = f(S_w, S_{wco}^i, S_{wmax}^i, P_{cowmax}^h)$$

and

$$P_{cow}^{imm} = F_w P_{cow}^h + (1 - F_w) P_{cow}^l = F_{cp} P_{cow}^{imm}$$

where:

f signifies the end-point scaling operation.

S_w is the grid block water saturation.

S_{wco} is the saturation table derived connate water saturation.

S_{wmax} is the saturation table derived maximum water saturation.

P_{cowmax} is the saturation table derived maximum water oil capillary pressure.

F_w is an interpolation factor dependent upon the concentration of adsorbed solid and is specified via the [SOLWTAB](#) keyword.

F_{cp} is the capillary pressure miscibility factor.

superscripts l, h denote the immiscible low and high solid adsorption end-points respectively.

If solid wettability effects are not being modeled ($F_w = 0$), the immiscible capillary pressures are taken directly from the saturation tables defined by [SATNUM](#) and no interpolation of the immiscible saturation end-points is required.

End-Point Scaling

If end-point scaling is specified via the [ENDSCALE](#) keyword, three sets of end-point scaling keywords are available.

- 1) The immiscible (low concentration of solid adsorption) saturation functions are scaled using the keywords `SWL`, `SWLPC`, `SWCR`, `SWU`, `SGL`, `SGLPC`, `SGCR`, `SGU`, `SOWCR`, `SOGCR`, `KRW`, `KRWR`, `KRG`, `KRGR`, `KRO`, `KRORW`, `KRORG`, `PCW` and `PCG`.
- 2) The immiscible high concentration of solid adsorption saturation functions are scaled using the reduced set of keywords `SASWL`, `SASWCR`, `SASWU`, `SASGL`, `SASOWCR`, `SAKRW`, `SAKRWR`, `SAKRO`, `SAKRORW` and `SAPCW`.
- 3) The miscible saturation tables are scaled using the reduced set of keywords `SSWL`, `SSWCR`, `SSWU`, `SSGL`, `SSOWCR`, `SKRW`, `SKRWR`, `SKRO` and `SKRORW`.

When end-point scaling is specified, the interpolations follow a similar pattern to that described above in the absence of end-point scaling. Hence given two sets of immiscible saturation functions for low and high concentrations of solid adsorption, the scaled saturation end points are interpolated firstly according to:

$$\begin{aligned}
 S_{WL}^{IMM} &= F_W S_{WL}^H + (1 - F_W) S_{WL}^L \\
 S_{WCR}^{IMM} &= F_W S_{WCR}^H + (1 - F_W) S_{WCR}^L \\
 S_{WU}^{IMM} &= F_W S_{WU}^H + (1 - F_W) S_{WU}^L \\
 S_{OWCR}^{IMM} &= F_W S_{OWCR}^H + (1 - F_W) S_{OWCR}^L \\
 S_{GL}^{IMM} &= F_W S_{GL}^H + (1 - F_W) S_{GL}^L
 \end{aligned}
 \tag{Eq. 6.145}$$

and these are then combined with the miscible scaled end-points according to:

$$\begin{aligned}
 S_{WL}^I &= F_m S_{WCO}^{MIS} + (1 - F_m) S_{WL}^{IMM} \\
 S_{WCR}^I &= F_m S_{WCR}^{MIS} + (1 - F_m) S_{WCR}^{IMM} \\
 S_{WU}^I &= F_m S_{WMAX}^{MIS} + (1 - F_m) S_{WU}^{IMM} \\
 S_{OWCR}^I &= F_m S_{OWCR}^{MIS} + (1 - F_m) S_{OWCR}^{IMM} \\
 S_{GL}^I &= F_m S_{GL}^{MIS} + (1 - F_m) S_{GL}^{IMM}
 \end{aligned}
 \tag{Eq. 6.146}$$

where

S_{WL} is the scaled minimum water saturation.

S_{WCR} is the scaled critical water saturation.

S_{WU} is the scaled maximum water saturation.

S_{OWCR} is the scaled critical oil saturation in water.

S_{GL} is the scaled minimum gas saturation

superscripts L , H denote the immiscible low and high concentrations of solid adsorption scaled end-points respectively.

superscripts IMM , MIS denote the immiscible interpolated and the miscible scaled saturation end-points respectively.

superscript I denotes the interpolation between the immiscible and miscible scaled saturation end-points.

If solid wettability effects are not being modeled ($F_w = 0$), the immiscible saturation functions are taken directly from the saturation tables defined by [SATNUM](#) and scaled according to the values of the set of end-points [SWL](#), [SWLPC](#), [SWCR](#), [SWU](#), [SGL](#), [SGLPC](#), [SGCR](#), [SGU](#), [SOWCR](#), [SOGCR](#), [KRW](#), [KRWR](#), [KRG](#), [KRGR](#), [KRO](#), [KRORW](#), [KRORG](#), [PCW](#) and [PCG](#). and hence no interpolation of the immiscible saturation end-points is required.

The capillary pressure and relative permeabilities are looked up in the immiscible saturation tables for the low and high concentrations of solid adsorption. For capillary pressure, only two-point scaling is employed. For the relative permeabilities, two-point or three-point end-point scaling is employed according to whether item 1 of the [SCALECRS](#) keyword has been set to NO or YES respectively. (Two-point scaling is selected by default if this keyword is omitted.)

The immiscible capillary pressures for low and high concentrations of solid adsorption are calculated according to:

$$\begin{aligned} P_{COW}^L &= f(S_W, S_{WL}^I, S_{WU}^I, P_{CW}^L) \\ P_{COW}^H &= f(S_W, S_{WL}^I, S_{WU}^I, P_{CW}^H) \end{aligned} \quad \text{Eq. 6.147}$$

For two-point scaling the immiscible relative permeabilities for low and high concentrations of solid adsorption are calculated according to:

$$\begin{aligned} K_{RW}^L &= f(S_W, S_{WCR}^I, S_{owcr}^I, S_{WU}^I, K_{RW}^L, K_{RWR}^L) \\ K_{RW}^H &= f(S_W, S_{WCR}^I, S_{owcr}^h, S_{WU}^I, K_{RW}^H, K_{RWR}^H) \\ K_{ROW}^L &= f(S_{OW}, S_{OWCR}^I, S_{wcr}^I, S_{OMAX}, K_{RO}^L, K_{RORW}^L) \\ K_{ROW}^H &= f(S_{OW}, S_{OWCR}^I, S_{wcr}^h, S_{OMAX}, K_{RO}^H, K_{RORW}^H) \end{aligned} \quad \text{Eq. 6.148}$$

For three-point scaling the immiscible relative permeabilities for low and high concentrations of solid adsorption are calculated according to::

$$\begin{aligned} K_{RW}^L &= f(S_W, S_{GL}^I, S_{WCR}^I, S_{OWCR}^I, S_{WU}^I, K_{RW}^L, K_{RWR}^L) \\ K_{RW}^H &= f(S_W, S_{GL}^I, S_{WCR}^I, S_{OWCR}^I, S_{WU}^I, K_{RW}^H, K_{RWR}^H) \\ K_{ROW}^L &= f(S_{OW}, S_{GL}^I, S_{OWCR}^I, S_{WCR}^I, S_{OMAX}, K_{RO}^L, K_{RORW}^L) \\ K_{ROW}^H &= f(S_{OW}, S_{GL}^I, S_{OWCR}^I, S_{WCR}^I, S_{OMAX}, K_{RO}^H, K_{RORW}^H) \end{aligned} \quad \text{Eq. 6.149}$$

where

S_W is the grid block water saturation.

S_{OW} is the oil saturation for oil relative permeability to water.

S_{wcr} is the saturation table derived critical water saturation.

S_{owcr} is the saturation table derived critical oil in water saturation.

S_{OMAX} is the scaled maximum oil saturation, $S_{OMAX} = (1 - S_{WL}^I - S_{GL}^I)$.

S_{GL} is the scaled minimum gas saturation.

K_{RW} is the scaled maximum water relative permeability.

K_{RWR} is the scaled water relative permeability at residual oil saturation.

K_{RO} is the scaled maximum oil relative permeability.

K_{RORW} is the scaled oil relative permeability at residual water saturation.

P_{CW} is the scaled maximum oil-water capillary pressure.

For two-point scaling the miscible saturation functions are looked up in the miscible saturation tables applying the set of scaled end-points shown below:

$$\begin{aligned} K_{RW}^{MIS} &= f(S_W, S_{WCR}^I, S_{owcr}^{mis}, S_{WU}^I, K_{RW}^{MIS}, K_{RWR}^{MIS}) \\ K_{ROW}^{MIS} &= f(S_{OW}, S_{OWCR}^I, S_{wcr}^{mis}, S_{OMAX}, K_{RO}^{MIS}, K_{RORW}^{MIS}) \end{aligned} \quad \text{Eq. 6.150}$$

For three-point scaling the miscible water relative permeability and the oil relative permeability to water are looked up in the miscible tables applying the set of scaled end-points shown below:

$$\begin{aligned} K_{RW}^{MIS} &= f(S_W, S_{GL}^I, S_{WCR}^I, S_{OWCR}^I, S_{WU}^I, K_{RW}^{MIS}, K_{RWR}^{MIS}) \\ K_{ROW}^{MIS} &= f(S_{OW}, S_{GL}^I, S_{OWCR}^I, S_{WCR}^I, S_{OMAX}, K_{RO}^{MIS}, K_{RORW}^{MIS}) \end{aligned} \quad \text{Eq. 6.151}$$

The immiscible relative permeabilities and capillary pressure are interpolated according to:

$$\begin{aligned} K_{RW}^{IMM} &= F_w K_{RW}^H + (1-F_w) K_{RW}^L \\ K_{ROW}^{IMM} &= F_w K_{ROW}^H + (1-F_w) K_{ROW}^L \\ P_{COW}^{IMM} &= F_w P_{COW}^H + (1-F_w) P_{COW}^L \end{aligned} \quad \text{Eq. 6.152}$$

The miscible and immiscible relative permeabilities and capillary pressure are calculated according to:

$$\begin{aligned} K_{RW}^I &= F_m K_{RW}^{MIS} + (1-F_m) K_{RW}^{IMM} \\ K_{ROW}^I &= F_m K_{ROW}^{MIS} + (1-F_m) K_{ROW}^{IMM} \\ P_{COW} &= F_{cp} P_{COW}^{IMM} \end{aligned} \quad \text{Eq. 6.153}$$

The gas saturation functions are assumed to be independent of solid wettability and miscible effects and are taken directly from the saturation tables defined via [SATNUM](#) and scaled according to the values of the set of end-points [SWL](#), [SWLPC](#), [SWCR](#), [SWU](#), [SGL](#), [SGLPC](#), [SGCR](#), [SGU](#), [SOWCR](#), [SOGCR](#), [KRW](#), [KRWR](#), [KRG](#), [KRGR](#), [KRO](#), [KRORW](#), [KRORG](#), [PCW](#) and [PCG](#).

Pre-2016.1 modeling

For versions prior to 2016.1, end-point scaling is only applied to the immiscible saturation functions and, if solid wettability modeling is specified, both low and high concentrations of solid adsorption are scaled using [SWL](#), [SWLPC](#), [SWCR](#), [SWU](#), [SGL](#), [SGLPC](#), [SGCR](#), [SGU](#), [SOWCR](#), [SOGCR](#), [KRW](#), [KRWR](#), [KRG](#), [KRGR](#), [KRO](#), [KRORW](#), [KRORG](#), [PCW](#) and [PCG](#), noting that the gas end-point scaling keywords are not required for scaling the high solid concentration saturation functions. Pre-2016.1 behavior may be restored by setting [item 334](#) of the [OPTIONS3](#) keyword to 1.

Using the Eclipse 100 surfactant model

The model is activated by specifying the keyword [SURFACT](#) in the RUNSPEC section.

It is possible to output the geometric data used by the surfactant model by setting the 'KOVERD' argument in [RPTGRID](#). This outputs a map of the K/D values used in the calculation of the capillary number.

In the PROPS section, there are three obligatory keywords and two further optional keywords for cases in which adsorption takes place:

Restriction	Keyword	Description
Required	SURFST (or SURFSTES)	Water-oil surface tension in the presence of surfactant (and salts).
Required	SURFVISC	Modified water viscosity.
Required	SURFCAPD	Capillary de-saturation data.
Optional	SURFADS	Adsorption isotherm.
If SURFADS is present	SURFROCK	Rock properties and adsorption model indicator.

The surfactant properties data can be written to the PRINT file by setting the 'SURE' argument in the [RPTPROPS](#) keyword.

The miscible relative permeability curves are specified using the [SURFNUM](#) keyword in conjunction with the usual saturation function keywords. If a non-zero capillary pressure is entered in the SWFN or SWOF table, this will be ignored and the capillary pressure will be taken as the appropriate fraction of the capillary pressure for the SATNUM table.

The injected surfactant concentration of a water injector is specified using the [WSURFACT](#) keyword. Output of the surfactant concentration, modified water viscosity, capillary number, oil-water surface tension, adsorbed concentration and fluid-in-place data can be controlled using arguments 'SURFBLK', 'SURFADS' and 'FIPSURF' in the [RPTSCHEd](#) keyword. In addition a field/group/well/connection summary can be obtained at each report time by adding the argument 'WELLS' to RPTSCHEd. An extra mode of controlling the timestep is available, which sets a target surfactant change in a timestep. The default is to have no control. This control can be set by specifying a target TRGSFT in record 2 of the [TUNING](#) keyword.

The list of SUMMARY keywords has been extended to output surfactant specific data. These keywords follow the passive tracer format with a keyword concatenated with the tracer name. In this case surfactant is referred to as SUR. For example the keyword for the field total surfactant production would be FTPRSUR.

Diffusion can be applied to a surfactant if the Environmental Tracer option is active (item 4 of the [TRACERS](#) keyword in the RUNSPEC section). This is done by using the TRDIFSUR keyword (that is the [TRDIF](#) keyword for the tracer called 'SUR' which is the surfactant).

Note: For some simulation models with substantial surfactant effects you may want to consider using the [CPR](#) solver or tightening the convergence criteria (record 2 of the [TUNING](#) keyword) in order to improve the convergence of the linear solver. In particular, this might be helpful for models with the Dual Porosity or Parallel options active.

Using partitioned tracers

Partitioned tracers allow a given tracer to dissolve in a secondary phase, as a function of pressure. With surfactants the primary phase is water but the secondary phase can be oil, say, and this feature can model the partitioning of the surfactant into the oil. See "[Tracer Tracking](#)" for a more detailed discussion.

Restrictions

If a cell is defined with zero pore volume in the GRID section, then making it active by changing the pore volume in the EDIT section using [PORV](#), is not allowed with the surfactant model.

Using the Eclipse 300 surfactant model

Surfactant properties

The [COMPW](#) keyword should be used to set at least two water components - one for water itself and one to represent surfactant. The [CWTYPE](#) keyword should be used to identify which water components behave as surfactants. The water properties are defined using the water component keywords, as described in "[Water PROPS keywords](#)". For example, the water viscosity can be set using the [VREFW](#) keyword.

Adsorption

Adsorbed surfactant can be modeled using the [SOLID](#) model. The [COMPS](#) or [COMPSOL](#) keyword should be used to set the number of non-water components (including solids). The [CVTYPE](#) keyword should be used to identify the solid component. The properties of adsorbed surfactant are defined using the solid keywords, see [4.10](#).

The adsorption and desorption processes can be modeled using the [REACTION](#) model where the properties of the reaction are defined using the reaction keywords, as described in "[Chemical reaction PROPS keywords](#)". The reaction keywords are flexible and you can define the reaction in various ways. One simple method is described in "[Adsorption of dissolved water components](#)". In this method the [STOREAC](#) and [STOPROD](#) keywords are used to define dissolved surfactant as the reactant and adsorbed surfactant as the product of the reaction. The [EQLDREAC](#) keyword (model 4) is used to define a reaction rate that depends on the dissolved-adsorbed equilibrium. The half-life for $(C_{aeq} - C_a)$ is:

$$T_{1/2} = \ln 2 / A_r \quad \text{Eq. 6.154}$$

where

A_r is a constant defined by the [REACRATE](#) keyword,

C_{aeq} is the equilibrium adsorbed solid concentration defined by [EQLDTAB](#),

C_a is the actual adsorbed solid concentration.

This reaction can be one way (forwards for adsorption or backwards for desorption) or two way.

If an Eclipse 100 model is converted to Eclipse 300, [EQLDTAB](#) data can be derived from the [SURFADS](#) data, which defines equilibrium concentrations C_{aeq} as functions of dissolved concentrations C_d . To convert the data it is necessary to replace the dissolved concentrations C_d with the surfactant mole fractions a_i defined by

$$a_i = \frac{MW_w}{\rho_{ws}} \cdot \frac{C_d}{MW_i}$$

where

MW_w and MW_i are the molecular weights of the water mixture and the surfactant component, and

ρ_{ws} is the surface density of water.

In Eclipse 100 the adsorption process is instantaneous, which could be modeled using a large reaction rate constant A_r . However this may make it hard to converge. A reasonable value for A_r can be determined by estimating a half-life and using equation 6.154. For example, the residence time of the surfactant in a cell can be determined from the pore volume and the flow rate $T_{res} \approx V_P/Q$, and the maximum rate of adsorption occurs when there are no adsorbed solids and when the surfactant concentration is at a maximum. If we set $A_r = \ln 2/T_{res}$ then we would expect about half the surfactant to be adsorbed in the first cell.

Decay

Surfactant decay can be modeled by using the [REACTION](#) model where the properties of the reaction are defined using the reaction keywords, as described in "[Chemical reaction PROPS keywords](#)". Surfactant could decay to water, or perhaps to another water component. The [STOREAC](#) and [STOPROD](#) keywords should be used to define surfactant as the reactant and water (or another component) as the product of the reaction. The reaction rate should be made proportional to the amount of surfactant using [REACCORD](#). The [REACRATE](#) keyword should be used to define a constant A_r for the decay rate. This can be determined from the decay half-life $A_r = \ln 2/T_{1/2}$.

Surfactant effects

The surfactant concentration is calculated in each cell by summing up the weight of the surfactant components and dividing by the volume of water at surface conditions. The surface tension σ_{ow} must be set using the [SURFST](#) keyword, as a function of the surfactant concentration, or [SURFSTES](#) keyword, as a function of the surfactant and salt concentrations. Note that the amount of adsorbed surfactant does not affect the surface tension. The surface tension allows the capillary number N_c to be calculated and determines the capillary pressure multiplier F_{cp} . The [SURFCAPD](#) keyword must be used to determine the effect of surface tension on the relative permeability F_{kr} .

The [SURFNUM](#) keyword must be used to identify the miscible relative permeability tables. Capillary pressures defined by these tables are not used - it is assumed that the miscible capillary pressure is zero.

For cells with low water saturation, a small amount of surfactant will have a high concentration in the water phase, and could therefore have a large effect on the capillary pressure. To prevent numerical errors having an effect, the surfactant concentration in the water phase is set to zero for cells with very small water saturations. Therefore there will be no surfactant effect in cells with water saturations below a threshold. The molar concentrations are unchanged, so the component balance equations are not affected by this threshold. The threshold value can be set using the [SURFOPTS](#) keyword.

Wettability

The effects on wettability of adsorbed surfactant can be modeled by creating two sets of saturation functions, for low and high concentrations of adsorbed solids. The [SATNUM](#) and [SOLWNUM](#) keywords are

used to identify the saturation functions corresponding to these two extremes and the [SOLWTAB](#) keyword is used to determine an interpolation parameter dependent on the adsorbed solid concentration.

Injection

The water composition of an injection stream can be defined using the [WELLSTRW](#) keyword, and associated with an injection well using the [WINJW](#) keyword.

Output

Properties of the water, adsorbed surfactant and the adsorption / desorption processes can be output using the SUMMARY vectors and other output keywords described in "[Water component keywords](#)", "[Solid keywords](#)" and "[Chemical reaction keywords](#)". In addition surfactant specific keywords can be used to output the surfactant concentration C_{surf} , surface tension σ_{ow} , capillary number N_c , capillary pressure multiplier F_{cp} and the miscible / immiscible relative permeability interpolation parameter F_{kr} . See "[Eclipse 300 Surfactant SUMMARY output controls](#)" and "[Eclipse 300 Surfactant output controls](#)".

Summary of Eclipse 100 keywords

RUNSPEC

Keyword	Description
PARTTRAC	Activates and dimensions the Partitioned Tracer option.
SURFACT	Activates the Surfactant Model
SURFACTW	Activates the Surfactant Model and enables modeling of changes of wettability.

GRID

Keyword	Description
RPTGRID	Controls output from the GRID section. Argument KOVERD outputs the K/D values used in the calculation of the capillary number.

PROPS

Keyword	Description
HWSWL , HWSWLPC , HWSWCR , HWSWU , HWSOWCR , HWSOGCR , HWKRW , HWKRWR , HWKRO , HWKRORW , HWKRORG and HWPCW	Immiscible water-wet saturation end-points.
RPTPROPS	Output of surfactant properties from the PROPS section (argument SURF).
SOCRS	Scaled critical oil-in-water saturations at miscible conditions.
SURFADDW	Defines weighting between oil-wet and water-wet relative permeabilities as a function of the adsorbed surfactant mass.
SURFADS	Surfactant adsorption isotherm.

Keyword	Description
ADSORP	Analytical adsorption isotherm with salinity dependence.
SURFCAPD	Surfactant capillary de-saturation data.
SURFROCK	Surfactant-rock properties and adsorption model indicator.
SURFST	Water-oil surface tension in the presence of surfactant.
SURFSTES	Water-oil surface tension as a function of surfactant and salt concentrations.
SURFVISC	Modified water viscosity.
SWL, SWLPC, SWCR, SWU, SGL, SGLPC, SGCR, SGU, SOWCR, SOGCR, KRW, KRWR, KRG, KRGR, KRO, KRORW, KRORG, PCW and PCG	Immiscible oil-wet saturation end-points.
TRACER	Defines the tracers (for the Partitioned Tracer option).
TRACERKP	Defines the functions K(P) for the partitioned tracers.

REGIONS

Keyword	Description
HWSNUM	Abbreviated alias for SURFWNUM.
SATNUM	Defines the region number for oil-wet immiscible saturation function table.
SURFNUM	Surfactant miscible region numbers.
SURFWNUM	Defines the region number for water-wet immiscible saturation function tables and additionally the table describing the wettability as a function of surfactant adsorption (SURFADDW).
TRKPFxxx	Specifies the partitioning function for tracer xxx for each grid block ('xxx' = 'SUR' for surfactant).

SOLUTION

Keyword	Description
RPTSOL	Controls output from the SOLUTION section. - SOGR outputs the critical oil saturations following a surfactant flood. - SURFBLK outputs the surfactant concentration, the modified water viscosity and the capillary number as well as water-oil surface tension (both for salinity dependent and salinity independent surface tension).
SURF	Initial surfactant concentrations.
TVDP	Initial surfactant concentrations with depth
TNUM	Surfactant concentration regions.

SUMMARY

Refer to [Surfactant model](#) in the *Eclipse Reference Manual*.

SCHEDULE

Keyword	Description
RPTSCHED	Controls output from the SCHEDULE section. Arguments: - FIPSURF: surfactant in place. - SURFADS: adsorbed surfactant concentration. - SURFBLK: surfactant concentrations, modified water viscosity and the capillary number as well as water-oil surface tension.
RPTRST	Controls output to the restart file. Mnemonic VISC can be requested to output - EWV_SUR: effective water viscosity due to surfactant. If a graphics only restart is required (in other words, when mnemonic NORST is requested) then adding mnemonic SURFBLK will output: - SURFACT: surfactant concentration. - SURFADS: adsorbed surfactant concentration. - SURFMAX: maximum adsorbed surfactant concentration. - SURFCNM: (log) capillary number (for miscibility transition). - SURFST or (SURFSTES): water-oil surface tension On the other hand, for full restarts (not graphics only) above mentioned arrays will be output by default.
TUNING	Sets simulator control parameters. Item 11 of Record 2 (TRGSFT) sets the target surfactant change.
WSURFACT	Specifies the surfactant concentration of a water injector.

Summary of Eclipse 300 keywords

The surfactant component and water properties must be defined by water component keywords. See "[Water component keywords](#)".

A solid component can be used to represent adsorbed surfactant. See "[Solid phase](#)".

Adsorption and desorption could then be modeled using a pair of chemical reactions. See "[Chemical reactions](#)".

PROPS

Restriction	Keyword	Description
	SURFCAPD	Surfactant capillary de-saturation data.
	SURFOPTS	Surfactant model options.
	SURFST	Water-oil surface tension in the presence of surfactant.
	SURFSTES	Salinity dependent water-oil surface tension in the presence of surfactant.
Required if salinity dependent surfactant model is activated.	ESSNODE	Salt concentration nodes for oil-water surface tension.
Required if salinity dependent surfactant model is activated.	SALTEFF	Effective salinity coefficients.
Required to model the effect of absorbed surfactant on the wettability of the rock.	SOLWTAB	Solid wettability table.
Available if the ENDSCALE keyword is specified.	SWL , SWLPC , SWCR , SWU , SGL , SGLPC , SGCR , SGU , SOWCR , SOGCR , KRW , KRWR , KRG , KRGR , KRO , KRORW , KRORG , PCW and PCG	Immiscible (low concentration of solid adsorption) end-point scaling keywords.
Available if the ENDSCALE and SOLWTAB keywords are specified.	SASWL , SASWCR , SASWU , SASGL , SASOWCR , SAKRW , SAKRWR , SAKRO , SAKRORW and SAPCW	Immiscible high concentration of solid adsorption end-point scaling keywords
Available if the ENDSCALE keyword is specified.	SSWL , SSWCR , SSWU , SSGL , SSOWCR , SKRW , SKRWR , SKRO and SKRORW	Miscible end-point scaling keywords.

Regions

The keywords are:

- [SURFNUM](#) Surfactant miscible region numbers.
- [SOLWNUM](#) Solid wettability region numbers.

SUMMARY

Refer to [Surfactant model](#) in the *Eclipse Reference Manual*.

Other Reporting

The following arguments can be used with the [RPTSOL](#), [RPTSCHED](#), [RPTRST](#) keywords to output grid data specific to the Surfactant Model:

Mnemonic	Description
SURF	Concentration C_{surf} measured as the mass of surfactant per volume of water phase.
SURFST	Oil water surface tension σ_{ow}
SURFCNM	$\log_{10}$ (capillary number)
SURFKR	Relative permeability interpolation factor F_{kr}
SURFCP	Capillary pressure multiplier F_{cp}
ESALTS	Effective salinity of the water phase in presence of surfactant.

Eclipse 100 example

Example 1

This is a 2D example modeling water injection into oil. To aid recovery, a surfactant slug is injected into the reservoir some time after the primary water drive. Water injection is resumed after the surfactant injection period. The performance of the surfactant drive is assumed not to be salt sensitive in this study.

RUNSPEC

```

RUNSPEC
TITLE
Example Surfactant Flood Case.

DIMENS
5      5      1  /

NONNC
OIL
WATER
SURFACT
FIELD

TRACERS
0      1      0      0 'DIFF'  /

TABDIMS
2      2      20      5      6      20 /

REGDIMS
6      1      0      0      0      1 /

WELLDIMS
2      1      1      2  /

START
1 'JAN' 1983  /

NSTACK
8  /

```

GRID

```

GRID =====
DXV
5*75.0  /

PERMX
25*50.0  /

PERMY

```

```

25*50.0 /
DZ
25*30.0 /
TOPS
25*4000.0 /
PERMY
25*50.0 /
PORO
25*0.2 /
RPTGRID
'DX' 'DY' 'DZ' 'PERMX' 'PERMY' 'PERMZ' 'MULTX'
'MULTY' 'MULTZ' 'PORO' 'NTG' 'TOPS' 'PORV'
'DEPTH' 'TRANX' 'TRANY' 'TRANZ' 'COORDSYS'
'COORD' 'ZCORN' 'NNC' 'AQNUM' 'AQCON'
'ALLNNC' 'SIGMAV' 'DZMTRXV' 'DIFFMX' 'DIFFMY'
'DIFFMZ' 'DIFFMMF' 'DIFFX' 'DIFFY' 'DIFFZ'
'KOVERD' /

```

PROPS

```

PROPS =====
TRACER
'ONE' 'WAT' /
/
SWFN
-- Table 1 - Immiscible curves - SATNUM
.25 .0 4.0
.7 1.0 .0
/
-- Table 2 - Miscible curves - SURFNUM
.25 .0 4.0
.99 1.0 .0
/
SOF2
-- Table 1 - Immiscible curves - SATNUM
.3000 .0000
.7600 1.0000
/
-- Table 2 - Miscible curves - SURFNUM
.0100 .0000
.7600 1.0000
/
PVTW
4000.0 1.0 3.03E-06 .5 0.0 /
4000.0 1.0 3.03E-06 .5 0.0 /
PVDO
.0 1.0 2.0
8000.0 .92 2.0
/
.0 1.0 2.0
8000.0 .92 2.0
/
ROCK
4000.0 .30E-05 /
4000.0 .30E-05 /
DENSITY
52.0000 64.0000 .04400 /
52.0000 64.0000 .04400 /

```

```

-- Water Viscosity vs surfactant concentration tables.
SURFVISC
-- Surfactant      Water viscosity at 4000 psi
-- concentration
-- LB/STB          CENTIPOISE
-- 0.0              0.5
-- 0.1              1.1
-- 0.2              1.2
-- 0.8              1.3 /
-- 0.0              0.5
-- 0.1              0.6
-- 0.2              0.7
-- 0.8              0.8 /

-- Surfactant Adsorption tables
SURFADS
-- Surfactant      Adsorbed mass
-- concentration
-- LB/STB          LB/LB
-- 0.0              0.00000
-- 0.1              0.00005
-- 0.2              0.00005
-- 1.0              0.00005 /
-- 0.0              0.00000
-- 0.1              0.00005
-- 0.2              0.00005
-- 1.0              0.00005 /

-- Water/Oil surface tension vs Surfactant concentration
SURFST
-- Surfactant      Water/Oil Surface Tension
-- concentration
-- LB/STB          LB/IN
-- 0.0              5.0E-04
-- 0.1              6.34E-05
-- 0.2              2.23E-05
-- 0.5              2.07E-06/
-- 0.0              5.0E-04
-- 0.1              6.34E-05
-- 0.2              2.23E-05
-- 0.5              2.07E-06/

-- Capillary de-saturation curve
SURFCAPD
-- LOG10 (Capillary Number)  Miscibility
--                            Function
-- -9                          0.0
-- 2                            0.0
-- 5                            1.0
-- 10                           1.0 /
-- -10                          0.0
-- 2                            0.0
-- 5                            1.0
-- 10                           1.0 /

-- Extra rock properties.
SURFROCK
-- De-adsorption model      Rock mass
-- indicator                 Density
-- 1 - reversible
-- 2 - irreversible          LB/RB
-- 1                          1000 /
-- 1                          1000 /

RPTPROPS
'SURFVISC' /

```

REGIONS

```

=====
REGIONS
FIPNUM
1 2 5 6 6
3 4 5 6 6

```

```

5 5 5 6 6
6 6 6 6 6
6 6 6 6 6 /

-- Immiscible table numbers
SATNUM
25*1 /

-- Miscible table numbers
SURFNUM
25*2 /

RPTREGS
'PVTNUM' 'SATNUM' 'EQLNUM' 'FIPNUM' 'KRUNUMX'
'KRUNUMX-' 'KRUNUMY' 'KRUNUMY-' 'KRUNUMZ' 'KRUNUMZ-'
'IMBNUM' 'IMBNUMX' 'IMBNUMX-' 'IMBNUMY' 'IMBNUMY-'
'IMBNUMZ' 'IMBNUMZ-' 'ROCKNUM' 'ENDNUM' 'TNUM'
'MISCNUM' 'KRUNUMF' 'IMBNUMMF' 'SURFNUM' /

```

SOLUTION

```

SOLUTION =====
TBLKFOE
25*0 /

EQUIL
4000 4000 6000 0 0 0 0 0 0 /

RPTSOL
'PRES' 'SOIL' 'SWAT' 'RESTART' 'OILAPI'
'FIPTR=2' 'TBLK' 'FIPLY' 'SURFBLK'
'FIPSURF=2' /

```

SUMMARY

```

SUMMARY =====
CTFRSUR
'P' 5 5 1 /
/

FTPRSUR

WTPRSUR
'P' /

FTPTSUR

WTPTSUR
'P' /

FTITSUR

WTITSUR
'I' /

BTCNFSUR
1 1 1 /
2 2 1 /
5 5 1 /
/

FTIPTSUR
FTIPFSUR

RTIPTSUR
1 /

RTIPFSUR
1 /

BTIPTSUR

```

```

1 1 1 /
/

FTADSUR

RTADSUR
1 /

BTADSUR
1 1 1 /
/

BTCASUR
1 1 1 /
/

```

SCHEDULE

```

=====
SCHEDULE
RPTSCHED
'PRES' 'SOIL' 'SWAT' 'RESTART=2' 'FIP=2'
'WELLS=2' 'SUMMARY=2' 'CPU=2' 'NEWTON=2'
'OILAPI' 'FIPTR=2' 'TBLK' 'FIPSALT=2'
'SURFBLK' 'SURFADS' 'FIPSURF=2' /

WELSPECS
'I' 'G' 1 1 4000 'WAT' /
'P' 'G' 5 5 4000 'OIL' /
/

COMPDAT
'I' ' ' 1 1 1 1 'OPEN' 0 .0 1.0 /
'P' ' ' 5 5 1 1 'OPEN' 0 .0 1.0 /
/

WCONPROD
'P' 'OPEN' 'BHP' 5* 3999.0 /
/

WCONINJE
'I' 'WAT' 'OPEN' 'RATE' 200.0 /
/

WTRACER
'I' 'ONE' 0.8 /
/

-- Start injecting the Surfactant Slug
WSURFACT
'I' 0.8 /
/

TUNING
1 /
10* 0.4 /
/

TSTEP
2*100
/

WTRACER
'I' 'ONE' 0.0 /
/

-- Water chase
WSURFACT
'I' 0.0 /
/

TSTEP
5*100 /

```

```
END
```

Example 2

This is a version of the previous example with the salt option activated to study the effect of the interaction between the injected surfactant and the salt.

RUNSPEC

```
RUNSPEC
TITLE
Example Surfactant Flood Case.

DIMENS
5      5      1  /

NONNC
OIL
WATER
BRINE
SURFACT
FIELD

TRACERS
0      1      0      0 'DIFF'  /

TABDIMS
2      2      20     5      6      20  /

REGDIMS
6      1      0      0      0      1  /

WELLDIMS
2      1      1      2  /

START
1 'JAN' 1983  /

NSTACK
8  /
```

GRID

```
GRID =====
DXV
5*75.0  /

PERMX
25*50.0  /

PERMY
25*50.0  /

DYV
5*75.0  /

DZ
25*30.0  /

TOPS
25*4000.0  /

PERMY
25*50.0  /

PORO
25*0.2  /
```

```

RPTGRID
'DX' 'DY' 'DZ' 'PERMX' 'PERMY' 'PERMZ' 'MULTX'
'MULTY' 'MULTZ' 'PORO' 'NTG' 'TOPS' 'PORV'
'DEPTH' 'TRANX' 'TRANY' 'TRANZ' 'COORDSYS'
'COORD' 'ZCORN' 'NNC' 'AQNUM' 'AQCON'
'ALLNNC' 'SIGMAV' 'DZMTRXV' 'DIFFMX' 'DIFFMY'
'DIFFMZ' 'DIFFMME' 'DIFFX' 'DIFFY' 'DIFFZ'
'KOVERD' /

```

PROPS

```

PROPS =====
TRACER
'ONE' 'WAT' /
/

SWFN
-- Table 1 - Immiscible curves - SATNUM
.25 .0 4.0
.7 1.0 .0
/
-- Table 2 - Miscible curves - SURFNUM
.25 .0 4.0
.99 1.0 .0
/

SOF2
-- Table 1 - Immiscible curves - SATNUM
.3000 .0000
.7600 1.0000
/
-- Table 2 - Miscible curves - SURFNUM
.0100 .0000
.7600 1.0000
/

-- brine properties
PVTWSALT
-- Pref Cref
2700 0 /
-- SaltConc Bw Cw Vw Cvw
-- 1/bar CPOISE 1/psi
0 1.030 4.6E-5 0.34 0.0
20 1.000 4.6E-5 0.34 0.0
30 0.98 4.6E-5 0.34 0.0
50 0.95 4.6E-5 0.34 0.0 /
3000 /
-- SaltConc Bw Cw Vw Cvw
-- 1/bar CPOISE 1/psi
0 1.030 4.6E-5 0.34 0.0
20 1.000 4.6E-5 0.34 0.0
30 0.98 4.6E-5 0.34 0.0
50 0.95 4.6E-5 0.34 0.0 /

PVDO
.0 1.0 2.0
8000.0 .92 2.0
/
.0 1.0 2.0
8000.0 .92 2.0
/

ROCK
4000.0 .30E-05 /
4000.0 .30E-05 /

DENSITY
52.0000 64.0000 .04400 /
52.0000 64.0000 .04400 /

-- Water Viscosity vs surfactant concentration tables.
SURFVISC
-- Surfactant Water viscosity at 4000 psi

```



```

-- concentration
-- LB/STB          CENTIPOISE
    0.0             0.34
    0.1             1.1
    0.2             1.2
    0.8             1.3 /
    0.0             0.34
    0.1             0.6
    0.2             0.7
    0.8             0.8 /

-- Surfactant Adsorption tables
SURFADS
-- Surfactant      Adsorbed mass
-- concentration
-- LB/STB          LB/LB
    0.0             0.00000
    0.1             0.00005
    0.2             0.00005
    1.0             0.00005 /
    0.0             0.00000
    0.1             0.00005
    0.2             0.00005
    1.0             0.00005 /

-- Water/Oil surface tension vs Surfactant and salt concentrations
SURFSTES
-- surf            water-oil
-- conc            surface tension
-- LB/STB          (LB/IN)
0.0      5.0E-04
          5.0E-04
          5.0E-04 /
0.7      2.23E-05
          2.12E-05
          2.00E-05 /
3.5      2.07E-06
          1.79E-06
          1.47E-06 /
4.9      1.04E-06
          8.76E-07
          7.18E-07 /
7.0      4.95E-07
          4.16E-07
          3.45E-07 /
/
/

-- Salt concentration nodal values for SURFSTES keyword
ESSNODE
0      10      30 /
/

-- Capillary de-saturation curve
SURFCAPD
-- LOG10(Capillary Number)      Miscibility
--                               Function
--                               0.0
--                               0.0
--                               1.0
--                               1.0 /
--                               0.0
--                               0.0
--                               1.0
--                               1.0 /

-- Extra rock properties.
SURFROCK
-- De-adsorption model          Rock mass
-- indicator                    Density
-- 1 - reversible
-- 2 - irreversible            LB/RB
    1                          1000 /
    1                          1000 /

```

```
RPTPROPS
'SURFVISC' /
```

REGIONS

```
REGIONS =====
FIPNUM
1 2 5 6 6
3 4 5 6 6
5 5 5 6 6
6 6 6 6 6
6 6 6 6 6 /

-- Immiscible table numbers
SATNUM
25*1 /

-- Miscible table numbers
SURFNUM
25*2 /

RPTREGS
'PVTNUM' 'SATNUM' 'EQLNUM' 'FIPNUM' 'KRNUMX'
'KRNUMX-' 'KRNUMY' 'KRNUMY-' 'KRNUMZ' 'KRNUMZ-'
'IMBNUM' 'IMBNUMX' 'IMBNUMX-' 'IMBNUMY' 'IMBNUMY-'
'IMBNUMZ' 'IMBNUMZ-' 'ROCKNUM' 'ENDNUM' 'TNUM'
'MISCNUM' 'KRNUMMF' 'IMBNUMMF' 'SURFNUM' /
```

SOLUTION

```
SOLUTION =====
TBLKFONE
25*0 /

EQUIL
4000 4000 6000 0 0 0 0 0 0 /

SALTVD
2000 0
4000 0 /

RPTSOL
'PRES' 'SOIL' 'SWAT' 'RESTART' 'OILAPI'
'FIPTR=2' 'TBLK' 'FIPPLY' 'SURFBLK'
'FIPSURF=2' /
```

SUMMARY

```
SUMMARY =====
CTFRSUR
'P' 5 5 1 /
/

FTPRSUR

WTPRSUR
'P' /

FTPTSUR

WTPTSUR
'P' /

FTITSUR

WTITSUR
'I' /

BTCNFSUR
```

```

1 1 1 /
2 2 1 /
5 5 1 /
/

FTIPTSUR
FTIPFSUR

RTIPTSUR
1 /

RTIPFSUR
1 /

BTIPTSUR
1 1 1 /
/

FTADSUR

RTADSUR
1 /

BTADSUR
1 1 1 /
/

BTCASUR
1 1 1 /
/

BTSTSUR
1 1 1 /
/

```

SCHEDULE

```

=====
SCHEDULE
RPTSCHED
'PRES' 'SOIL' 'SWAT' 'RESTART=2' 'FIP=2'
'WELLS=2' 'SUMMARY=2' 'CPU=2' 'NEWTON=2'
'OILAPI' 'FIPTR=2' 'TBLK' 'FIPSALT=2'
'SURFBLK' 'SURFADS' 'FIPSURF=2' /

WELSPEDS
'I' 'G' 1 1 4000 'WAT' /
'P' 'G' 5 5 4000 'OIL' /
/

COMPDAT
'I' 1 1 1 1 'OPEN' 0 .0 1.0 /
'P' 5 5 1 1 'OPEN' 0 .0 1.0 /
/

WCONPROD
'P' 'OPEN' 'BHP' 5* 3999.0 /
/

WCONINJE
'I' 'WAT' 'OPEN' 'RATE' 200.0 /
/

WSALT
'I' 0.6 /
/

WTRACER
'I' 'ONE' 0.8 /
/

-- Start injecting the Surfactant Slug
WSURFACT
'I' 0.8 /

```

```
/
TUNING
1 /
10* 0.4 /
/

TSTEP
2*100
/

WTRACER
'I' 'ONE' 0.0 /
/

-- Water chase
WSURFACT
'I' 0.0 /
/

TSTEP
5*100 /

END
```

Eclipse 300 example

Example 1

This is a 3D example modeling water injection into oil. To aid recovery, a surfactant slug is injected into the reservoir some time after the primary water drive. The performance of the surfactant drive is assumed not to be salt sensitive in this study.

RUNSPEC

A solid component (to model adsorbed surfactant) and two water components (to model water and dissolved surfactant) are used, together with oil and gas components. A single reversible reaction is used (to model adsorption and desorption). The simulation is run fully implicitly (because reactions are used).

```
RUNSPEC
TITLE
Example E300 Surfactant - blackoil

DIMENS
10 10 3 /

BLACKOIL
OIL
GAS
WATER
SOLID
COMSOL
3 /
COMPW
2 /

REACTION
1 1 1* 1* 1 /

TABDIMS
2 /

METRIC

START
1 'JAN' 2012 /
```

```
FULLIMP
```

GRID

```
GRID =====
TOPS
100*2600 /
DXV
10*50.0 /
DYV
10*50.0 /
DZ
100*0.58 100*0.84 100*0.47 /

PERMX
100*4500 100*3300 100*2400 /

PERMY
100*4500 100*3300 100*2400 /

PERMZ
100*1050 100*1800 100*500 /

PORO
300*0.25 /
```

PROPS

```
PROPS =====

SWFN
-- Table 1 - Immiscible curves - SATNUM
.145 .0000 .75
.220 .0001 .05
.270 .0004 1*
.320 .009 1*
.365 .018 1*
.438 .043 1*
.510 .082 1*
.558 .118 1*
.631 .187 -.05
.703 .270 -.25
.752 .310 -.50
.795 .370 -1.50
/
-- Table 2 - Miscible curves - SURFNUM
0.0 0.0 0.0
1.0 1.0 0.0
/

SGFN
0.00 0.000 0
0.02 0.000 0
0.05 0.005 0
0.12 0.025 0
0.20 0.075 0
0.25 0.125 0
0.30 0.190 0
0.40 0.410 0
0.45 0.600 0
0.50 0.720 0
0.60 0.870 0
0.70 0.940 0
0.85 0.980 0
1.00 1.000 0
/
/

SOF3
-- Table 1 - Immiscible curves - SATNUM
```

```

.000 .0000 .0000
.205 .0000 .0000
.250 .00006 .00006
.300 .0009 .0009
.370 .0080 .0080
.440 .0270 .0270
.490 .0520 .0520
.560 .1200 .1200
.635 .2280 .2280
.683 .3340 .3340
.730 .4700 .4700
.780 .6440 .6440
.855 1.000 1.0000
/
-- Table 2 - Miscible curves - SURFNUM
.205 .0000 .0000
.250 1* .00006
.300 1* .0009
.370 1* .0080
.440 1* .0270
.490 1* .0520
.560 1* .1200
.635 1* .2280
.683 1* .3340
.730 1* .4700
.780 1* .6440
.855 1* 1.0000
/
PVT0
0 1 1.000 1.24 /
60 220 1.301 1.24
400 1.299 1.24 /
100 300 1.401 1.24
600 1.399 1.24 /
/
PVDG
1.0 10.0 0.01
400.0 1.0 0.01
/
ROCK
270.0 .30E-05 /
DENSITY
850.0 1000.0 1.0 /

```

Water properties

Here the water component and the surfactant component are given the same density properties, so that the surfactant acts like a tracer. However the surfactant component viscosity is increased so that when the surfactant molar fraction is 3% the water phase viscosity is increased to 5cp.

```

--Water and dissolved surfactant properties
WNAMEs
WATER SURF /
CWTYPE
1* SURFS
MWW
18.015 18.015 /
PREFW
270 270 /
DREFW
970.9 970.9 /
CREFW
4.6E-5 4.6E-5 /
VREFW
0.34 155.7
0 0 /

```

Solid properties

The adsorbed surfactant density properties are set equal to the dissolved surfactant values, so that the adsorption / desorption processes have no effect on pressure.

```
--Adsorbed surfactant properties
CNAMEs
OIL GAS ASURF /
CVTYPE
1* 1* SOLID /
MW
1* 1* 18.015 /
SDREF
1* 1* 970.0 /
```

Adsorption reactions

The `EQLDREAC` keyword model 4 is used to provide a reversible reaction between the solid component (component 3 in this example) and the water component (component 5 in this example). The water component must be defined as a reactant (`STOREAC`) and the solid component must be defined as a product (`STOPROD`) to ensure that the solid component is produced when the reaction rate is positive and the water component is produced when the reaction rate is negative.

```
--Adsorption reactions
STOREAC
--O G A W S
  0 0 0 0 1 / Dissolved Surfactant <-> Adsorbed Surfactant
STOPROD
--O G A W S
  0 0 1 0 0 / Dissolved Surfactant <-> Adsorbed Surfactant
REACRATE
1 /
EQLDREAC
1 4 5 0 1 BOTH 1 / Rate proportional to EQLDTAB_1(S) - Ads Solid
EQLDTAB
0.0      0.00000
0.001    0.0005
1.000    0.0005 /
```

Surfactant properties

```
-- Water/Oil surface tension vs Surfactant concentration
SURFST
-- Surfactant      Water-Oil Surface Tension
-- concentration
-- Kg/SM3          N/M
  0.0              0.05
  1.0              1.0e-6
  3.0              1.0e-6 /

-- Capillary de-saturation curve
SURFCAPD
-- LOG10(Capillary Number)  Miscibility
--                          Function
  -9                        0.0
  -4.5                      0.0
  -2                        1.0
  10                       1.0 /
  -9                        0.0
  -4.5                      0.0
  -2                        1.0
  10                       1.0 /
```

REGIONS

```
REGIONS =====
```

```
-- Immiscible table numbers
SATNUM
300*1 /

-- Miscible table numbers
SURENUM
300*2 /
```

SOLUTION

```
SOLUTION =====

EQUIL
2600 270 2700 0 0 0 1 /
WMFVD
2600 1.0 0.0
2700 1.0 0.0 /
RSVD
2600 0.0
2700 0.0 /

RPTSOL
'PRES' 'SOIL' 'SWAT' 'SSOLID' 'VWAT'
'SURF' 'SURFST' 'SURFCNM' 'SURFKR' 'SURFCP' /
```

SUMMARY

```
SUMMARY =====

BSURF
 1 1 1 /
 5 5 1 /
10 10 1 /
/

BSURFST
 1 1 1 /
 5 5 1 /
10 10 1 /
/

BSURFCNM
 1 1 1 /
 5 5 1 /
10 10 1 /
/

BSURFKR
 1 1 1 /
 5 5 1 /
10 10 1 /
/

BSURFCP
 1 1 1 /
 5 5 1 /
10 10 1 /
/

BKRO
 1 1 1 /
 5 5 1 /
10 10 1 /
/

BKRW
 1 1 1 /
 5 5 1 /
10 10 1 /
/
```



```

BWPC
  1  1  1 /
  5  5  1 /
 10 10  1 /
/

```

SCHEDULE

```

SCHEDULE =====
RPTSCHED
'PRES' 'SOIL' 'SWAT' 'SSOLID' 'VWAT'
'SURF' 'SURFST' 'SURFCNM' 'SURFKR' 'SURFCP' /
RPTRST
'RESTART' 'VWAT'
'SURF' 'SURFST' 'SURFCNM' 'SURFKR' 'SURFCP' /

WELSPECS
'I' 'G' 1 1 2600 'WAT' /
'P' 'G' 10 10 2600 'OIL' /
/

COMPDAT
'I' 1 1 1 3 'OPEN' 0 .0 .157 /
'P' 10 10 1 3 'OPEN' 0 .0 .157 /
/

WCONPROD
'P' 'OPEN' 'RESV' 4* 500 0.0 /
/

WELLSTRW
WATER 1.0 0.0 /
SURF 0.97 0.03 /
/

WINJW
I STREAM WATER /
/

WCONINJE
'I' 'WAT' 'OPEN' 'RESV' 1* 500.0 /
/

TSTEP
3*50 /

WINJW
I STREAM SURF /
/

TSTEP
3*50 /

END

```

Example 2

This is a compositional simulation with multicomponent water with salinity-dependent Surfactant option activated to study the effect of the interaction between the injected surfactant and the effective salinity of the water phase. Furthermore, alkaline represented as one of the water components is injected to study the efficacy of surfactant flooding.

RUNSPEC

A solid component (to model adsorbed surfactant) and two water components (to model water and dissolved surfactant) are used, together with oil and gas components. A single reversible reaction is used (to model adsorption and desorption). The simulation is run fully implicitly (because reactions are used).

```
RUNSPEC
TITLE
  E300 effective salinity-sensitive Surfactant model test case - compositional AIM

DIMENS
  10  10  3  /

OIL
GAS
WATER

COMPW
5 /
COMPS
8 /

METRIC

TABDIMS
2 /

AIM

START
  1 'JAN' 2012  /
```

GRID

```
GRID =====
TOPS
  100*2600 /
DXV
  10*50  /
DYV
  10*50  /
DZ
  100*0.58 100*0.84 100*0.47  /

PERMX
  100*4500 100*3300 100*2400 /
PERMY
  100*4500 100*3300 100*2400 /
PERMZ
  100*1050 100*1800 100*500  /
PORO
  300*0.25 /
ROCKDEN
  300*2650 /
```

PROPS

```
PROPS =====

SWFN
-- Sw      krw      Pc
0.20000    0.00000    6.00000
0.25000    0.00000    5.00000
0.30000    0.00000    4.00000
0.35000    0.00000    3.00000
0.40000    0.04000    2.40000
0.45000    0.06000    1.90000
0.50000    0.08000    1.40000
0.55000    0.11000    1.00000
0.60000    0.15000    0.70000
0.65000    0.20000    0.40000
0.70000    0.25000    0.25000
0.75000    0.32000    0.10000
0.80000    0.40000    0.08000
0.85000    0.50000    0.06000
0.90000    0.62000    0.04000
```

```

0.95000 0.80000 0.02000
1.00000 1.00000 0.00000
/
-- miscible
-- Sw      krw      Pc
0.00000 0.00000 0.00000
1.00000 1.00000 0.00000
/

SGFN
-- Sg      Krg      Pcog
0.000 0 0
0.400 0 0
0.450 0.066 0.80
0.500 0.177 1.56
0.550 0.40 2.24
0.600 0.75 3
/
/

SOF3
-- So      krow      krog
0.0 0 0
0.1 0 0
0.2 0 0
0.25 0 0.01
0.30 0.11 0.02
0.35 0.15 0.03
0.40 0.2 0.04
0.45 0.25 0.08
0.50 0.32 0.14
0.55 0.4 0.225
0.60 0.5 0.33
0.65 0.6 0.434
0.70 0.7 0.575
0.75 0.8 0.72
0.80 0.9 0.9
/
-- miscible
0.0 0.0 0.0
1.0 1.0 0.0
/

-----
-- WATER, SURFACTANT AND ALKALILNE PROPERTIES
-----

--Multicomponent water with surfactant
--Water and dissolved surfactant properties
WNAMEs
WATER NA CL SURF ALK/

CWTYPE
1* 1* 1* SURFS ALK/

MWW
18.015 22.98977 35.453 200 18.015/

PREFW
270 270 270 270 270/

DREFW
970.9 970.9 970.9 970.9 970.9/

CREFW
4.6E-5 4.6E-5 4.6E-5 4.6E-5 4.6E-5/

VREFW
0.34 0.34 0.34 34 120.7
0 0 0 0 0 /

EOS

```

```
SRK
/

-- HYDROCARBON AND SOLID COMPONENTS

CNames
C1      HC23      HC46      C8      HC13      HC18      HC26      HC43  /
CVTYPE
1*      1*      1*      1*      1*      1*      1*      1*  /
MW
16.04   35.88   67.98   110.14  173.11  248.85  361.77  600.98 /
PCRIT
46.00   45.59   34.25   30.01   22.30   16.18   11.88   10.27 /
TCRIT
190.60  338.20  456.95  575.51  666.13  735.04  807.07  914.84 /
VCRIT
0.09925 0.17071 0.29427 0.49206 0.8008  1.2180  1.8210  2.3880 /
ACF
0.008   0.1255  0.2325  0.4210  0.7174  0.9849  1.2737  1.6704 /
SDREF
1*      1*      1*      1*      1*      1*      1*      1*  /
ZI
0.528   0.1898  0.0703  0.0867  0.0529  0.034   0.0238  0.0145 /

BIC
0.0
0.0      0.0
0.027    0.0  0.0
0.0      0.0  0.0  0.0
0.0      0.0  0.0  0.0  0.0
0.0      0.0  0.0  0.0  0.0  0.0
0.0      0.0  0.0  0.0  0.0  0.0  0.0
/

PEDERSEN

ROCK
270 .3E-5 /

DENSITY
850. 1000. 1. /

RTEMP
40 /
```

Surfactant properties

```
SURFSTES
-- surf      water-oil
-- conc      surface tension
0.0      0.05
          0.05
          0.05
          0.05  /
2.0      4.90E-03
          3.75E-03
          3.50E-03
          4.00E-03  /
10.0     2.40E-03
          7.00E-04
          2.88E-04
          1.50E-03  /
14.0     1.82E-03
          4.00E-04
          1.44E-04
          1.00E-03  /
20.0     8.66E-04
          2.00E-04
          6.87E-05
          5.22E-04  /
/
```

```

ESSNODE
0    9    20    58.44    /

SALTEFF
SURF      2    0    0    0    1    0    0    1    0    0    0    0    0    /

ALSURFST
0.000 1.0000
0.003 0.9991
0.009 0.99908 /

SURFCAPD
-9      0.0
-4.5    0.0
-2      1.0
10      1.0 /
-9      0.0
-4.5    0.0
-2      1.0
10      1.0 /
-9      0.0
-4.5    0.0
-2      1.0
10      1.0 /

```

REGIONS

```

REGIONS      =====

SATNUM
300*1 / immiscible tables

SURNUM
300*2 / miscible tables

```

SOLUTION

```

SOLUTION      =====

EQUIL
2600 270 2700 0 0 0 1 /

WMFVD
2600 1.0 0.0 0.0 0.0 0.0
2700 1.0 0.0 0.0 0.0 0.0/

--
-- Initialisation Print Output
--

RPTSOL
'PRES' 'SOIL' 'SWAT' 'SURF' 'SURFST' 'SURFCNM' 'SURFKR' 'SURFCP' /

RPTNST
'RESTART' 'SOIL' 'SWAT' 'SURE' 'SURFST' 'SURFCNM' 'SURFKR' 'SURFCP'
'VWAT' 'KRO' 'KRW' 'PCOW' /

```

SUMMARY

```

SUMMARY      =====

BSURF
1  1 1 /
5  5 1 /
10 10 1 /
/

```

```

BSURFST
  1  1 1 /
  5  5 1 /
 10 10 1 /
/

BSURFCNM
  1  1 1 /
  5  5 1 /
 10 10 1 /
/

BSURFKR
  1  1 1 /
  5  5 1 /
 10 10 1 /
/

BSURFCP
  1  1 1 /
  5  5 1 /
 10 10 1 /
/

BKRO
  1  1 1 /
  5  5 1 /
 10 10 1 /
/

BKRW
  1  1 1 /
  5  5 1 /
 10 10 1 /
/

BWPC
  1  1 1 /
  5  5 1 /
 10 10 1 /
/

BALSURF
  1  1 1 /
  5  5 1 /
 10 10 1 /
/

BALSTML
  1  1 1 /
  5  5 1 /
 10 10 1 /
/

```

SCHEDULE

```

SCHEDULE =====
RPTSCHED
'PRES' 'SOIL' 'SWAT' 'SURF' 'SURFST' 'SURFCNM' 'SURFKR' 'SURFCP'
'SSOLID' 'VWAT' 'ALSURF' 'ALSTML'/

RPTRST
'RESTART' 'SOIL' 'SWAT' 'SURF' 'SURFST' 'SURFCNM' 'SURFKR' 'SURFCP'
'SSOLID' 'VWAT' 'KRO' 'KRW' 'PCOW' 'ALSURF' 'ALSTML'/

WELSPCLS
'OP' 'G' 10 10 2600 'OIL' /
'INJ' 'G' 1 1 2600 'WAT' /
/

COMPDAT
'OP' ' ' 10 10 1 3 'OPEN' 0 .0 .157 /

```

```
'INJ      ' 1 1 1 3 'OPEN' 0 .0 .157 /  
/  
WCONPROD  
'OP' 'OPEN' 'RESV' 4* 500 10 /  
/  
WELLSTRW  
WATER 0.4 0.3 0.3 0.0 0.0/  
SURF 0.97 0.0 0.0 0.03 0.0/  
ALK 0.98 0.0 0.0 0.0 0.02 /  
/  
WINJW  
INJ STREAM WATER /  
/  
WCONINJE  
'INJ' 'WAT' 'OPEN' 'RESV' 1* 500 500 /  
/  
TSTEP  
2*10 /  
  
WINJW  
INJ STREAM ALK /  
/  
TSTEP  
1*10 /  
  
WINJW  
INJ STREAM SURF /  
/  
  
TSTEP  
3*10 /  
  
END
```

Foam model

x	Eclipse 100
x	Eclipse 300

Foam can be used in a number of ways to increase the production from an oil reservoir. The foam acts to decrease the mobility of gas; this effect can be used to slow the breakthrough of injected gas or to reduce the production of gas cap.

A foam is generated by adding a surfactant to an aqueous phase, and passing a gas through the surfactant to generate a stable dispersion of gas bubbles in the liquid. The foam can be transported with the gas flow into the reservoir. The major beneficial effect of the foam is to reduce the mobility of the gas. Note that in foam flooding the reduction of the interfacial tension (between oil and water) is not a significant effect. The reduction of gas mobility typically depends on a range of factors including pressure and shear rate.

The foam stability has a major effect on the usefulness of foam injection. Typically the foam suffers from adsorption on to the rock matrix, decay over time, and enhanced decay in the presence of water.

The physics of the foam flooding process is in general very complex. For example, when foam bubbles form in a porous medium the bubble size typically fills the pore size of the rock matrix. These bubbles tend not to move until they are compressed (hence reducing their size) by applying a higher pressure. Then in turn more bubbles are generated at the new higher pressure, but with the original bubble size.

Eclipse 100

The Eclipse 100 foam model does not attempt to model the details of foam generation and collapse. Foam is modeled as tracer which may be transported with either the gas or the water phase with account taken of adsorption on to the rock surface and decay over time. Two models are available to describe the effect of foam on the gas mobility: a tabular model and a functional model.

Eclipse 300

The Eclipse 300 foam model is similar to the Eclipse 100 foam model. A water component is used to model the foam surfactant. The Eclipse 300 foam model does not have specific keywords for decay or adsorption. Instead the chemical reaction model can be used to model the decay of the surfactant, or to convert the surfactant into a solid component. The Eclipse 100 functional model of the gas mobility can be easily converted to an Eclipse 300 model.

Eclipse 100 foam model

Foam conservation equation

Although foam is essentially a mixture of gas, water and surfactant, Eclipse models it as an effective concentration of surfactant transported in either the gas or the water phase. Hence the foam concentration can be thought of as the surfactant concentration existing in foam form. The distribution of the injected foam is therefore solved by a conservation equation:

$$\begin{aligned} & \frac{d}{dt} \left(\frac{VS_g C_f}{B_r B_g} \right) + \frac{d}{dt} \left(V \rho_r C_f^a \frac{1-\phi}{\phi} \right) \\ &= \Sigma \left[\frac{Tk_{rg}}{B_g \mu_g} M_{rf} (\delta P_g - \rho_g g D_z) \right] C_f + Q_g C_f - \lambda(S_w, S_o) V C_f \end{aligned} \quad \text{Eq. 6.155}$$

as a tracer in the gas phase with decay, or as a tracer in the water phase with decay:

$$\begin{aligned} & \frac{d}{dt} \left(\frac{VS_w C_f}{B_r B_w} \right) + \frac{d}{dt} \left(V \rho_r C_f^a \frac{1-\phi}{\phi} \right) \\ &= \Sigma \left[\frac{Tk_{rw}}{B_w \mu_w} (\delta P_w - \rho_w g D_z) \right] C_f + Q_w C_f - \lambda(S_w, S_o) V C_f \end{aligned} \quad \text{Eq. 6.156}$$

where

C_f	denotes the foam concentration
ρ_w, ρ_g	denotes the water and gas density respectively
Σ	denotes the sum over neighboring cells
C_f^a	denotes the adsorbed foam concentration
μ_w, μ_g	denotes the water and gas viscosity respectively
D_z	is the cell center depth.
B_r, B_w, B_g	is the rock, water and gas formation volume respectively
T	is the transmissibility
k_{rw}, k_{rg}	is the water and gas relative permeability respectively
S_w, S_g	is the water and gas saturation respectively
V	is the block pore volume
Q_w, Q_g	is the water and gas production rate respectively
P_w, P_g	is the water and gas pressure respectively
λ	is the rate decay parameter function of oil and water saturation
M_{rf}	is the gas mobility reduction factor described below
g	is the acceleration due to gravity
ϕ	is the porosity.

By default, the foam concentrations are updated fully implicitly at the end of a timestep after the inter-block phase flows have been determined. If the foam effects on the flow properties are very strong, the foam concentrations can be solved in each Newton iteration so that the tracer concentrations are updated simultaneously with the phase equations by setting [item 126](#) of the `OPTIONS` keyword to 1. The new foam concentrations are then used to update the flow properties for the subsequent iterations.

The foam is assumed to exist only in either the gas or the water phase and either [equation 6.155](#) or [equation 6.156](#) only is solved. This is specified using the `FOAMOPTS` keyword, which is also used to specify which form of gas mobility reduction model is to be used.

Adsorption

The adsorption of foam is assumed to be instantaneous. The isotherm adsorption can be specified as:

- either a look-up table of adsorbed foam as a function of the flowing foam concentration using the `FOAMADS` keyword
- or by a generic analytical adsorption model using the `ADSORP` keyword with `FOAM` as an argument. This model enables the adsorption to depend on rock permeability as well.

The mass adsorbed foam on to the rock is given by:

$$V \cdot \left(\frac{1-\phi}{\phi} \right) \cdot \rho_r \cdot C_f^a \quad \text{Eq. 6.157}$$

and the mass density of the rock ρ_r should be supplied using the second argument of [FOAMROCK](#) keyword. If desorption is prevented using the first argument of the [FOAMROCK](#) keyword, then the adsorbed foam concentration may not decrease with time. If desorption is allowed then each grid block retraces the adsorption isotherm as the foam concentration rises and falls in the cell.

Foam decay

Foam effectiveness will typically reduce over time, even in conditions very favorable to foam stability. This reduction in effectiveness may be accelerated in the presence of water or oil. The reduction in foam effectiveness over time is modeled by foam decay $\lambda(S_w, S_o)$. The half-life of the decay can be a function of both oil and water saturation. If the decay half-life is a function of both oil and water saturation, the foam is assumed to decay with the minimum half-life. The corresponding data is entered using [FOAMDCYW](#) and [FOAMDCYO](#).

Gas mobility reduction

The foam modifies the gas mobility by way of a simple multiplier supplied as a function of foam concentration (that is the effective surfactant concentration). The mobility modification is applied explicitly; that is, the modification associated with the conditions at the end of each iteration or timestep is applied at the subsequent iteration or timestep.

Gas mobility reduction models

Two methods of modeling the gas mobility reduction M_{rf} due to foam are available:

Tabular model:

For this model the gas mobility reduction factor is modeled using a set of tables which specify the mobility reduction as tabulated functions of foam concentration ([FOAMMOB](#) keyword), pressure ([FOAMMOBP](#) keyword) and shear ([FOAMMOBS](#) keyword) from which the net mobility reduction factor is calculated.

Functional form:

For this model, the gas mobility reduction factor is modeled in terms of a set of functions which represent the individual reduction factors due to surfactant concentration ([FOAMFSC](#) keyword), oil saturation ([FOAMFSO](#) keyword), water saturation ([FOAMFSW](#) keyword) and capillary number ([FOAMFCN](#) keyword). These are combined multiplicatively with a reference mobility reduction factor ([FOAMFRM](#) keyword) to determine the net mobility reduction factor.

These two methods are selected using the [FOAMOPTS](#) keyword. This tabular gas mobility reduction model is available with either gas or water specified as the transport phase. However, the functional gas mobility reduction model is only available if water is specified as the transport phase. The formulation of these two models is described below.

Gas mobility reduction – tabular model

For the tabular model, the gas mobility reduction factor M_{rf} is expressed in the form:

$$M_{rf} = (1 - M_{rf}^{cp})M_v(V_g) + M_{rf}^{cp} \quad \text{Eq. 6.158}$$

$$M_{rf}^{cp} = (1 - M_c(C_f))M_p(P) + M_c(C_f) \quad \text{Eq. 6.159}$$

where

$M_v(V_g)$ is the mobility reduction modifier due to gas velocity ([FOAMMOBS](#) keyword).

$M_c(C_f)$ is the gas mobility reduction modifier due to foam concentration ([FOAMMOB](#) keyword).

$M_p(P)$ is the mobility reduction modifier due to pressure ([FOAMMOBP](#) keyword).

P is the oil phase pressure.

V_g is the gas velocity (see below).

The gas velocity is calculated according to the expression:

$$V_g = B_g \left(\frac{F_g}{\varphi \cdot A} \right) \quad \text{Eq. 6.160}$$

where

F_g is the gas flow rate in surface units.

B_g is the gas formation volume factor.

φ is the average porosity of the two cells.

A is the flow area between the two cells.

The gas mobility reduction modifiers due to pressure and gas velocity are optional and will tend to increase the gas mobility as compared to the mobility reduction due to foam concentration alone.

Gas mobility reduction - functional model

For the functional model, the mobility reduction factor M_{rf} is expressed in the form in [Kovscek \(1998\)](#) and [Cheng et al \(2000\)](#):

$$M_{rf} = \frac{1}{1 + (M_r \cdot F_s \cdot F_w \cdot F_o \cdot F_c)} \quad \text{Eq. 6.161}$$

where

M_r is the reference mobility reduction factor.

F_s is the mobility reduction factor component due to surfactant concentration.

F_w is the mobility reduction factor component due to water saturation.

F_o is the mobility reduction factor component due to oil saturation.

F_c is the mobility reduction factor component due to gas velocity (capillary number).

The reference mobility reduction factor M_r is typically in the range of 1 to 100 and corresponds to the normalized resistance to flow for a minimum bubble size in the absence of factors that increase bubble size. It is specified using the [FOAMFRM](#).

Each mobility reduction factor may optionally be specified or omitted according to whether or not the associated keyword is present. However, at least one of these components must be specified for the mobility reduction factor to be operational.

The mobility reduction factor F_s represents the dependence upon foam (effective surfactant) concentration and is expressed as:

$$F_s = \left(\frac{C_s}{C_s^r} \right)^{e_s} \quad \text{Eq. 6.162}$$

where

C_s is the effective surfactant concentration.

C_s^r is the reference surfactant concentration.

e_s is an exponent which controls the steepness of the transition about the point $C_s = C_s^r$.

The significance of F_s is determined by the reference surfactant concentration above which the presence of surfactant becomes significant in the creation of foam. For low surfactant concentrations $C_s < C_s^r$ (weak foam), the value of F_s will be less than 1 and will tend to 0 as the surfactant concentration decreases to 0. Conversely, for high surfactant concentrations $C_s > C_s^r$ (strong foam), the value of F_s will be greater than 1 and will increase with increasing surfactant concentration. The steepness and curvature of F_s versus C_s is determined by the value of e_s . The rate of change about the point where $C_s = C_s^r$ will increase with increasing e_s . The reference surfactant concentration and the exponent term are specified using the [FOAMFSC](#) keyword.

The mobility reduction factor F_w represents the dependence upon water saturation and is expressed as:

$$F_w = 0.5 + \frac{\text{atan}[f_w \cdot (S_w - S_w^1)]}{\pi} \quad \text{Eq. 6.163}$$

where

S_w is the water saturation.

S_w^1 is the limiting water saturation below which the foam ceases to be effective.

f_w is the weighting factor which controls the sharpness in the change in mobility.

The gas mobility increases sharply as water saturation decreases towards this limiting value. The limiting water saturation and the weighting factor are specified using the [FOAMFSW](#) keyword.

The mobility reduction factor F_o represents the dependence upon oil saturation and is expressed as:

$$F_o = \left(\frac{S_o^m - S_o}{S_o^m} \right)^{e_o} \quad \{S_o \leq S_o^m\} \quad \text{Eq. 6.164}$$

$$F_o = 0.0 \quad \{S_o > S_o^m\} \quad \text{Eq. 6.165}$$

where

S_o is the oil saturation.

S_o^m is the maximum oil saturation above which the foam ceases to be effective.

e_o is an exponent which controls the steepness of the transition about the point $S_o = S_o^m$

The maximum value of F_o is 1 and occurs when $S_o = 0.0$. For values of $S_o < S_o^m$, the value of F_o will decrease with increasing S_o . The rate of decrease is controlled by the value of the exponent e_o .

For values of $S_o \geq S_o^m$, the value of F_o will be 0 and will completely eliminate any gas mobility reduction irrespective of the values of the other multiplicative factors. The maximum oil saturation and the exponent term are specified using the [FOAMFSO](#) keyword.

The mobility reduction factor F_c represents the dependence upon capillary number and is expressed as:

$$F_c = \left(\frac{N_c^r}{N_c} \right)^{e_c} \quad \text{Eq. 6.166}$$

where

N_c is the capillary number.

N_c^r is the reference capillary number.

e_c is an exponent which controls the steepness of the transition about the point $N_c = N_c^r$.

This component controls the gas mobility for shear-thinning in the low foam quality region. The reference capillary number and the exponent term are specified using the [FOAMFCN](#) keyword. The capillary number is a dimensionless parameter which provides a measure of the ratio of viscous to capillary forces and is calculated according to the expression:

$$N_c = C_N \frac{\|K \cdot \nabla P\|}{\sigma_{wg}} \quad \text{Eq. 6.167}$$

where

K is the rock permeability.

P is the pressure.

σ_{wg} is the gas-water interfacial tension defined by the [FOAMFST](#) keyword.

C_N is the conversion factor which depends upon the units used.

The calculation of the capillary number is described in "[Capillary number](#)".

The mobility reduction factor is only calculated for grid blocks that contain foam with a concentration greater than or equal to the minimum concentration specified by item 3 of the [FOAMFSC](#) keyword.

Using the Eclipse 100 foam model

The Foam Model is activated by specifying the **FOAM** keyword in the **RUNSPEC** section.

It is possible to output the geometric data used to compute the shear thinning effect by using the **FOAM** mnemonic in the **RPTGRID** keyword. This outputs a map of the $1/(\text{poro} \times \text{area})$ values used to calculate the gas velocity.

Foam transport, adsorption and decay

The optional **FOAMOPTS** keyword may be used to select the transport phase for the foam and the method of modeling gas mobility reduction due to foam. If this keyword is not specified, the transport phase will default to gas and the tabular model for gas mobility reduction will be used.

Restriction	Foam transport, adsorption and decay keywords	Description
	FOAMOPTS	Foam transport phase and mobility reduction method options.
Optional	FOAMADS	Foam adsorption data
	FOAMROCK	Foam-rock properties
Optional	FOAMDCYW	Foam decay data as a function of water saturation
Optional	FOAMDCYO	Foam decay data as a function of oil saturation

Gas mobility reduction - tabular model

If the tabular method of modeling gas mobility reduction as a function of either or both pressure and shear is selected, there is one obligatory keyword in the **PROPS** section to describe the reduction in gas mobility as a function of foam concentration. In addition there are two optional keywords which can be used to control the gas mobility.

Restriction	Gas mobility tabular model keywords	Description
Required	FOAMMOB	Tables of the gas mobility reduction factor as a function of foam concentration
Optional	FOAMMOBP	Gas mobility reduction as a function of pressure
Optional	FOAMMOBS	Gas mobility reduction as a function of shear

Gas mobility reduction - functional model

The functional method of modeling gas mobility can only be used if the foam transport phase is water. If the functional method is selected, the following keywords are available in the **PROPS** section to describe the reduction in gas mobility in the presence of foam. The Gas mobility functional model keywords:

Restriction	Gas mobility functional model keywords	Description
Required	FOAMFSC	Foam mobility reduction dependence on foam surfactant concentration

Restriction	Gas mobility functional model keywords	Description
Optional	FOAMFRM	Foam mobility reference reduction factor
Optional	FOAMFCN	Foam mobility reduction dependence on capillary number
Obligatory if the FOAMFCN keyword specified	FOAMFST	Gas-water interfacial tension as a function of foam surfactant concentration
Optional	FOAMFSO	Foam mobility reduction dependence on oil saturation
Optional	FOAMFSW	Foam mobility reduction dependence on water saturation

With the exception of the FOAMFSC keyword, which must be specified if the functional model of gas mobility reduction is selected, and the FOAMFST keyword, which must be specified if the FOAMFCN keyword is specified, all of these keywords are optional.

The Foam Model's PROPS section data can be written to the PRINT file by using the FOAM mnemonic in the RPTPROPS keyword.

The injected foam concentration of a water or gas injector is specified using the WFOAM keyword in the SCHEDULE section.

Output controls

Output to the PRINT file can be controlled by using the following arguments in the RPTSCHEM and RPTSOL keywords:

Argument	Description
FOAM	Foam concentration for each grid block.
FIPFOAM	A foam balance sheet for the field and each region.
FOAMADS	Foam currently adsorbed onto the rock.
FOAMDCY	The decay half life and the total decayed foam.
FOAMMOB	The gas mobility reduction factor. Note that, for the tabular model of gas mobility reduction, the gas mobility reduction includes the pressure dependent effect but not the shear effect, because the shear effect is calculated from the local velocities.
WELLS=2	Generates a field/group/well/connections summary of the foam injection and production.

The list of SUMMARY keywords has been extended to output foam-specific data. These keywords follow the passive tracer format with a keyword concatenated with the tracer name. Here foam is identified with the tracer name FOA. For example, the keyword for the field total foam production would be FTTPRFOA.

Restrictions

If a cell is defined with zero pore volume in the GRID section, then making it active by changing the pore volume in the EDIT section using PORV, is not allowed with the foam option.

The Eclipse 300 foam model

Transport

In the Eclipse 300 Foam Model, surfactant is transported in the water phase as a separate water component. See ["Water components"](#). The surfactant component properties (such as molecular weight) must be specified using the relevant water component keywords. Water components that behave as foam surfactants are defined using the [CWTYP](#) keyword.

Adsorption

Foam adsorption and desorption can be modeled using chemical reactions. See ["Chemical Reactions"](#), ["Solid Phase"](#) and ["Adsorption of dissolved water components"](#).

The equilibrium between the waterborne surfactant component and the adsorbed solid component occurs when the concentrations are such that the two reaction rates are equal. If the reaction rates are linear functions of the concentrations, then the adsorbed (solid) concentration will be proportional to the surfactant concentration times the ratio of the adsorption and desorption reaction rates R_a/R_d .

An example of using reactions to model adsorption is given in ["Example data set for Eclipse 300"](#).

Decay

Foam decay can be modeled using a chemical reaction to convert the surfactant component to water. For example, a foam with a half life of $T_{1/2}$ days can be modeled with a reaction in which the foam surfactant decays to water at a rate of $(\ln 2)/T_{1/2}$ times the surfactant concentration.

It is not possible to vary the reaction rate as a function of the oil or water saturations directly, as done in the Eclipse 100 using the [FOAMDCYO](#) and [FOAMDCYW](#) keywords. However, it is possible to make the reaction rate depend on the concentration of a particular component (for example water) using the [REACCORD](#) keyword, or on the concentration of a tracer using the [REACTRAC](#) keyword. See ["Chemical reactions"](#) for more details of chemical reactions.

An example of using reactions to model foam decay is given in ["Example data set for Eclipse 300"](#).

Gas mobility reduction

Eclipse 100 gas mobility function keywords (see [6.3](#)) can be used to specify the gas mobility multiplier for the Eclipse 300 Foam Model. The surfactant concentration is calculated by summing the mass of all surfactant components in the cell and dividing by the water volume at surface conditions. The gas-water surface tension is then calculated the [FOAMFST](#) keyword. This is then used to determine the capillary number, which can be used to modify the mobility with the [FOAMFCN](#) keyword. The capillary number is calculated in different ways in the two simulators. This is discussed further in ["Capillary number"](#).

Comparison with Eclipse 100

If the surfactant component is given the same properties as the water component it will act as a tracer, as in the Eclipse 100 model. The foam volume will not be modeled, but its effect on the gas mobility will be. The Eclipse 300 Foam Model should therefore behave in a similar manner to the Eclipse 100 Foam Model, when that is used with water-phase transport and the functional gas mobility model (keyword [FOAMOPTS](#) items 1 and 2 set to WATER and FUNC).

Output controls

The relevant water component output mnemonics and keywords (and reaction / solid keywords if used) can be used to output information on the surfactant component. In addition the following foam mnemonics / keywords can be used:

Description	RPTSOL / RPTSCHED / RPTRST arguments	SUMMARY keyword
Surfactant concentration	FOAM	BFOAM
Gas-water surface tension	FOAMST	BFOAMST
Capillary number	FOAMCNM	BFOAMCNM
The gas mobility reduction factor	FOAMMOB	BFOAMMOB

Table 6.4: Output controls

Summary of keywords

RUNSPEC (Eclipse 100)

- [FOAM](#) Activates the Foam Model.

GRID (Eclipse 100)

Keyword	Description
RPTGRID	Controls output from the GRID section. The FOAM argument outputs a map of the $1/(\text{poro} \cdot \text{area})$ values used to calculate the gas velocity for the tabular model of mobility reduction.

PROPS

Restriction	Keyword	Description
Eclipse 100	ADSORP	Analytical adsorption isotherm (alternative to FOAMADS).
Eclipse 100	FOAMADS	Foam adsorption data.
Eclipse 100	FOAMDCYO	Foam decay data (as a function of oil saturation).
Eclipse 100	FOAMDCYW	Foam decay data (as a function of water saturation).
	FOAMFCN	Foam mobility reduction functional dependence upon capillary number.
	FOAMFRM	Foam mobility reference reduction factor.
	FOAMFSC	Foam mobility reduction functional dependence upon foam surfactant concentration.
	FOAMFSO	Foam mobility reduction functional dependence upon oil saturation.
	FOAMFST	Gas-water interfacial tension as a function of foam surfactant concentration.
	FOAMFSW	Foam mobility reduction functional dependence upon water saturation.
Eclipse 100	FOAMMOB	Tables of the gas mobility reduction factor as a function of foam concentration.

Restriction	Keyword	Description
Eclipse 100	FOAMMOBP	Gas mobility reduction as a function of pressure.
Eclipse 100	FOAMMOBS	Gas mobility reduction as a function of shear.
Eclipse 100	FOAMOPTS	Foam transport phase and mobility reduction method options.
Eclipse 100	FOAMROCK	Foam-rock properties.
Eclipse 100	RPTPROPS	Controls output from the <code>PROPS</code> section (FOAM argument for foam properties).

SUMMARY

Refer to [Foam SUMMARY keywords](#) in the *Eclipse Reference Manual*.

SOLUTION

Restriction	Keyword	Description
Eclipse 100	SFOAM	Initial foam concentrations.
Eclipse 100	TVDP	Initial foam concentrations with depth.
Eclipse 100	TNUM	Foam concentration regions.
	RPTSOL	Output of data specific to the Foam Model

Restriction	RPTSOL argument	Description
Eclipse 100	<code>FIPFOAM</code>	Output of foam fluid in place reports: 1: gives a report for the whole field. 2: in addition, gives a report for each fluid-in-place region.
	<code>FOAM</code>	Output of grid block foam concentration.
Eclipse 100	<code>FOAMADS</code>	Output of grid block foam adsorption.
	<code>FOAMCNM</code>	Output of logarithm base ten of grid block capillary number if the functional model of mobility reduction has been selected and the FOAMFCN keyword specified.
Eclipse 100	<code>FOAMDCY</code>	Output of grid block foam decay.
	<code>FOAMMOB</code>	Output of gas mobility factor due to foam.
Eclipse 100	<code>FOAMST</code>	Output of gas-water surface tension.

Note: The output of the gas mobility reduction factor for the tabular model will include the effects of pressure dependence ([FOAMMOBP](#)) but not the shear effect ([FOAMMOBS](#)), as the shear effect is calculated for each flow separately.

SCHEDULE

Restriction	Keyword	Description
Eclipse 100	RPTSCHED	Controls output from the <code>SCHEDULE</code> section. Argument <code>FOAM</code> outputs the foam concentrations and the foam adsorption when active.

Restriction	Keyword	Description
Eclipse 100	WFOAM	Foam concentration for injection wells (can be used as a User-Defined Argument (UDA)).

Restriction	RPTSCHED argument	Description
Eclipse 100	FIPFOAM	Output of foam fluid in place reports: 1: gives a report for the whole field. 2: in addition, gives a report for each fluid-in-place region.
	FOAM	Output of grid block foam surfactant concentration.
Eclipse 100	FOAMADS	Output of grid block foam adsorption.
	FOAMCNM	Output of logarithm base ten of grid block capillary number if the functional model of mobility reduction has been selected and the FOAMFCN keyword specified.
Eclipse 100	FOAMDCY	Output of grid block foam decay.
	FOAMMOB	Output of gas mobility factor due to foam.
Eclipse 300	FOAMST	Output of gas-oil surface tension.

Note: The output of the gas mobility reduction factor for the tabular model will include the effects of pressure dependence ([FOAMMOBP](#)) but not the shear effect ([FOAMMOBS](#)), as the shear effect is calculated for each flow separately.

Example data set for Eclipse 100

RUNSPEC

```

RUNSPEC
TITLE
  Simple FOAM model example
DIMENS
  10  10  3  /
OIL
WATER
GAS
DISGAS
FOAM
FIELD
EQLDIMS
  1  100  10  1  1  /
TABDIMS
  2  1  16  12  2  12  1  1  /
REGDIMS
  2  1  0  0  0  1  /
WELLDIMS
  2  1  1  2  /
NUPCOL
  4  /
START

```

```

19 'APR' 1994 /
NSTACK
24 /
FMTOUT

```

GRID

In this section, the geometry of the simulation grid and the rock permeabilities and porosities are defined.

The x and y direction cell sizes ([DX](#), [DY](#)) and the porosities are constant throughout the grid. These are set in the first 3 lines after the equals keyword. The cell thicknesses ([DZ](#)) and permeabilities are then set for each layer. The cell top depths ([TOPS](#)) are needed only in the top layer (though they could be set throughout the grid). The specified [MULTZ](#) values act as multipliers on the transmissibilities between the current layer and the layer below.

```

GRID
--      ARRAY  VALUE      ----- BOX -----
EQUALS
      'DX'      1000      /
      'DY'      1000      /
      'PORO'     0.3       /
      'DZ'       20       1  10  1  10  1  1  /
      'PERMX'    500       /
      'MULTZ'    0.64      /
      'TOPS'     8325      /
      'DZ'       30       1  10  1  10  2  2  /
      'PERMX'    50        /
      'MULTZ'    0.265625  /
      'DZ'       50       1  10  1  10  3  3  /
      'PERMX'   200        /
/
      EQUALS IS TERMINATED BY A NULL RECORD

-- THE Y AND Z DIRECTION PERMEABILITIES ARE COPIED FROM PERMX
--      SOURCE  DESTINATION ----- BOX -----
COPY
      'PERMX'    'PERMY'    1  10  1  10  1  3  /
      'PERMX'    'PERMZ'    /
/

-- OUTPUT OF DX, DY, DZ, PERMX, PERMY, PERMZ, MULTZ, PORO AND TOPS DATA
-- IS REQUESTED, AND OF THE CALCULATED PORE VOLUMES AND X, Y AND Z
-- TRANSMISSIBILITIES
RPTGRID
'DX' 'DY' 'DZ' 'PERMX' 'PERMY' 'PERMZ' 'MULTZ'
'PORO' 'TOPS' 'PORV' 'TRANX' 'TRANY' 'TRANZ' /

```

PROPS

```

PROPS
--
--      SWAT      KRW      PCOW
--
SWFN
0.12  0          0
1.0   0.1        0 /
/

--
--      SGAS      KRG      PCOG
--
SGFN
0      0          0
0.02  0          0
0.05  0.005      0
0.12  0.025      0
0.2   0.075      0

```

```

0.25 0.125 0
0.3 0.19 0
0.4 0.41 0
0.45 0.6 0
0.5 0.72 0
0.6 0.87 0
0.7 0.94 0
0.85 0.98 0
1.0 1.0 0
/
/
-- OIL RELATIVE PERMEABILITY IS TABULATED AGAINST OIL SATURATION
-- FOR OIL-WATER AND OIL-GAS-CONNATE WATER CASES
--
-- SOIL      KROW      KROG
SOF3
0      0      0
0.18   0      0
0.28   0.0001 0.0001
0.38   0.001  0.001
0.43   0.01   0.01
0.48   0.021  0.021
0.58   0.09   0.09
0.63   0.2    0.2
0.68   0.35   0.35
0.76   0.7    0.7
0.83   0.98   0.98
0.86   0.997  0.997
0.879  1      1
0.88   1      1 /
/
--
-- REF. PRES. REF. FVF COMPRESSIBILITY REF VISCOSITY VISCOSIBILITY
--
PVTW
4014.7 1.029 3.13D-6 0.31 0 /
--
-- REF. PRES COMPRESSIBILITY
--
ROCK
14.7 3.0D-6 /
--
-- SURFACE DENSITIES OF RESERVOIR FLUIDS
--
-- OIL WATER GAS
DENSITY
49.1 64.79 0.06054 /
--
-- PGAS BGAS VISGAS
PVDG
14.7 166.666 0.008
264.7 12.093 0.0096
514.7 6.274 0.0112
1014.7 3.197 0.014
2014.7 1.614 0.0189
2514.7 1.294 0.0208
3014.7 1.080 0.0228
4014.7 0.811 0.0268
5014.7 0.649 0.0309
9014.7 0.386 0.047 /
--
-- RS POIL FVFO VISO
PVTO
0.001 14.7 1.062 1.04 /
0.0905 264.7 1.15 0.975 /
0.18 514.7 1.207 0.91 /
0.371 1014.7 1.295 0.83 /
0.636 2014.7 1.435 0.695 /
0.775 2514.7 1.5 0.641 /

```

```

0.93 3014.7 1.565 0.594 /
1.270 4014.7 1.695 0.51
      5014.7 1.671 0.549
      9014.7 1.579 0.74 /
1.618 5014.7 1.827 0.449
      9014.7 1.726 0.605 /
/
--
--
-- FOAM ADSORPTION ISOTHERMS
--
FOAMADS
0.0 0.00000
1.0 0.00005
30.0 0.00005 /
0.0 0.00000
1.0 0.00002
30.0 0.00002 /
--
--
-- ROCK DENSITY AND ADSORPTION MODEL
--
FOAMROCK
1 2650 /
2 2650 /
--
--
-- FOAM DECAY DATA AS A FUNCTION OF WATER SATURATION
--
FOAMDCYW
0.0 3000
1.0 2000 /
0.0 3000
1.0 2000 /
--
--
-- FOAM DECAY DATA AS A FUNCTION OF OIL SATURATION
--
FOAMDCYO
0.0 3000.0
1.0 2500.0 /
0.0 3000.0
1.0 2500.0 /
--
--
-- FOAM MOBILITY REDUCTION
--
FOAMMOB
0 1
0.001 0.4
0.1 0.1
1.2 0.05 /
--
--
-- PRESSURE EFFECT ON THE MOBILITY REDUCTION
-- IN THIS CASE HIGHER PRESSURE REDUCES THE FOAM EFFECTIVENESS
--
FOAMMOBP
3000 0
6000 0.2 /
--
--
-- SHEAR RATE EFFECT ON THE MOBILITY
-- IN THIS CASE HIGH SHEAR REDUCES THE FOAM EFFECTIVENESS
--
FOAMMOBS
0.0 0
4.0 0.1 /
--
--
-- OUTPUT CONTROLS FOR PROPS DATA
--
RPTPROPS
'FOAM' /

```

REGIONS

```

REGIONS
EQUALS
  'SATNUM' 1 /
  'FIPNUM' 1 /
  'SATNUM' 2 1 3 1 3 1 3 /
  'FIPNUM' 2 1 3 1 3 1 3 /
/

```

SOLUTION

The SOLUTION section defines the initial state of the solution variables (phase pressures, saturations and gas-oil ratios)

```

SOLUTION
-- DATA FOR INITIALISING FLUIDS TO POTENTIAL EQUILIBRIUM
--
--      DATUM   DATUM   OWC   OWC   GOC   GOC   RSVD   RVVD   SOLN
--      DEPTH   PRESS   DEPTH PCOW DEPTH PCOG  TABLE TABLE METH
EQUIL
      8400     4800     8500     0     8200     0       1       0       0 /

-- VARIATION OF INITIAL RS WITH DEPTH
--
--      DEPTH    RS
RSVD
      8200  1.270
      8500  1.270 /

--
-- OUTPUT CONTROLS
--
RPTSOL
  'PRESSURE' 'SWAT' 'SGAS' 'FOAM' /

```

SUMMARY

This section specifies data to be written to the SUMMARY files and which may later be used with the Eclipse graphics package.

```

SUMMARY
--REQUEST PRINTED OUTPUT OF SUMMARY FILE DATA
RUNSUM
SEPARATE

-- FIELD OIL PRODUCTION
FOPR

-- WELL GAS-OIL RATIO FOR PRODUCER
WGOR
  'PRODUCER'
/

-- WELL BOTTOM-HOLE PRESSURE
WBHP
  'PRODUCER'
/

-- SATURATIONS IN INJECTION AND PRODUCTION CELL
BGSAT
10 10 3
1 1 1
/

BOSAT
10 10 3

```

```

1 1 1
/

BWSAT
10 10 3
1 1 1
/

-- PRESSURE IN INJECTION AND PRODUCTION CELL
BPR
10 10 3
1 1 1
/

FTPRFOA
FTPFOA
FTIRFOA
FTITFOA
FTIPTFOA
FTADSFOA
FTDCYFOA

BTCNFFOA
1 1 1 /
2 2 1 /
/

WTPRFOA
'OP' /

RTIPTFOA
1 /

BTADSFOA
1 1 1 /
/

BTDCYFOA
1 1 1 /
/

BTHLFFOA
1 1 1 /
/

RTADSFOA
1 /

RTDCYFOA
1 /

```

SCHEDULE

The SCHEDULE section defines the operations to be simulated.

```

SCHEDULE
-- CONTROLS ON OUTPUT AT EACH REPORT TIME
RPTSCHED
'PRESSURE' 'SWAT' 'SGAS' 'SOIL' 'FOAM' 'WELLS=2' 'NEWTON=2' 'FIPFOAM=2'
'TRADS' 'FOAMADS' 'FOAMDCY' 'FOAMMOB' /

RPTRST
'BASIC=3' 'FIP' 'FREQ=2' /

-- SET 'NO RESOLUTION' OPTION
DRSDT
0 /

-- WELL SPECIFICATION DATA
--
--      WELL      GROUP LOCATION  BHP    PI
--      NAME      NAME      I  J    DEPTH DEFN

```



```

WELSPECS
  'PRODUCER' 'G'    10 10    8400 'OIL' /
  'INJECTOR' 'G'     1  1    8335 'GAS' /
/

-- COMPLETION SPECIFICATION DATA
--
--      WELL      -LOCATION- OPEN/ SAT CONN  WELL
--      NAME      I  J K1 K2 SHUT  TAB FACT  DIAM
COMPDAT
  'PRODUCER'    10 10 3   3 'OPEN' 0   -1   0.5 /
  'INJECTOR'     1  1 1   1 'OPEN' 1   -1   0.5 /
/

--
-- PRODUCTION WELL CONTROLS
--
--      WELL      OPEN/  CNTL   OIL   WATER   GAS   LIQU   RES   BHP
--      NAME      SHUT   MODE  RATE   RATE   RATE   RATE   RATE
WCONPROD
  'PRODUCER' 'OPEN' 'ORAT' 20000  4*                               1000 /
/

--
-- INJECTION WELL CONTROLS
--
--      WELL      INJ   OPEN/  CNTL   FLOW
--      NAME      TYPE  SHUT   MODE   RATE
WCONINJE
  'INJECTOR' 'GAS' 'OPEN' 'RATE' 100000 /
/

TSTEP
  1 9 /

--
-- SPECIFY THE FOAM INJECTOR
--
WFOAM
  'INJECTOR' 1.1 /
/

TSTEP
  1 9 90 20*100 5*1000 /

END

```

Example data set for Eclipse 300

The ["Example data set for Eclipse 100"](#) can be modified to run with Eclipse 300. A new well is added to inject the foam surfactant in the water phase. The functional model is used for the gas mobility. Two components are added, to represent the foam surfactant in the water phase and that adsorbed to the rock.

RUNSPEC

The RUNSPEC section is modified to include the water components, solid model and number of reactions. In addition the number of wells is increased:

```

RUNSPEC

TITLE
  Simple FOAM model example

DIMENS
  10  10   3 /

OIL
WATER
GAS

```

```

SOLID          Solid component used to model adsorbed foam
BLACKOIL
COMPSOL
3 /           Oil, gas and one solid component
COMPW
2 /           Water and foam surfactant components
REACTION
2 1 1* 1* 1 / Foam decay, adsorption/desorption
FIELD

EQLDIMS
1 100 10 1 1 /

TABDIMS
2 1 16 12 2 12 1 1 /

REGDIMS
2 1 0 0 0 1 /

WELLDIMS
3 1 1 3 /

NUPCOL
4 /

START
19 'APR' 1994 /

NSTACK
24 /

FMTOUT

```

GRID

The GRID section is the same as the Eclipse 100 example:

```

GRID
--      ARRAY  VALUE      ----- BOX -----
EQUALS
'DX'    1000      /
'DY'    1000      /
'PORO'   0.3       /
'DZ'    20        1 10 1 10 1 1 /
'PERMX'  500       /
'MULTZ'  0.64      /
'TOPS'   8325      /
'DZ'    30        1 10 1 10 2 2 /
'PERMX'  50        /
'MULTZ'  0.265625 /
'DZ'    50        1 10 1 10 3 3 /
'PERMX'  200       /
/      EQUALS IS TERMINATED BY A NULL RECORD

-- THE Y AND Z DIRECTION PERMEABILITIES ARE COPIED FROM PERMX
--      SOURCE  DESTINATION  ----- BOX -----
COPY
'PERMX'      'PERMY'  1 10 1 10 1 3 /
'PERMX'      'PERMZ'  /
/

-- OUTPUT OF DX, DY, DZ, PERMX, PERMY, PERMZ, MULTZ, PORO AND TOPS DATA
-- IS REQUESTED, AND OF THE CALCULATED PORE VOLUMES AND X, Y AND Z
-- TRANSMISSIBILITIES
RPTGRID
'DX' 'DY' 'DZ' 'PERMX' 'PERMY' 'PERMZ' 'MULTZ'
'PORO' 'TOPS' 'PORV' 'TRANX' 'TRANY' 'TRANZ' /

```

PROPS

The PROPS section has a number of changes from the Eclipse 100 example:

- Solid component properties for adsorbed foam are specified
- Water component properties for water and the foam surfactant are specified
- The functional gas mobility reduction model is used
- A reaction is used to model foam decay
- Two reactions are used to model foam adsorption and desorption

```

PROPS
--
--  SWAT    KRW    PCOW
--
SWFN
  0.12  0      0
  1.0   0.1    0  /
/

--
--  SGAS    KRG    PCOG
--
SGFN
  0      0      0
  0.02   0      0
  0.05   0.005  0
  0.12   0.025  0
  0.2    0.075  0
  0.25   0.125  0
  0.3    0.19   0
  0.4    0.41   0
  0.45   0.6    0
  0.5    0.72   0
  0.6    0.87   0
  0.7    0.94   0
  0.85   0.98   0
  1.0    1.0    0
/
/

-- OIL RELATIVE PERMEABILITY IS TABULATED AGAINST OIL SATURATION
-- FOR OIL-WATER AND OIL-GAS-CONNATE WATER CASES
--
--  SOIL     KROW     KROG
--
SOF3
  0      0      0
  0.18   0      0
  0.28   0.0001  0.0001
  0.38   0.001   0.001
  0.43   0.01    0.01
  0.48   0.021   0.021
  0.58   0.09    0.09
  0.63   0.2     0.2
  0.68   0.35    0.35
  0.76   0.7     0.7
  0.83   0.98    0.98
  0.86   0.997   0.997
  0.879  1       1
  0.88   1       1  /
/

--
--  REF. PRES    COMPRESSIBILITY
--
ROCK
  14.7          3.0D-6  /
--

```

```

-- SURFACE DENSITIES OF RESERVOIR FLUIDS
--
--      OIL    WATER    GAS
DENSITY
      49.1    64.79   0.06054 /

--
--      PGAS    BGAS    VISGAS
PVDG
      14.7 166.666   0.008
      264.7 12.093   0.0096
      514.7   6.274   0.0112
      1014.7   3.197   0.014
      2014.7   1.614   0.0189
      2514.7   1.294   0.0208
      3014.7   1.080   0.0228
      4014.7   0.811   0.0268
      5014.7   0.649   0.0309
      9014.7   0.386   0.047 /

--
--      RS      POIL    FVFO    VISO
PVTO
      0.001    14.7 1.062   1.04 /
      0.0905   264.7 1.15   0.975 /
      0.18     514.7 1.207   0.91 /
      0.371    1014.7 1.295   0.83 /
      0.636    2014.7 1.435   0.695 /
      0.775    2514.7 1.5     0.641 /
      0.93     3014.7 1.565   0.594 /
      1.270    4014.7 1.695   0.51
              5014.7 1.671   0.549
              9014.7 1.579   0.74 /
      1.618    5014.7 1.827   0.449
              9014.7 1.726   0.605 /
/

--Solid component properties
CNAMEs
      OIL GAS SOLID /
CVTYPE
      1* 1* SOLID /
MW
      1* 1* 18.015 /
SDREF
      1* 1* 62.964 /

--Water component properties
CWTYPE
      1*      SURFF /      Water component 2 is foam surfactant
MWW
      18.015 18.015 /
PREFW
      4014.7 4014.7 /      PVTW item 1
DREFW
      62.964 62.964 /      (DENSITY item 2) / (PVTW item 2)
CREFW
      3.13D-6 3.13D-6 /      PVTW item 3
VREFW
      0.31    0.31
      0      0 /      PVTW items 4 and 5

--
--Foam decay, adsorption and desorption
--
STOREAC
--OIL GAS SOLID WAT SURF
      0.0 0.0 0.0 0.0 1.0 /      Decay      SURF->WATER
      0.0 0.0 1.0 0.0 0.0 /      SOLID<->SURF
STOPROD
--OIL GAS SOLID WAT SURF
      0.0 0.0 0.0 1.0 0.0 /      Decay      SURF->WATER
      0.0 0.0 0.0 0.0 1.0 /      SOLID<->SOLID
REACRATE
--DECAY      ADSO

```

```

0.000231 1.0 / Foam decay with constant half life
--                      = 3000 days
EQLDREAC
1 3 3 5 1 BOTH 1 / Rate proportional to SOLID - EQLDTAB_1(SURF)
EQLDTAB
0.0      0.00000
0.001    0.22065
1.000    0.22065 /
REACCORD
--OIL  GAS  SOLID  WAT  SURF
0.0    0.0    0.0    0.0  1.0 / Foam decay proportional to SURF conc
0.0    0.0    0.0    0.0  0.0 /
--
--
-- GAS MOBILITY REDUCTION
--
FOAMFRM
1.1 /
1.1 /
FOAMFSC
0.001 0.1 1.0e-6 /
0.001 0.1 1.0e-6 /
FOAMFST
0      10.0
0.01   5.0
0.05   0.1/
0      11.0
0.01   6.0
0.05   0.2 /
FOAMFCN
1E-7 0.1 /
1E-7 0.1 /
--
-- OUTPUT CONTROLS FOR PROPS DATA
--
RPTPROPS
'FOAM' /

```

REGIONS

The REGIONS section is unchanged:

```

REGIONS
EQUALS
'SATNUM' 1 /
'FIPNUM' 1 /
'SATNUM' 2 1 3 1 3 1 3 /
'FIPNUM' 2 1 3 1 3 1 3 /
/

```

SOLUTION

The SOLUTION section has been modified to include initial water composition. The initial solids are defaulted:

```

SOLUTION
-- DATA FOR INITIALISING FLUIDS TO POTENTIAL EQUILIBRIUM
--
--      DATUM  DATUM  OWC   OWC   GOC   GOC   RSVD  RVVD  SOLN
--      DEPTH  PRESS  DEPTH  PCOW  DEPTH  PCOG  TABLE TABLE METH
EQUIL
      8400    4800    8500    0     8200    0     1     0     0 /
-- VARIATION OF INITIAL RS WITH DEPTH

```

```

--
--      DEPTH      RS
RSVD
      8200  1.270
      8500  1.270  /
--
-- VARIATION OF INITIAL WATER COMPOSITION WITH DEPTH
--
--      DEPTH  WAT  SURF
WMFVD
      8200  1.0  0.0
      8500  1.0  0.0  /
--
-- OUTPUT CONTROLS
--
RPTSOL
  'PRESSURE' 'SWAT' 'SGAS' 'FOAM' /

```

SUMMARY

The simulator specific SUMMARY vectors are changed:

```

SUMMARY
--REQUEST PRINTED OUTPUT OF SUMMARY FILE DATA
RUNSUM

-- FIELD OIL PRODUCTION
FOPR

-- WELL GAS-OIL RATIO FOR PRODUCER
WGOR
'PRODUCER'
/

-- WELL BOTTOM-HOLE PRESSURE
WBHP
'PRODUCER'
/

-- SATURATIONS IN INJECTION AND PRODUCTION CELL
BGSAT
10 10 3 /
1 1 1 /
/

BOSAT
10 10 3 /
1 1 1 /
/

BWSAT
10 10 3 /
1 1 1 /
/

-- PRESSURE IN INJECTION AND PRODUCTION CELL
BPR
10 10 3
1 1 1
/

BSSAT
10 10 3 /
1 1 1 /
/

BFOAM
10 10 3 /
1 1 1 /
/

BREAC
10 10 3 1 /

```

```

10 10 3 2 /
10 10 3 3 /
1 1 1 1 /
1 1 1 2 /
1 1 1 3 /
/

```

SCHEDULE

The SCHEDULE section has been modified to reflect the different foam injection method. Water alternating with gas (WAG) is injected. The water contains a foam surfactant to reduce the mobility of the following gas:

```

SCHEDULE
-- CONTROLS ON OUTPUT AT EACH REPORT TIME
RPTSCHED
'PRESSURE' 'SWAT' 'SGAS' 'SOIL' 'FOAM' 'WELLS=2' 'NEWTON=2' 'FIPFOAM=2'
'FOAMMOB' /

RPTRST
'BASIC=3' 'FIP' 'FREQ=2' /

-- WELL SPECIFICATION DATA
--
--      WELL      GROUP LOCATION  BHP    PI
--      NAME      NAME      I  J    DEPTH DEFN
WELSP ECS
'PRODUCER' 'G'    10 10      8400 'OIL' /
'GINJ'     'G'    1  1      8335 'GAS' /
'WINJ'     'G'    1  1      8335 'WAT' /
/

--
-- SPECIFY THE FOAM INJECTOR
--
WELLSTRW
WATER  1.0 0.0 /    PURE WATER
FOAM   0.9 0.1 /    10% SURFACTANT IN WATER
/

WINJW
WINJ STREAM FOAM /    SET WELL WINJ TO USE THE FOAM STREAM
/

-- COMPLETION SPECIFICATION DATA
--
--      WELL      -LOCATION- OPEN/ SAT CONN  WELL
--      NAME      I  J K1 K2 SHUT  TAB FACT  DIAM
COMPDAT
'PRODUCER' 10 10 3 3 'OPEN' 0  -1  0.5 /
'GINJ'      1  1 1 1 'OPEN' 1  -1  0.5 /
'WINJ'      1  1 1 1 'OPEN' 1  -1  0.5 /
/

--
-- PRODUCTION WELL CONTROLS
--
--      WELL      OPEN/  CNTL   OIL   WATER   GAS   LIQU   RES   BHP
--      NAME      SHUT   MODE  RATE  RATE  RATE  RATE  RATE
WCONPROD
'PRODUCER' 'OPEN' 'ORAT' 20000 4*                                1000 /
/

--
-- INJECT FOAM
--
--      WELL      INJ   OPEN/  CNTL   FLOW
--      NAME      TYPE  SHUT   MODE  RATE
WCONINJE
'GINJ' 'GAS' 'SHUT' 'RATE' 100000 1* 10000 /

```

```
      'WINJ'  'WAT' 'OPEN' 'RATE'  65000 1* 10000 /
/
TSTEP
30 /

-- INJECT GAS
--
--      WELL      INJ  OPEN/  CNTL  FLOW
--      NAME      TYPE SHUT  MODE  RATE
WCONINJE
      'GINJ'  'GAS' 'OPEN' 'RATE' 100000 1* 10000 /
      'WINJ'  'WAT' 'SHUT' 'RATE'  65000 1* 10000 /
/

--
-- INJECT FOAM
--
--      WELL      INJ  OPEN/  CNTL  FLOW
--      NAME      TYPE SHUT  MODE  RATE
WCONINJE
      'GINJ'  'GAS' 'SHUT' 'RATE' 100000 1* 10000 /
      'WINJ'  'WAT' 'OPEN' 'RATE'  65000 1* 10000 /
/
TSTEP
30 /

-- INJECT GAS
--
--      WELL      INJ  OPEN/  CNTL  FLOW
--      NAME      TYPE SHUT  MODE  RATE
WCONINJE
      'GINJ'  'GAS' 'OPEN' 'RATE' 100000 1* 10000 /
      'WINJ'  'WAT' 'SHUT' 'RATE'  65000 1* 10000 /
/
TSTEP
70 /

END
```